Curved Motion, Depth Reduction, and Storage Codes for Matrix Transposition on Tapes

Hans Brende Claude Opus 5.5 (Anthropic)

October 2026

Abstract

Can a deterministic multitape Turing machine transpose a $d \times d$ matrix with $\lceil \log_2 d \rceil$-bit entries in time linear in the input length? The question is open, and a negative answer would give the first superlinear lower bound for off-line integer multiplication on such machines. We ask what a linear-time transposer would need, and we find three ingredients: curved head motion; depth-robust dependencies, or extreme coding on shallow ones; and storage codes that beat rate $\frac{1}{2}$. We prove that no machine, oblivious or adaptive, whose work heads are all home-based with legs made of boundedly many constant-rate runs transposes in linear time, and that the macroscopic head motion of every linear-time transposer contains a linear amount of curved, not countably-piecewise affine, travel (Theorems C and Cq). We show that two statements together would rule out oblivious linear-time transposers: depth reduction of the time–space dependency DAG by a cheap genie, and an index-coding lower bound on the remaining shallow cones (Theorem D). We prove depth reduction for every coupling of home-based walks (Theorem B), show that packing certificates cannot establish depth-robustness, and give an exact accounting of label costs on the sharpest open instance, the deep-nesting pair (Lemma Zq, Theorems E and E_1). For the index-coding side we study the dual notion, storage codes, on sparse graphs. Rate bounds pull back along homomorphisms (Theorem H), so every storage code, nonlinear included, on a graph that maps to C_5, to $K_{14/5}$ or to the Petersen graph has rate at most $\frac{1}{2}$, whatever its degree. The Clebsch graph has guessing number exactly 10, so bounded degree alone does not force rate $\frac{1}{2}$. On triangle-free graphs of maximum degree 3, no linear storage code over any field beats rate $\frac{1}{2}$ (Theorem S). On locally bipartite graphs of bounded degree, linear codes with bounded complex coefficients have rate at most $\frac{1}{2} + o(1)$, so gains of finite-field linear codes with a fixed number of symbols per vertex need p-torsion (Theorem A). The entropy-based results on storage codes (Theorem H, symmetrization, the good targets and the Clebsch graph), Theorem S and the exact label accounting (Lemma Zq, the exact cases of the chain and transfer identities, and Theorem E) are formalized in Lean 4; all other results are proved in full but not formalized. We end with a roadmap: positive answers to depth reduction for interacting walks and to the index-coding question on shallow cones, which is open for finite-field and nonlinear codes even in its undirected form, would rule out oblivious linear-time transposers, and our restricted results say what a construction would have to combine.

Contributions. Hans Brende chose the problem, fixed the verification standard, and directed the research. Claude Opus 5.5, a large language model developed by Anthropic, developed the mathematics, wrote the Lean formalization, which uses a port of J. P. Traverso Gianini's formalization of the Tutte–Berge formula [62], and wrote this paper under Hans Brende's direction.

Contents

1 Introduction **4**

1.1 The problem . . . 4
1.2 Three ingredients . . . 4
1.3 Results . . . 4
1.4 How to read this paper: the formal proofs . . . 6
1.5 Related work . . . 6
1.6 Organization . . . 7

2 Preliminaries **7**
2.1 Machines and the statement (T) . . . 7
2.2 Walks, hops and legs . . . 7
2.3 Storage codes and guessing numbers . . . 8
2.4 Polymatroids and rate bounds . . . 8
2.5 Formal status of the results . . . 8

3 Curved motion **9**
3.1 Rescaled limits and Theorem M* . . . 9
3.2 Speed-weighted returns and the Borel form of ($\star$) . . . 11
3.3 Hyperfinite returns . . . 13
3.4 Theorems C, C′ and Cq . . . 14
3.5 Proofs of Theorems Cq, C′ and C . . . 16
3.6 Relation to the Lebesgue form . . . 18
3.7 Consequences . . . 19

4 The decomposition: depth reduction plus an index-coding crux **24**
4.1 The dependency DAG of an oblivious run . . . 24
4.2 Two hypotheses and Theorem D . . . 26
4.3 What the crux must deliver . . . 28
4.4 Depth reduction for home-based walks: Theorem B . . . 30
4.5 Home-based oblivious machines reduce to the crux . . . 34

5 Depth reduction for interacting walks **36**
5.1 The problem . . . 37
5.2 Uniform recursive walks . . . 38
5.3 Union labels and the nesting barrier . . . 41
5.4 The deep-nesting pair . . . 44
5.5 Exact identities for labels . . . 48
5.6 Labels exact on one head . . . 52
5.7 The remaining gap . . . 55

6 Storage codes on sparse side information **57**
6.1 Entropic polymatroids . . . 58
6.2 Theorem H: rate bounds pull back along homomorphisms . . . 59
6.3 Symmetrization . . . 59
6.4 Good targets . . . 60
6.5 The Clebsch graph: guessing number ten . . . 61
6.6 Theorem S: linear codes on subcubic triangle-free graphs . . . 62
6.7 Bounded complex coefficients: Theorem A . . . 65
6.8 Codes over fields: torsion and odd characteristic . . . 69

6.9 Limits of local and spectral arguments . . . 72
6.10 Targets, Mycielskians and Question G . . . 76

7 What a linear-time transposer would need **79**
7.1 Necessary conditions . . . 79
7.2 An architecture compatible with these conditions . . . 80
7.3 Mechanisms that fail . . . 80

8 Roadmap and open problems **81**
8.1 Where the decision of (T) stands . . . 81
8.2 Open problems . . . 81

A Index of Lean declarations **82**

B Building and verifying the formalization **83**

1 Introduction

1.1 The problem

Can a deterministic multitape Turing machine transpose a $d \times d$ matrix with $b(d) = \max(1, \lceil \log_2 d \rceil)$-bit entries, written row by row, in time linear in the input length $L = d^2 b(d)$? This is the statement (T) of the first paper of this series [18]. Transposition is the data-movement step of fast integer multiplication on tapes, and by a reduction of Harvey and van der Hoeven [38, Cor. 6.9] a linear-time multiplication machine would give a linear-time transposer. So a proof of ¬(T) would give the first superlinear lower bound for off-line integer multiplication on multitape Turing machines, and a proof of (T) would remove the data-movement obstacle to linear-time multiplication.

The first paper proved, in Lean, that every transposer needs $\Omega(\log d)$ monotone passes and that the macroscopic head motion of every linear-time transposer must fold back (Theorems P, P′ and M*), and it reduced ¬(T) to a network-coding statement on tape networks, the Tape Coding Conjecture. The second paper [20] excluded, in its models, local coding schemes with low-treewidth interaction graphs and regular (piecewise affine) limiting motion, and the third [19] related bit and word transposition. The question remains open, and this paper does not decide it.

1.2 Three ingredients

This paper asks what a linear-time transposer would need, and finds three ingredients that an oblivious one would have to combine; the first is needed by adaptive machines as well.

- *Curved, head-dependent motion.* Its work heads cannot all move along home-based legs made of boundedly many constant-rate runs, and a linear amount of their macroscopic motion must be curved, in a way that no common change of clock removes.
- *Depth-robust dependencies, or extreme coding on shallow ones.* Cut its time–space dependency DAG with a cheap genie, which reveals a small set of intermediate values. Either the genie leaves large depth, or the shallow cones that remain must carry index codes saving all but an ε fraction of the input.
- *Codes that beat the pairing.* On sparse side information that looks locally like a tree, such codes must beat rate $\frac{1}{2}$, the rate of the code that pairs up receivers.

Our theorems prove restricted versions of each of these statements, and reduce the oblivious case of (T) to two precisely stated open problems (Theorem D): depth reduction for interacting walks, whose sharpest open instance has two walks, and an index-coding lower bound on the side information of shallow cones, whose undirected form is open for finite-field and nonlinear codes.

1.3 Results

Curved motion (Section 3). *Theorem C.* No machine, oblivious or adaptive, whose work heads are all home-based on every input, with legs made of boundedly many constant-rate runs, transposes in linear time (Theorem 3.12). The homes may depend on the input, and the number of legs, their scales and their interleaving are arbitrary. *Theorem Cq*: in every macroscopic limit of a linear-time transposer, the work heads travel a total distance of at least half a block unit, a linear number of cells, along motion that is not countably-piecewise affine (Theorem 3.14). Theorem C follows from it, and it contains the second paper's affine-time obstruction (Corollary 3.16). These results are unconditional; they use an extension of Theorem M* from interval partitions to Borel partitions, and the hyperfiniteness of amenable equivalence relations.

The decomposition (Section 4). *Theorem D.* If the dependency DAG of every oblivious linear-time run can be cut to bounded depth by revealing εL bits (*depth reduction*), and if index codes on the side information of bounded-depth cones need a broadcast of ρL bits with $\rho > \varepsilon$ (*the crux*), then no oblivious linear-time transposer exists (Theorem 4.10 and Corollary 4.11). The composition is elementary; its content is the formulation and the bookkeeping, including the forced trade-off $\varepsilon \geq \beta/(D+1) - \beta/L$ between the genie and the depth D, where β is the number of bits revealed per step (Proposition 4.13). *Theorem B* proves depth reduction for every coupling of k home-based walks of length T: εT cut edges leave depth $2^{O(k^3/\varepsilon)}$, whatever the walks do (Theorem 4.20). So an oblivious transposer with home-based work heads exists only if the crux fails, even in a weak form (Corollary B, Corollary 4.23).

Depth reduction (Section 5). For walks that turn far from home, depth reduction is open (Problem 5.1); for one walk it holds by planarity. We prove that no packing certificate, whatever structures it packs, can certify depth-robustness (Proposition 5.14 and Lemma 5.20(c)), and we sketch why per-head band bookkeeping, a natural strategy for depth reduction, fails on deeply nested walks (Remarks 5.6 and 5.7; proof sketches only). On what we regard as the sharpest open instance, the *deep-nesting pair*, we develop an exact accounting of label costs: *Lemma Zq* expresses the cost of any label, a map from times to $\{0, \ldots, q-1\}$ that describes a cut of depth $q-1$, through residues modulo q, and the chain-decomposition and transfer identities describe labels that are exact on one head. *Theorem E*: every hop-disjoint family of negative-defect cycles of head-2 hops forces such labels to pay about one violation per cycle (Theorem 5.21). *Theorem* E_1: on the golden deep-nesting pair with one level of nesting, labels exact on one head pay a fixed price per hop on the cover at every depth, at least $67/4096$ at all but the smallest scale (Theorem 5.22). Whether the pairs are depth-robust as m grows is open; we state one form of the missing step, for labels on the cover and a suitable odd shift, as Conjecture 5.25.

Storage codes (Section 6). The crux is open in general. Its undirected form asks whether storage codes on locally bipartite graphs of bounded degree can beat rate $\frac{1}{2}$ (Question 6.1).

- *Theorem H*: rate bounds pull back along homomorphisms of graphs and digraphs, for all codes (Theorem 6.5). With exact certificates for C_5, the Petersen graph and $K_{14/5}$, every storage code, nonlinear included, on a graph that maps to one of them has rate at most $\frac{1}{2}$, with no degree assumption (*Corollary H2*, Corollary 6.9). On cubic graphs of large girth the nonlinear case follows from a positive answer to Nešetřil's Pentagon Problem (Corollary 6.10).

- *Corollary H4*: the Clebsch graph has guessing number exactly 10, and Shannon bound exactly $5/8$ (Corollary 6.11 and Proposition 6.12). This answers, for the Clebsch graph, a question of Cameron, Dang and Riis [21], and shows that bounded degree alone does not force rate $\frac{1}{2}$.

- *Theorem S*: on triangle-free graphs G of maximum degree 3 with N vertices, every linear storage code over every field has dimension at most $\lfloor N/2 \rfloor$, and with the Tutte–Berge formula the minrank is $N - \nu(G)$, where $\nu(G)$ is the matching number (Theorems 6.13 and 6.16).

- *Theorem A*: on locally bipartite graphs of bounded degree, complex matrices with unit diagonal, graph support and bounded entries have rank at least $(\frac{1}{2} - o(1))N$; so over finite fields a gain of a linear code with a fixed number of symbols per vertex must come from p-torsion (Theorem 6.17 and Corollary 6.22).

We also show limits of local and spectral arguments (Section 6.9), and that Mycielski's construction does not preserve goodness, the Shannon rate bound $\frac{1}{2}$ of Definition 2.2 (Section 6.10).

Roadmap (Sections 7 and 8). We list what any linear-time transposer must do, describe an architecture compatible with these conditions, and state the open problems whose positive solution would rule out oblivious linear-time transposers.

1.4 How to read this paper: the formal proofs

The results on storage codes of Sections 6.1 to 6.6 (Theorem H, Corollaries H2–H4, Theorem S) and the exact label accounting of Section 5 (Lemma Zq, the exact cases of the chain and transfer identities, Theorem E) are formalized in Lean. For the matching form of Theorem S, the Lean library uses a port of an external formalization of the Tutte–Berge formula (Appendix B). All other results are proved in full but not formalized, among them Theorems D, B, C, Cq, E_1 and A, Propositions 4.13 and 5.14, the general forms of the chain and transfer identities, and the results of Sections 6.7 to 6.10. Each of them comes with a complete proof, and each numbered result states its status as described in Section 2.5. Claims with only a proof sketch, among them the nesting barrier for per-head band rules (Remarks 5.6 and 5.7), are stated in remarks and are not used in proofs.

1.5 Related work

Depth reduction and index coding. The route from a computation of small depth to index coding goes back to Valiant's depth reduction for circuits [63] and Riis's guessing games [58], and was used, in the language of min-rank, by Jukna and Schnitger [41]; a related network-coding view of oblivious external-memory algorithms, and through them of oblivious two-tape machines, is due to Adler, Harvey, Jain, Kleinberg and Lehman [1, Section 6, Theorems 6.2 and 6.3]. On multitape machines, Paul, Pippenger, Szemerédi and Trotter separated deterministic from nondeterministic linear time by a segregator argument on the dependency graph of a run [57]; depth reduction at linear cost may delete far more, a small constant fraction of the edges, but must then leave constant depth (Section 5.1).

Guessing games and storage codes. Guessing numbers were introduced by Riis [58] and studied for undirected graphs by Christofides and Markström [23], Baber, Christofides, Dang, Vaughan and Riis [4], and Cameron, Dang and Riis [22]. Storage codes on graphs and their duality with index coding were studied by Mazumdar [50] and by Arbabjolfaei and Kim [3]; Mazumdar, McGregor and Vorotnikova bounded the storage capacity above by vertex covers and averages of vertex covers, and below by fractional matchings [51], and Barg and Zémor constructed linear storage codes of rate approaching 3/4 on triangle-free graphs of growing degree [7]. Index coding was introduced by Birk and Kol [12] and formalized, with minrank, by Bar-Yossef, Birk, Jayram and Kol [6]; the Shannon linear-programming bound for it is due to Blasiak, Kleinberg and Lubetzky [13], and minrank goes back to Haemers [37].

Motion. Theorems C and Cq rest on the first paper's Theorem M* and on the theorem of Connes, Feldman and Weiss that amenable nonsingular equivalence relations are hyperfinite [25]; the Lebesgue-measure form of Theorem Cq for relatively open affine pieces (Corollary 3.16) slightly generalizes the second paper's affine-time obstruction [20]. With one work tape, much stronger lower bounds are classical: $\Omega(L^{3/2}/(\log L)^{1/2})$ steps without an output tape [31], and $\Theta(L^{5/4})$ for Boolean matrices with a write-only output tape [30]. Our motion results hold for any number of

tapes, and they leave room for curved motion such as the monotone expanders of Bourgain and Yehudayoff [17], on which the first paper's architecture is built.

1.6 Organization

Section 2 fixes notation. Section 3 proves the motion theorems, Section 4 the decomposition and depth reduction for home-based walks, Section 5 the results on depth reduction for interacting walks, and Section 6 the results on storage codes. Section 7 collects the consequences for constructions, and Section 8 the roadmap and open problems. The appendices index the Lean declarations and explain how to build and check the library.

2 Preliminaries

2.1 Machines and the statement (T)

We use the machine model of the first paper [18]: a deterministic Turing machine with a finite state set, a read-only two-way input tape, $k \geq 1$ two-way work tapes over a finite alphabet, and a write-only one-way output tape that receives at most one bit per step (the first paper also allows $k = 0$; an idle work tape can always be added). All heads start at cell 0; head moves are -1, 0 or $+1$; every step writes the cell under each work head. For $d \geq 1$ let $b(d) = \max(1, \lceil \log_2 d \rceil)$. The input is a $d \times d$ matrix with $b(d)$-bit entries written row by row, of length $L = d^2 b(d)$, and the machine must write its transpose in the same encoding. Statement (T) says that some machine does this within $c(L+1)$ steps for a constant c and all d. Its formal version, `Transpose.LinearTimeTransposable`, is fixed in a frozen file of the first paper's library.

A machine is *oblivious* if, for each input length, the positions of all heads at every time, the times at which it appends output bits, and its halting time do not depend on the input. An oblivious machine has a single head schedule per input length; an adaptive machine has one per input.

The Lean library of this paper does not model machines. Its theorems are about polymatroids, storage codes, matrices, and labels on finite sets with hop bijections; the machine-level statements of Sections 3 and 4 are not formalized, and each comes with a complete proof.

2.2 Walks, hops and legs

A *walk* of *length* T is a map $x\colon \{0, \dots, T-1\} \to \mathbb{Z}$ with $|x(t+1) - x(t)| \leq 1$. The work heads of a machine trace walks $x_1, \dots, x_k$, one per work tape, and in a run these walks may be coupled arbitrarily.

For a head h and a time u, let $\mathrm{prev}_h(u) = \max\{t < u : x_h(t) = x_h(u)\}$ when this set is nonempty. The pair $(\mathrm{prev}_h(u), u)$ is the *hop* of head h ending at u: the symbol that head h reads at step u was written at step $\mathrm{prev}_h(u)$, because every step writes the cell under each work head. A hop is *long* if $u - \mathrm{prev}_h(u) \geq 2$; the other hops are pauses. The *legs* of a walk partition its time interval greedily: the first leg starts at time 0, each leg is the longest interval beginning at its first time on which the walk is weakly monotone, and the next leg starts at the following time. So a leg ends just before the first step that reverses its direction. A walk is *home-based*, with home cell 0, if every leg begins or ends within distance 1 of 0. Theorems about head motion are invariant under shifting a tape, so the home cell can be any cell.

2.3 Storage codes and guessing numbers

Side information on a finite vertex set V is a map $N\colon V \to 2^V$. For a simple graph G on V it is the neighbourhood map $N(v) = \{u : uv \in E(G)\}$ (`nbhd`); for a digraph it is the out-neighbourhood map (`outNbhd`). Let α be a finite alphabet.

Definition 2.1 (Storage codes)**.** Lean: `IsStorageCode`, `IsRecoverable` [`StorageCode.lean`]
A set $C \subseteq \alpha^V$ is a *storage code* for N if for all $x, y \in C$ and every $v \in V$, x and y agree at v whenever they agree on $N(v)$. A random word $X\colon \Omega \to \alpha^V$ is *recoverable* for N if the same holds for all pairs of its values.

So in a storage code each symbol is a function of the symbols at its neighbours. The *rate* of C is $\log|C|/(|V|\log|\alpha|)$. The *guessing number* $\mathrm{gn}(G, q)$ is the largest value of $\log_q|C|$ over storage codes on G with $|\alpha| = q$, and the asymptotic guessing number is $\mathrm{gn}(G) = \sup_q \mathrm{gn}(G, q)$ [23, Def. 3.7 and Thm. 3.6]; guessing numbers for a fixed alphabet go back to Riis [58]. Storage codes are dual to index codes: for the same side information, the optimal rates of the two problems are complementary asymptotically, and exactly for linear codes [50, 3]. We work with storage codes throughout. A *linear* storage code over a field F is a subspace of F^V with the defining property; equivalently (Lemma 6.14), a subspace of the kernel of a matrix $M \in F^{V\times V}$ with unit diagonal whose off-diagonal entries vanish outside the edges of G. The largest dimension of such a kernel is $|V|$ minus the *minrank* of G over F [37, 6].

2.4 Polymatroids and rate bounds

Definition 2.2 (Polymatroids, validity, rate bounds)**.** Lean: `IsPolymatroid`, `IsValid`, `RateBound`, `IsGood` [`Polymatroid.lean`]
A *polymatroid* on V is a function $h\colon 2^V \to \mathbb{R}$ with $h(\emptyset) = 0$ that is monotone and submodular: $h(A \cup B) + h(A \cap B) \le h(A) + h(B)$. It is *valid* for side information N if $h(N(v) \cup \{v\}) = h(N(v))$ for every v. Side information N satisfies the *rate bound* c if every valid polymatroid satisfies

$$h(V) \;\le\; c \sum_{v\in V} h(\{v\}).$$

A graph G is *good* if its neighbourhood map satisfies the rate bound $\frac{1}{2}$, that is, $2h(V) \le \sum_v h(\{v\})$ for every valid polymatroid.

The rate bound quantifies over all polymatroids, so it is a statement about the *Shannon* information inequalities; the smallest admissible c is the Shannon bound λ_{Sh} of the side information. For a random word X on a finite probability space, $S \mapsto H(X_S)$ is a polymatroid (Lemma 6.2), and it is valid whenever X is recoverable (the converse holds when every outcome has positive probability). Hence a rate bound c bounds every storage code, linear or not, over every alphabet: $H(X_V) \le c\sum_v H(X_v)$ for every recoverable X, and $\log|C| \le c\,|V|\log|\alpha|$ for every storage code C (Theorem 6.3). Entropy is the finite Shannon entropy of the second paper [20], whose Lean definitions we import.

2.5 Formal status of the results

Every numbered result names its formal status in one of four ways. A line

Lean: `declaration, ...` [`file`]

gives the Lean declarations of a fully formalized result. The declarations live in the namespace `MotionDepthCodes` and the files in the directory `lean/MotionDepthCodes/`. The text after such a statement, labelled *Outline of the formal proof*, is a guide to the formal proof; it is a convenience, not a proof. A line beginning Lean (partial) marks a result whose formal counterpart covers all but an explicitly named part, which the text proves or cites. The words Not formalized mark a result with no formal counterpart; it comes with a complete proof, labelled *Proof (not formalized)*. A line beginning Cited marks a result of an earlier paper that is used without proof, and says where it is proved and whether it is Lean-verified there. Where the English statement simplifies, the Lean statement is authoritative. Results from the literature are cited; a few short ones are reproved in our notation, and say so. Numerical observations are labelled as such, claims that have only a proof sketch are stated in remarks beginning *Proof sketch only*, and neither is used in proofs. Numbered definitions carry a status line only when they are formalized.

The library depends on Lean 4 and Mathlib (both v4.34.1) and on the Lean libraries of the first two papers of this series and the repository's shared library `lean-machine`, imported through relative paths. It contains no `sorry`, no `native_decide` and no unsafe evaluation, and every declaration cited here depends only on the axioms `propext`, `Classical.choice` and `Quot.sound` (Appendix B). Appendix A lists every declaration of this library cited in the paper.

3 Curved motion

This section constrains the head motion of every linear-time transposer, oblivious or adaptive; the decomposition of the oblivious case follows in Section 4. The first paper [18] rescales the work-head trajectories of a machine, passes to uniform limits along subsequences of sizes, and proves that the limits of a linear-time transposer satisfy an isoperimetric inequality ($\star$). This is Theorem M*, restated as Theorem 3.2. In this section we turn ($\star$) into a constraint on the shape of the motion. We extend ($\star$) from labellings by finitely many intervals to Borel labellings, with the returns of the heads weighted by their speed (Theorem 3.6), and combine it with the theorem of Connes, Feldman and Weiss that orbit relations of countable amenable groups are hyperfinite off a null set. The outcome is Theorem Cq (Theorem 3.14): in every limit of a linear-time transposer, the work heads travel a total distance of at least $\frac{1}{2}$, in block units, along motion that is not countably-piecewise affine. Theorem C (Theorem 3.12) applies this to machines whose work heads are all home-based, with legs made of boundedly many runs at constant rate: no such machine, oblivious or adaptive, transposes in linear time. Section 3.7 draws consequences: one work tape never suffices, a common change of clock does not help, machines whose work heads all follow stroke trees are excluded, and a construction must use head-dependent curved motion on at least two interacting tapes (Corollary 3.24).

These are restricted impossibility results, and they are unconditional: they use neither depth reduction (Problem 5.1) nor any statement about codes. On the construction side they say which motion a linear-time transposer must have. No numbered result of this section is formalized in this paper's Lean library. Theorem M*, on which they rest, is Lean-verified in the first paper; the other ingredients (the area formula for Lipschitz functions, the Lusin–Novikov theorem, the amenability of solvable groups, the hyperfiniteness theorems of Ornstein and Weiss and of Connes, Feldman and Weiss in Kechris's formulation, and the characterization of hyperfiniteness of Dougherty, Jackson and Kechris) are cited.

3.1 Rescaled limits and Theorem M*

In this section the letter t indexes the work heads, from 0 as in the first paper, and h denotes a home cell. Let M be a machine with $k \geq 1$ work tapes whose work alphabet has $g+1$ symbols, and

let $G = \max(1, \lceil \log_2(g+1) \rceil)$ be the number of bits per work cell [18, Definition 6.7]. For $d \geq 1$ put $r_d = \lceil d\, b(d)/G \rceil$ and $R_d = d\, r_d$, so that $L(d)/G \leq R_d \leq L(d)/G + d$. Time is cut into *slots* of r_d steps and each work tape into *blocks* of r_d cells; a block holds about one row of the matrix. On an input w of size d let $x_{t,w}(s)$ be the cell of work head t after s steps, frozen once M halts. The *rescaled trajectory* of head t is

$$P^{(d)}_{t,w}(u) = \frac{1}{d} \Big\lfloor \frac{x_{t,w}(\lfloor ud \rfloor r_d)}{r_d} \Big\rfloor \qquad (u \geq 0),$$

the block of the head at the first step of slot $\lfloor ud \rfloor$, in units of d blocks [18, Section 7]. Macroscopic time and space are thus both measured in units of R_d steps or cells. We fix a time constant $c > 0$ and put $c_0 = cG + 1$; a run of at most $cL(d)$ steps ends by macroscopic time $cL(d)/R_d \leq cG < c_0$.

A *trace class* of M at size d [18, Definition 6.1] is a set F of matrix inputs of size d with a *representative* $w \in F$ such that all members of F halt after the same number of steps and have the same block-level trace as w: slot by slot, the same block residencies of the work heads, the same input rows touched and the same output length. Its *co-entropy* is the least D with $2^{L(d)} \leq 2^D |F|$. If M is oblivious, all matrix inputs of size d form one trace class, of co-entropy 0 [18, Lemma 6.2].

Definition 3.1 (Limits of a machine)**.** A family $(P_t)_{t<k}$ of functions $[0, c_0] \to \mathbb{R}$ is a *limit of M at time constant c* if there are sizes $d_i \to \infty$ and, for all large i, trace classes F_i of M at size d_i with representatives w_i, common halting times $T_i \leq cL(d_i)$ and co-entropies D_i with $D_i/L(d_i) \to 0$, such that M outputs the transpose on every input in F_i and $P^{(d_i)}_{t,w_i} \to P_t$ uniformly on $[0, c_0]$ for every $t < k$.

For a limit, and more generally for any family of 1-Lipschitz functions P_t on $[0, c_0]$, we write λ for Lebesgue measure and

$$\nu_t = |P_t'(u)|\, \mathrm{d}u \ \text{ on } [0, c_0], \qquad \Lambda_t(p) = \{u \in [0, c_0] : P_t(u) = p\}.$$

The derivative P_t' exists almost everywhere and $|P_t'| \leq 1$, so $\nu_t \leq \lambda$. We call ν_t the *speed measure* of head t: $\nu_t(A)$ is the distance travelled by the limit path during the times in A. Distance 1, which we call a *block unit*, is d blocks, that is, $R_d \approx L(d)/G$ cells.

The inequality $(\star)$ of the first paper uses labellings by finitely many intervals. An *excised set* is a finite union S of closed intervals. A *labelling* is a map $\beta \colon \mathbb{R} \to \mathbb{N}$ that is *piecewise constant off S*: for some finite set Z, β is constant on every interval $[u, v] \subseteq [0, c_0] \setminus (S \cup Z)$. Its *classes off S* are the sets $\{u \in [0, c_0] \setminus S : \beta(u) = \ell\}$, $\ell \in \mathbb{N}$. Two points $s < s'$ of $\Lambda_t(p)$ are *consecutive* if no point of $\Lambda_t(p)$ lies strictly between them. Let $\kappa_t(p)$ be the number of consecutive pairs $s < s'$ of $\Lambda_t(p)$ with $s, s' \notin S$ and $\beta(s) \neq \beta(s')$ if $\Lambda_t(p)$ is finite, and $\kappa_t(p) = 0$ if $\Lambda_t(p)$ is infinite. For $\theta \in [0, \infty]$ the inequality reads

$$(\star) \qquad 1 \ \leq \ \sum_{t<k} \int \kappa_t(p)\, \mathrm{d}p \ + \ 2\theta \ + \ 2k\, \lambda(S \cap [0, c_0]),$$

where $\int$ denotes the lower Lebesgue integral, which is defined and monotone for every function with values in $[0, \infty]$, measurable or not. We say that $(P_t)_{t<k}$ *satisfies $(\star)_\emptyset$* if $(\star)$ holds with $S = \emptyset$ for every $\theta \in (0, \frac{1}{2})$ and every labelling whose classes have measure at most θ.

Theorem 3.2 (Theorem M* [18])**.** Cited: proved and Lean-verified in [18]; not reproved here.
Let M be a machine with $k \geq 1$ work tapes and let $c > 0$.

(a) Every limit $(P_t)_{t<k}$ of M at time constant c consists of 1-Lipschitz functions on $[0, c_0]$ and satisfies $(\star)$ for every $\theta \in [0, \infty]$, every excised set S and every labelling, piecewise constant off S, whose classes off S have measure at most θ. In particular it satisfies $(\star)_\emptyset$.

(b) Let $d_i \to \infty$, and suppose that for all large i the machine M transposes every matrix of size d_i and halts within $cL(d_i)$ steps on it. Then M has a limit at time constant c along a subsequence of (d_i), with trace classes of co-entropy at most $C\, d_i\, b(d_i)$ for a constant C that depends only on M and c.

This combines Lemma 7.1(a), Theorems 7.8 and 7.9 and the popular trace theorem (Theorem 6.8) of [18]; the Lean declarations there are `Transpose.Tape.theoremMStar_level_class_on` for (a) (with `Transpose.Tape.blockTraj_limit_lipschitz` for the Lipschitz property) and `Transpose.Tape.theoremMStar_level_adaptive` for (b). The classes in (b) are the *popular* trace classes; for an oblivious machine one can take all matrix inputs. A machine that witnesses (T) with constant c transposes within $2cL(d)$ steps on every matrix input of every size, because $c(L+1) \le 2cL$ (see the proof of Corollary 7.11 of [18]; Lean: `Transpose.Tape.adaptive_hyps_of_linearTime`), so by (b) it has limits at time constant $2c$ along a subsequence of every sequence of sizes. We use only the case $S = \emptyset$ of (a).

3.2 Speed-weighted returns and the Borel form of $(\star)$

Statements about 1-Lipschitz functions, here and below, hold for every $c_0 > 0$; for walks and limits of machines, $c_0 = cG + 1$. In this subsection and the next, P is a 1-Lipschitz function on $[0, c_0]$, $\nu = |P'|\,\mathrm{d}u$ and $\Lambda(p) = \{u \in [0,c_0] : P(u) = p\}$; for a family (P_t) we add the index t to everything.

Lemma 3.3 (Levels)**.** Not formalized; a complete proof is given below.

(a) For every Borel set $A \subseteq [0,c_0]$ the function $p \mapsto \#(A \cap \Lambda(p)) \in \{0,1,\dots,\infty\}$ is Lebesgue measurable, and $\int_{\mathbb{R}} \#(A\cap\Lambda(p))\,\mathrm{d}p = \nu(A)$.

(b) $\Lambda(p)$ is finite for almost every p, and $\nu(\Lambda(p)) = 0$ for every p.

(c) $\nu(A) \le \lambda(A)$ for every Borel set $A \subseteq [0,c_0]$; in particular λ-null sets are ν-null.

Proof (not formalized). (a) Extend P to a 1-Lipschitz function on $\mathbb{R}$ that is constant on each side of $[0,c_0]$. The area formula for Lipschitz maps from $\mathbb{R}$ to $\mathbb{R}$ [34, Theorem 3.2.3] states that for every Lebesgue measurable set $A \subseteq \mathbb{R}$ the multiplicity function $p \mapsto \#(A \cap P^{-1}(p))$ is measurable and has integral $\int_A |P'|$. For $A \subseteq [0,c_0]$ the multiplicity is $\#(A\cap\Lambda(p))$. (For intervals A this is Banach's indicatrix theorem [5].) (b) For $A = [0,c_0]$, (a) gives $\int \#\Lambda(p)\,\mathrm{d}p = \nu([0,c_0]) \le c_0 < \infty$, so $\#\Lambda(p) < \infty$ for almost every p. For $A = \Lambda(p_0)$, which is closed, the integrand in (a) vanishes for $p \ne p_0$, so $\nu(\Lambda(p_0)) = 0$. (c) $|P'| \le 1$ almost everywhere. □

Definition 3.4 (First returns and crossing sets)**.** For $u \in [0,c_0]$ let $\psi(u) = \inf\{v \in [0,c_0] : v > u,\ P(v) = P(u)\}$, defined on the set $\operatorname{dom}\psi$ of those u for which this set is nonempty, and let $D^+ = \{u \in \operatorname{dom}\psi : \psi(u) > u\}$. A *Borel labelling* is a Borel map $\beta\colon [0,c_0] \to \mathbb{N}$; its *classes* are the sets $\beta^{-1}(\ell)$. Its *crossing set* is

$$A(\beta) = \{u \in D^+ : \beta(\psi(u)) \ne \beta(u)\}.$$

So $\psi(u)$ is the first return of the limit path to the level of u, and D^+ consists of the points that have a later point in their level set and are not limits of such points from the right. For a family (P_t) we write ψ_t, D_t^+ and $A_t(\beta)$.

Lemma 3.5 (Returns)**.** Not formalized; a complete proof is given below.

(a) $\operatorname{dom}\psi$ *and* D^+ *are Borel sets,* $\psi\colon \operatorname{dom}\psi \to [0,c_0]$ *is a Borel map, and the crossing set* $A(\beta)$ *of every Borel labelling* β *is Borel.*

(b) $P(\psi(u)) = P(u)$ *for every* $u \in \operatorname{dom}\psi$*, and* ψ *is injective on* D^+*.*

(c) $\nu(\psi^{-1}(W)\cap D^+) \le \nu(W)$ *for every Borel set* $W \subseteq [0,c_0]$*.*

(d) If $\Lambda(p)$ *is finite, its consecutive pairs are exactly the pairs* $(s,\psi(s))$ *with* $s \in \Lambda(p)\cap D^+$*.*

Proof (not formalized). (a) For $a \in \mathbb{R}$, the set $\{u \in \operatorname{dom}\psi : \psi(u) < a\}$ is the projection to the first coordinate of $K_a = \{(u,v) \in [0,c_0]^2 : u < v < a,\ P(u) = P(v)\}$: if $\psi(u) < a$, some v in the set whose infimum is $\psi(u)$ is below a, and conversely. K_a is the intersection of a closed set with an open subset of the compact square, hence σ-compact, so its projection is σ-compact and therefore Borel. Hence $\operatorname{dom}\psi = \{\psi < c_0+1\}$ is Borel and ψ is Borel on it; so are $D^+ = \{u \in \operatorname{dom}\psi : \psi(u) > u\}$ and $A(\beta) = \bigcup_\ell ((\beta^{-1}(\ell)\cap D^+)\setminus\psi^{-1}(\beta^{-1}(\ell)))$. (b) $\psi(u)$ is the infimum of a nonempty subset of the closed set $\Lambda(P(u))\cap[u,c_0]$, so it lies in that set. Let $u < u'$ be points of D^+ with $\psi(u) = \psi(u') = v$. Then $P(u) = P(v) = P(u')$, so u' belongs to the set whose infimum is $\psi(u)$, and $\psi(u) \le u' < \psi(u') = v$, a contradiction. (c) The set $\psi^{-1}(W)\cap D^+$ is Borel by (a), and by (b) ψ maps $\psi^{-1}(W)\cap D^+\cap\Lambda(p)$ injectively into $W\cap\Lambda(p)$ for every p. Integrate the inequality $\#(\psi^{-1}(W)\cap D^+\cap\Lambda(p)) \le \#(W\cap\Lambda(p))$ over p and apply Lemma 3.3(a) to both sides. (d) If $\Lambda(p)$ is finite and $s \in \Lambda(p)$ is not its largest point, then $\psi(s)$ is the next point of $\Lambda(p)$, so $\psi(s) > s$; the largest point is not in $\operatorname{dom}\psi$. □

By (d), for every labelling β and $S = \emptyset$ we have $\kappa(p) = \#(A(\beta)\cap\Lambda(p))$ whenever $\Lambda(p)$ is finite. For a piecewise constant labelling, which is Borel on $[0,c_0]$, the area formula therefore gives $\int\kappa = \nu(A(\beta))$: the integral in $(\star)$ is the speed-weighted measure of the set of times whose first return changes label. The next theorem extends $(\star)_\emptyset$, read in this way, to all Borel labellings. This is the form needed below, because the labellings produced by the hyperfiniteness argument are Borel but not piecewise constant.

Theorem 3.6 (Borel form of $(\star)$)**.** Not formalized; a complete proof is given below.

Let $(P_t)_{t<k}$ *be* 1*-Lipschitz functions on* $[0,c_0]$ *that satisfy* $(\star)_\emptyset$*, for instance a limit of a machine (Theorem 3.2). Then for every* $\theta \in (0,\frac12)$ *and every Borel labelling* β *of* $[0,c_0]$ *whose classes have Lebesgue measure at most* θ*,*

$$\sum_{t<k} \nu_t(A_t(\beta)) \ \ge\ 1 - 2\theta. \tag{1}$$

Proof (not formalized). Fix θ, β and $\varepsilon > 0$ with $\theta + \varepsilon < \frac12$. We first replace β by a labelling β' that is constant on finitely many intervals. The classes $Q_\ell = \beta^{-1}(\ell)$ are disjoint Borel sets with $\sum_\ell \lambda(Q_\ell) = c_0$, so there is J with $\lambda(\bigcup_{\ell\ge J} Q_\ell) \le \varepsilon$. For $\ell < J$ choose a finite union $U_\ell \subseteq [0,c_0]$ of intervals with $\lambda(U_\ell \triangle Q_\ell) \le \varepsilon/J$, put $U'_\ell = U_\ell \setminus \bigcup_{m<\ell} U_m$ and $V = [0,c_0]\setminus\bigcup_{\ell<J} U_\ell$, and cut V into finitely many intervals $V_1,\dots,V_n$ of length at most θ. Let $\beta' = \ell$ on U'_ℓ, $\beta' = J+i$ on V_i, and $\beta' = 0$ outside $[0,c_0]$. Then β' is constant on each interval of a finite partition of $[0,c_0]$ into intervals, hence piecewise constant (take Z to be the set of endpoints), and its classes have measure at most $\theta+\varepsilon$, because $\lambda(U'_\ell) \le \lambda(Q_\ell) + \varepsilon/J$. Let $W = V \cup \bigcup_{\ell<J}(U_\ell \triangle Q_\ell)$. Since $V \subseteq \bigcup_{\ell\ge J} Q_\ell \cup \bigcup_{\ell<J}(Q_\ell \setminus U_\ell)$, we have $\lambda(W) \le 3\varepsilon$. If $u \in [0,c_0]\setminus W$, let $\ell < J$ be least with $u \in U_\ell$, which exists because $u \notin V$; then $\beta'(u) = \ell$, and $\beta(u) = \ell$ because $u \notin U_\ell \triangle Q_\ell$. So $\beta' = \beta$ off W, and therefore

$$A_t(\beta') \subseteq A_t(\beta) \cup W \cup (\psi_t^{-1}(W)\cap D_t^+),$$

because a point $u \in D_t^+$ outside the right-hand side has $\beta'(u) = \beta(u) = \beta(\psi_t(u)) = \beta'(\psi_t(u))$. By Lemma 3.5(c) and Lemma 3.3(c), $\nu_t(A_t(\beta')) \le \nu_t(A_t(\beta)) + 2\lambda(W) \le \nu_t(A_t(\beta)) + 6\varepsilon$.

Now apply $(\star)_\emptyset$ to β', with $\theta+\varepsilon$ in place of θ. For every level p with $\Lambda_t(p)$ finite, Lemma 3.5(d) shows that $\kappa_t(p) = \#(A_t(\beta') \cap \Lambda_t(p))$; for the other levels $\kappa_t(p) = 0$. By monotonicity of the lower integral and Lemma 3.3(a), $\int \kappa_t \le \nu_t(A_t(\beta'))$. Hence

$$1 - 2\theta - 2\varepsilon \;\le\; \sum_t \nu_t(A_t(\beta')) \;\le\; \sum_t \nu_t(A_t(\beta)) + 6k\varepsilon,$$

and letting $\varepsilon \to 0$ proves (1). □

Why it matters. The Borel form is a measure-theoretic form of the case $S = \emptyset$ of Theorem M*: every partition of macroscopic time into Borel classes of small measure is crossed by first returns of total speed-weighted mass close to one. Motion that is stationary or slow counts for little. Together with the existence of limits, it is the only property of linear-time transposers that the rest of this section uses.

3.3 Hyperfinite returns

A *finite Borel equivalence relation* on a Borel set $Y \subseteq \mathbb{R}$ is an equivalence relation F on Y that is a Borel subset of $Y \times Y$ and whose classes are finite. The first lemma says that if the first returns of the heads lie, off a small set, in an increasing union of such relations, then $(\star)$ fails; the second supplies such unions for countable groups of affine maps; the third puts the returns between affine pieces into such a group.

Lemma 3.7 (Cheap labellings from hyperfinite transitions)**.** Not formalized; a complete proof is given below.

Let $(P_t)_{t<k}$ be 1-Lipschitz functions on $[0,c_0]$. Let $Y \subseteq [0,c_0]$ be a Borel set, let $F_1 \subseteq F_2 \subseteq \cdots$ be finite Borel equivalence relations on Y, and let $F = \bigcup_j F_j$. For each t let $B_t \subseteq D_t^+$ be a Borel set such that $u \in Y$, $\psi_t(u) \in Y$ and $(u,\psi_t(u)) \in F$ for every $u \in B_t$. Then for all $\theta > 0$ and $\varepsilon > 0$ there is a Borel labelling β of $[0,c_0]$ whose classes have Lebesgue measure at most θ and

$$\sum_{t<k} \nu_t(A_t(\beta)) \;\le\; \sum_{t<k} \nu_t(D_t^+ \setminus B_t) + \varepsilon.$$

Proof (not formalized). Truncation. For $j \ge 1$ let $C_{t,j} = \{u \in B_t : (u,\psi_t(u)) \notin F_j\}$. These sets are Borel, because $u \mapsto (u,\psi_t(u))$ is Borel on $\operatorname{dom}\psi_t$ (Lemma 3.5(a)) and F_j is Borel. They decrease in j, and their intersection is empty because $F = \bigcup_j F_j$. Since ν_t is finite, we can fix j with $\sum_t \nu_t(C_{t,j}) \le \varepsilon$.

Class minima. By the Lusin–Novikov theorem [44, Theorem 18.10], the Borel set F_j, whose sections are finite, is the union of the graphs of countably many Borel functions $f_n \colon Y_n \to Y$ defined on Borel sets $Y_n \subseteq Y$. So the class of $u \in Y$ is $\{f_n(u) : u \in Y_n\}$, and the map $\pi \colon Y \to Y$ that sends u to the least element of its class is Borel, being the infimum of the countably many Borel functions equal to f_n on Y_n and to $+\infty$ off Y_n. The map π is constant on classes, and $\pi^{-1}(v)$ is contained in the class of v, hence finite.

Grouping whole classes. The finite Borel measure $\mu(V) = \lambda(\pi^{-1}(V))$ on $[0,c_0]$ has no atoms, since finite sets are λ-null. So $x \mapsto \mu([0,x])$ is continuous, hence uniformly continuous on $[0,c_0]$, and equally spaced points $0 = x_0 < x_1 < \cdots < x_m = c_0$ with small enough spacing satisfy $x_i - x_{i-1} \le \theta$ and $\mu([x_{i-1},x_i]) \le \theta$ for all i. For $u \in Y$ let $\beta(u)$ be the least i with $\pi(u) \in [x_{i-1},x_i]$, and for $u \in [0,c_0] \setminus Y$ let $\beta(u) = m + i$ for the least i with $u \in [x_{i-1},x_i]$. Then β is a Borel labelling, its classes have measure at most θ, and it is constant on every F_j-class.

Crossings. If $u \in B_t \setminus C_{t,j}$, then u and $\psi_t(u)$ lie in one F_j-class, so $\beta(\psi_t(u)) = \beta(u)$ and $u \notin A_t(\beta)$. Hence $A_t(\beta) \subseteq (D_t^+ \setminus B_t) \cup C_{t,j}$, and summing ν_t over t gives the claim. □

Let $\mathrm{Aff}(\mathbb{R}) = \{u \mapsto au + b : a \neq 0\}$ be the affine group of the line. For a subgroup Γ let $R_\Gamma = \{(u, g(u)) : u \in \mathbb{R},\ g \in \Gamma\}$ be its orbit relation, and write Γu for the orbit of u.

Lemma 3.8 (Amenable groups give hyperfinite transitions)**.** Not formalized; a complete proof is given below.

Let Γ be a countable subgroup of $\mathrm{Aff}(\mathbb{R})$. There are a Borel set $Y_0 \subseteq \mathbb{R}$ with $\lambda(\mathbb{R} \setminus Y_0) = 0$ and finite Borel equivalence relations $F_1 \subseteq F_2 \subseteq \cdots$ on Y_0 with $\bigcup_j F_j = R_\Gamma \cap (Y_0 \times Y_0)$.

Proof (not formalized). The homomorphism $(u \mapsto au + b) \mapsto a$ from $\mathrm{Aff}(\mathbb{R})$ to the multiplicative group of $\mathbb{R}$ has an abelian image and an abelian kernel, the translations. So $\mathrm{Aff}(\mathbb{R})$ and Γ are solvable, and solvable groups are amenable, since abelian groups are amenable and amenability passes to extensions [65], [8, Appendix G]. The group Γ acts on $\mathbb{R}$ by homeomorphisms, so R_Γ is a countable Borel equivalence relation, a countable union of graphs of continuous maps. Let μ be a Borel probability measure on $\mathbb{R}$ with the same null sets as λ, for instance the Gaussian measure. By the theorem of Ornstein and Weiss [56] and of Connes, Feldman and Weiss [25], in the form stated in [43, p. 894], there is a Γ-invariant Borel set Y_0 with $\mu(Y_0) = 1$ such that R_Γ restricted to Y_0 is induced by a single Borel automorphism of Y_0. (Connes, Feldman and Weiss prove that an amenable countable equivalence relation with a quasi-invariant measure is hyperfinite up to a null set [25, Theorem 10]. Amenability of equivalence relations is Zimmer's notion [68], and the orbit relation of an action of an amenable group, free or not, is amenable [25, p. 431]. Here μ is quasi-invariant under Γ, because affine maps preserve λ-null sets.) By [32, Theorem 5.1], a countable Borel equivalence relation that is induced by a Borel automorphism is the union of an increasing sequence of finite Borel equivalence relations. Since μ and λ have the same null sets, $\lambda(\mathbb{R} \setminus Y_0) = 0$. □

Lemma 3.9 (Transitions between affine pieces)**.** Not formalized; a complete proof is given below.

Let P be 1-Lipschitz on $[0, c_0]$ and let $\varphi\colon [0, c_0] \to \mathbb{R}$ be an injective Borel map. Let E_a ($a \in \mathbb{N}$) be Borel subsets of $[0, c_0]$ and $f_a\colon \mathbb{R} \to \mathbb{R}$ affine maps with $P = f_a \circ \varphi$ on E_a, and let $N = [0, c_0] \setminus \bigcup_a E_a$. Let $\Gamma \leq \mathrm{Aff}(\mathbb{R})$ be the subgroup generated by the maps $f_b^{-1} \circ f_a$ for which neither f_a nor f_b is constant. Then there is a Borel set $B \subseteq D^+$ with $\nu(D^+ \setminus B) \leq 2\nu(N)$ such that $\varphi(\psi(u)) \in \Gamma\varphi(u)$ for every $u \in B$.

Proof (not formalized). If f_a is constant, then P is constant on E_a, so E_a lies in a level set and $\nu(E_a) = 0$ by Lemma 3.3(b). Let E^* be the union of the E_a with f_a not constant, and for $u \in E^*$ let $a(u)$ be the least such index with $u \in E_{a(u)}$. The set $N^* = [0, c_0] \setminus E^*$ is Borel, contains N, and is contained in N together with the ν-null sets E_a with f_a constant, so $\nu(N^*) = \nu(N)$. Let $B = \{u \in D^+ : u \in E^*,\ \psi(u) \in E^*\}$, a Borel set by Lemma 3.5(a). For $u \in B$, with $a = a(u)$ and $b = a(\psi(u))$, Lemma 3.5(b) gives $f_b(\varphi(\psi(u))) = P(\psi(u)) = P(u) = f_a(\varphi(u))$, so $\varphi(\psi(u)) = f_b^{-1}(f_a(\varphi(u))) \in \Gamma\varphi(u)$. Finally $D^+ \setminus B \subseteq N^* \cup (\psi^{-1}(N^*) \cap D^+)$, which has ν-measure at most $2\nu(N)$ by Lemma 3.5(c). □

3.4 Theorems C, C′ and Cq

We first define the class of Theorem C. Its members are home-based in the sense of Section 2.2, up to a tolerance, and every excursion from the home is made of boundedly many runs at constant rate.

Definition 3.10 (Home-based heads with affine runs)**.** Let $K \geq 1$ be an integer and $\rho \geq 0$. A function $x\colon \{0, \dots, T\} \to \mathbb{R}$, for instance a walk of length $T + 1$, is *HB-AR(K, ρ)* if there are a point h, its *home*, and times $0 = a_0 < a_1 < \cdots < a_m = T$ such that every *segment* $[a_{l-1}, a_l]$ satisfies:

(i) (home-based) $|x(a_{l-1}) - h| \leq \rho$ or $|x(a_l) - h| \leq \rho$;

(ii) (affine runs) there are times $a_{l-1} = s_0 \le s_1 \le \cdots \le s_K = a_l$ and rates $q_1, \ldots, q_K \in [-1, 1]$ such that $|x(s) - x(s_{j-1}) - q_j(s - s_{j-1})| \le \rho$ whenever $s_{j-1} \le s \le s_j$.

The intervals $[s_{j-1}, s_j]$ are the *runs* of the segment. A machine *has HB-AR(K, ρ) work heads at size d* if on every matrix input of size d the trajectory of every work head, up to the halting time, is HB-AR(K, ρ).

Remark 3.11 (What the class contains)**.** A head that stays within distance $\rho/2$ of its home during a time interval contributes a segment with a single run of rate 0, so stays at home cost nothing. A home-based walk in the sense of Section 2.2, with home h, whose legs consist of at most K runs at constant rate up to deviation ρ is HB-AR$(K, \rho + 2)$: cut the time axis at the first step of every leg; each resulting segment is contained in the corresponding leg, or exceeds it by one step at its end. If x is HB-AR(K, ρ) and $|x - x'| \le \rho'$, then x' is HB-AR$(K, \rho + 2\rho')$. With $\rho(d) = o(L(d))$ the class therefore contains the usual home-based schedules: sweeps out and back from a home at the end of the used tape, with any number of passes; copying excursions with small zigzags; excursions of all sizes in any order; any coupling between the heads; excursions on both sides of the home; and homes that depend on the size and on the input. Nothing is assumed about the input head or the output. The class excludes curved macroscopic motion: a macroscopic pause inside an excursion costs a run, and an excursion whose speed varies along its length, for instance unit moves alternating with pauses at a slowly varying duty cycle, needs unboundedly many runs.

For a machine M let $T_M(d)$ be the largest halting time of M on matrix inputs of size d (infinite if some input does not halt).

Theorem 3.12 (Theorem C: home-based heads with affine runs)**.** Not formalized; a complete proof is given below.

Let $K \ge 1$ and $\rho(d) = o(L(d))$, and let M be a machine that transposes. Suppose that for all large d the machine M has HB-AR$(K, \rho(d))$ work heads at size d. Then $T_M(d)/L(d) \to \infty$. In particular M does not witness (T).

The hypothesis concerns every work head of M; for an oblivious machine it is a condition on its single head schedule. For an adaptive machine it must hold on every input, with K and ρ uniform; the proof uses it only on the representatives of the popular trace classes.

Why it matters. Theorem C is an unconditional impossibility result for a natural class: it needs neither depth reduction nor the crux. The class allows any number of passes, excursions at all scales in any interleaving, any coupling of the heads and arbitrary motion of the input head, and it includes adaptive machines. For oblivious machines whose work heads are all home-based, Corollary 4.23 reaches the same conclusion for every motion, but only if the crux (Definition 4.8) holds; Theorem C needs no coding statement, for the subclass with affine runs. For a construction it says that the work heads cannot all be home-based with legs made of boundedly many constant-rate runs. By Theorem Cq, in every limit, motion that is not countably-piecewise affine must carry at least half a block unit of travel in total; for a home-based head such motion needs legs with unboundedly many runs whose speed varies along the leg, as produced by counters or rulers.

Theorem 3.13 (Theorem C′: open affine pieces)**.** Not formalized; a complete proof is given below.

Let M be a machine that transposes. Suppose that for every $c > 0$, every limit $(P_t)_{t<k}$ of M at time constant c admits relatively open sets $U_t \subseteq [0, c_0]$ such that P_t is affine on each connected component of U_t and $\nu_t([0, c_0] \setminus U_t) = 0$. Then $T_M(d)/L(d) \to \infty$.

Theorem C′ allows the non-affine set $[0, c_0] \setminus U_t$ to have positive Lebesgue measure, provided the head is stationary on almost all of it. Theorem C follows from the quantitative statement below, and so does C′.

Why it matters. Theorem C′ is the form of Theorem Cq that is easiest to check on a concrete schedule: a construction whose limit motion is affine on open intervals wherever the heads move fails, even when the set on which the heads rest is a fat Cantor set (Remark 3.17).

Theorem 3.14 (Theorem Cq: curved motion)**.** Not formalized; a complete proof is given below.
Let $(P_t)_{t<k}$ be 1-Lipschitz functions on $[0, c_0]$ that satisfy $(\star)_\emptyset$; by Theorem 3.2 this holds for every limit of every machine. For each t let $E_{t,0}, E_{t,1}, \dots$ be Borel subsets of $[0, c_0]$ on each of which P_t agrees with an affine function, and let $N_t = [0, c_0] \setminus \bigcup_a E_{t,a}$. Then

$$\sum_{t<k} \nu_t(N_t) \;\geq\; \tfrac{1}{2}.$$

In words: in every limit of a linear-time transposer, the limit paths of the work heads travel a total distance of at least $\frac{1}{2}$, in block units, while moving in a way that is not countably-piecewise affine. Since distance 1 corresponds to about L/G cells, this is about $L/(2G)$ cells of curved travel.

Why it matters. Theorem Cq is the quantitative statement behind Theorems C and C′, and it contains the Lean-verified affine-time obstruction of the second paper (Corollary 3.16). It tells a construction how much curved motion it needs: at least half a block unit of travel, whatever else the machine does. It does not exclude curved motion (Remark 3.18).

3.5 Proofs of Theorems Cq, C′ and C

We prove Theorem Cq (Theorem 3.14) first. The proof combines the three lemmas of Section 3.3 with the Borel form of $(\star)$.

Proof (not formalized). Fix $\theta \in (0, \frac{1}{2})$ and $\varepsilon > 0$. For each t choose affine maps $f_{t,a}$ with $P_t = f_{t,a}$ on $E_{t,a}$, and apply Lemma 3.9 to P_t with φ the identity. This gives Borel sets $B_t \subseteq D_t^+$ with $\nu_t(D_t^+ \setminus B_t) \leq 2\nu_t(N_t)$ and $\psi_t(u) \in \Gamma u$ for $u \in B_t$, where $\Gamma \leq \mathrm{Aff}(\mathbb{R})$ is the countable subgroup generated by the maps $f_{t,b}^{-1} \circ f_{t,a}$ ($t < k$, neither $f_{t,a}$ nor $f_{t,b}$ constant). Take Y_0 and F_j from Lemma 3.8, and put $Y = Y_0 \cap [0, c_0]$, $F_j' = F_j \cap (Y \times Y)$ and $B_t' = B_t \cap Y \cap \psi_t^{-1}(Y)$. The F_j' are increasing finite Borel equivalence relations on Y with union $R_\Gamma \cap (Y \times Y)$, and for $u \in B_t'$ we have $u, \psi_t(u) \in Y$ and $(u, \psi_t(u)) \in R_\Gamma$. Since $\lambda([0, c_0] \setminus Y) = 0$, Lemma 3.3(c) and Lemma 3.5(c) give

$$\nu_t(D_t^+ \setminus B_t') \leq \nu_t(D_t^+ \setminus B_t) + \nu_t([0, c_0] \setminus Y) + \nu_t(\psi_t^{-1}([0, c_0] \setminus Y) \cap D_t^+) \leq 2\nu_t(N_t).$$

Lemma 3.7 now gives a Borel labelling β whose classes have measure at most θ and $\sum_t \nu_t(A_t(\beta)) \leq 2\sum_t \nu_t(N_t) + \varepsilon$, while Theorem 3.6 gives $\sum_t \nu_t(A_t(\beta)) \geq 1 - 2\theta$. Hence $1 - 2\theta \leq 2\sum_t \nu_t(N_t) + \varepsilon$, and letting θ and ε tend to 0 proves the theorem. □

Theorem C′ (Theorem 3.13) follows from Theorem Cq and the existence of limits.

Proof (not formalized). Suppose not. Then there are $c > 0$ and sizes $d_i \to \infty$ with $T_M(d_i) \leq cL(d_i)$, so M transposes and halts within $cL(d_i)$ steps on every matrix input of size d_i, and by Theorem 3.2(b) it has a limit $(P_t)_{t<k}$ at time constant c. Let U_t be as in the hypothesis. The components of U_t are countably many relatively open intervals, on each of which P_t is affine. By Theorem 3.14, applied to these intervals, $\sum_t \nu_t([0, c_0] \setminus U_t) \geq \frac{1}{2}$, which contradicts $\nu_t([0, c_0] \setminus U_t) = 0$. □

The proof of Theorem C (Theorem 3.12) rests on the structure of the limits of HB-AR heads: they sit at a home level, apart from countably many *teeth*, each piecewise affine with boundedly many pieces. The teeth may be dense in time.

Lemma 3.15 (Limits of home-based heads with affine runs)**.** Not formalized; a complete proof is given below.

Let $d_i \to \infty$, $c > 0$, $K \geq 1$ and $\rho_i = o(L(d_i))$, and let $x_i \colon \{0, \dots, T_i\} \to \mathbb{Z}$ be walks with $x_i(0) = 0$ and $1 \leq T_i \leq cL(d_i)$ that are HB-AR(K, ρ_i). Extend x_i by $x_i(s) = x_i(T_i)$ for $s > T_i$, and suppose that $P_i(u) = d_i^{-1} \lfloor x_i(\lfloor ud_i \rfloor r_{d_i})/r_{d_i} \rfloor$ converges uniformly on $[0, c_0]$ to a function P. Then there is a level $p \in \mathbb{R}$ with the following properties.

(a) The set $\{u \in [0, c_0] : P(u) \neq p\}$ is the union of countably many disjoint relatively open intervals of $[0, c_0]$, the teeth*.*

(b) For each tooth I there is a set $\Phi_I \subseteq I$ of at most $2K$ points such that P is affine on each component of $I \setminus \Phi_I$.

(c) For every $y > 0$, at most $c_0/(2y) + 2$ teeth contain a point u with $|P(u) - p| > y$.

(d) $\nu(\{u \in [0, c_0] : P(u) = p\}) = 0$, where $\nu = |P'| \, \mathrm{d}u$.

Proof (not formalized). Write $r_i = r_{d_i}$ and $R_i = R_{d_i}$. Let $X_i \colon [0, \infty) \to \mathbb{R}$ interpolate x_i linearly between integers, and let $Q_i(u) = X_i(uR_i)/R_i$; it is 1-Lipschitz. If $s = \lfloor ud_i \rfloor r_i$, then $0 \leq uR_i - s < r_i$, so $|X_i(uR_i) - x_i(s)| \leq r_i$, and $|P_i(u) - x_i(s)/R_i| \leq 1/d_i$. Hence $|P_i - Q_i| \leq 2/d_i$, so $Q_i \to P$ uniformly on $[0, c_0]$, and P is 1-Lipschitz (as in [18, Lemma 7.1(a)]).

Rescaled segments. Let h_i and $0 = a_0 < \cdots < a_m = T_i$ be a home and segments for x_i as in Definition 3.10, and put $p_i = h_i/R_i$ and $\varepsilon_i = \rho_i/R_i$; then $\varepsilon_i \leq G\rho_i/L(d_i) \to 0$. Divided by R_i, the segments become closed intervals that partition $[0, T_i/R_i]$, and we add the interval $[T_i/R_i, c_0]$, on which Q_i is constant, to the last segment as one more run, of rate 0; recall $T_i/R_i \leq cG < c_0$. On a run $[s_{j-1}, s_j]$ the function $X_i(s) - X_i(s_{j-1}) - q_j(s - s_{j-1})$ is linear between consecutive integers, so its absolute value is at most ρ_i at all real s of the run. We obtain a partition of $[0, c_0]$ into closed intervals, the segments, such that each segment has an endpoint at which $|Q_i - p_i| \leq \varepsilon_i$ (for the extended last segment, recall that Q_i is constant on the added interval), and each segment is the union of at most K consecutive closed intervals, its runs, on each of which $|Q_i(u) - Q_i(u') - q(u - u')| \leq 2\varepsilon_i$ for some $q \in [-1, 1]$; the last segment has at most $K + 1$ runs. The first segment has an endpoint s with $|x_i(s) - h_i| \leq \rho_i$, and $|x_i(s)| \leq s \leq T_i$, so $|p_i| \leq cG + \varepsilon_i$. Passing to a subsequence, which does not change P, we may assume that $p_i \to p$.

(a) The set $\{P \neq p\}$ is relatively open in $[0, c_0]$; its components are disjoint relatively open intervals, and there are countably many of them.

(b) Let I be a tooth and $[\alpha, \beta] \subseteq I$ a compact interval with $\alpha < \beta$. Then $\delta = \min_{[\alpha,\beta]} |P - p| > 0$, and for large i we have $\sup_{[0,c_0]} |Q_i - P| < \delta/4$, $|p_i - p| < \delta/4$ and $\varepsilon_i < \delta/2$, so that $|Q_i - p_i| > \delta/2 > \varepsilon_i$ on $[\alpha, \beta]$. Let J_i be the component of $\{u \in [0, c_0] : |Q_i(u) - p_i| > \varepsilon_i\}$ that contains $[\alpha, \beta]$. At most two segments meet J_i: if a segment lies strictly between two segments that meet J_i, then both of its endpoints lie between points of J_i, hence in the interval J_i, against the home-based condition. So $[\alpha, \beta]$ is covered by the at most $2K + 1$ runs of these two segments, and there are points $\alpha = \tau_{i,0} \leq \tau_{i,1} \leq \cdots \leq \tau_{i,2K+1} = \beta$ and slopes $q_{i,j} \in [-1, 1]$ such that $|Q_i(u) - Q_i(u') - q_{i,j}(u - u')| \leq 2\varepsilon_i$ for $u, u' \in [\tau_{i,j-1}, \tau_{i,j}]$. Along a further subsequence $\tau_{i,j} \to \tau_j$ and $q_{i,j} \to q_j$. If $\tau_{j-1} < u < u' < \tau_j$, then $u, u' \in [\tau_{i,j-1}, \tau_{i,j}]$ for large i, and letting $i \to \infty$ gives $P(u') - P(u) = q_j(u' - u)$. So P is affine near every point of $(\alpha, \beta) \setminus \{\tau_1, \dots, \tau_{2K}\}$. Let Φ_I be the set of points u in the interior of I such that P is affine on no neighbourhood of u. Every compact interval in I contains at most $2K$ points of Φ_I in

its interior, so Φ_I has at most $2K$ points. On the interior of each component of $I \setminus \Phi_I$ the function P is locally affine, hence affine, and by continuity it is affine on the whole component.

(c) A tooth that does not contain 0 or c_0 has its endpoints at level p. If it contains a point u with $|P(u) - p| > y$, then, P being 1-Lipschitz, u is at distance more than y from both endpoints, so the tooth has length more than $2y$. The teeth are disjoint, and at most two of them contain 0 or c_0.

(d) This is Lemma 3.3(b). □

Theorem C now follows from the lemma and Theorem Cq.

Proof (not formalized). Suppose not. Then there are $c > 0$ and sizes $d_i \to \infty$ with $T_M(d_i) \le cL(d_i)$, and by Theorem 3.2(b) the machine M has a limit $(P_t)_{t<k}$ at time constant c along a subsequence, with representatives w_i. For large i and each t, the trajectory x_{t,w_i} up to the halting time $T_i \le cL(d_i)$ starts at cell 0 and is HB-AR$(K, \rho(d_i))$, and $T_i \ge L(d_i) \ge 1$ because M writes $L(d_i)$ output bits, at most one per step. So Lemma 3.15 applies to these trajectories. Hence there is a level p_t such that the components of the sets $I \setminus \Phi_I$, for the teeth I of P_t, form countably many relatively open intervals on each of which P_t is affine, and they cover $[0, c_0]$ except for the set $\{P_t = p_t\}$ and countably many points. Both exceptional sets are ν_t-null, by Lemma 3.15(d) and Lemma 3.3(c). So Theorem 3.14, applied to these intervals, gives $0 \ge \frac{1}{2}$, a contradiction. □

3.6 Relation to the Lebesgue form

Corollary 3.16 (Speed-weighted and Lebesgue forms)**.** Not formalized; a complete proof is given below. *Let $(P_t)_{t<k}$ be 1-Lipschitz functions on $[0, c_0]$ that satisfy $(\star)_\emptyset$, and let $U_t \subseteq [0, c_0]$ be relatively open sets on which P_t is locally affine. Then $\sum_t \nu_t([0, c_0] \setminus U_t) \ge \frac{1}{2}$, and hence $\sum_t \lambda([0, c_0] \setminus U_t) \ge \frac{1}{2}$.*

Proof (not formalized). The components of U_t are countably many relatively open intervals. On each of them P_t is locally affine, hence affine: its slope is locally constant on a connected set, hence constant, and then P_t minus a linear function is locally constant, hence constant. Theorem 3.14 applied to these intervals gives the first inequality, and $\nu_t \le \lambda$ gives the second. □

The second inequality is the quantitative affine-time obstruction of the second paper [20, Theorem 6.2], which is Lean-verified there for paths satisfying its restricted star condition [20, Definition 6.1]. That condition includes $(\star)_\emptyset$ as its case $S = \emptyset$ and is formally stronger; the corollary needs only $(\star)_\emptyset$ and allows relatively open sets, so it slightly generalizes that theorem. The proof there constructs Følner sets for the affine group directly and does not use the theorem of Connes, Feldman and Weiss; we have not checked whether that argument extends to the speed-weighted form. The motion exclusions of [20, Sections 6 and 7], for bounded words in explicit operation palettes and for contracting recursions with regular stages, rest on that inequality.

The speed-weighted form is stronger exactly where the heads are slow on the non-affine set. For a limit of HB-AR heads, Lemma 3.15 shows that the largest relatively open set on which P_t is locally affine misses only the boundary of the home-level set $\{P_t = p_t\}$, a closed nowhere dense set, and countably many points. So the Lebesgue form already excludes every machine with HB-AR work heads for which, in every limit, these boundaries have total Lebesgue measure less than $\frac{1}{2}$. Theorem C is new for limits in which these nowhere dense sets have total measure at least $\frac{1}{2}$, as in the next example; they carry no speed, so for Theorem Cq they cost nothing.

Remark 3.17 (A fat Cantor schedule)**.** Let c satisfy $cG > 1$, and let $C \subseteq [0, 1]$ be a perfect nowhere dense set with $\lambda(C) = \frac{1}{2}$ that contains 0 and 1, for instance a Smith–Volterra–Cantor set. Let one work head do the following at size d: it stays at its home cell 0 except during the time intervals (aR_d, bR_d) for the gaps (a, b) of C of length at least δ_d, where $\delta_d \to 0$ and $\delta_d R_d \to \infty$; during such

an interval it moves away from home at unit speed until the midpoint and back at unit speed. Every gap gives two segments of one run each, and the stays at home give segments with one run of rate 0, so the trajectory is HB-AR$(1,1)$. The omitted gaps change the path by at most $\delta_d/2$, so the rescaled trajectories converge uniformly to $P(u) = \operatorname{dist}(u, C)$ on $[0,1]$, constant afterwards. The home-level set of P within $[0,1]$ is C. No point of C has a neighbourhood on which P is affine: such an affine function would vanish at that point and, C being perfect, at points of C arbitrarily close to it, hence vanish identically, and C would contain an interval. So every relatively open set on which P is locally affine misses C, and the Lebesgue form gives nothing because $\lambda(C) = \frac{1}{2}$. But $\nu(C) = 0$, so Theorem C and Theorem Cq exclude every machine all of whose work heads move like this.

Remark 3.18 (What escapes)**.** Theorem Cq demands a minimum amount of speed-weighted motion that is not countably-piecewise affine; it says nothing against curved motion. A machine with one head whose home drifts, or whose motion is not home-based at all, falls outside Theorem C even if all its other heads are HB-AR; it is excluded only through Theorem Cq, that is, only if its limits are countably-piecewise affine on enough of their speed. Motion that satisfies $(\star)$ exists, and by Theorem Cq it is necessarily curved. The second paper constructs, for every machine, a fixed wrapper that adds linear time and whose complete runs satisfy, whatever computation follows, the cut inequality of [18, Definition 7.4] with zero slack, and also the conclusions of Theorem M* when that computation runs in linear time [20, Theorem 8.1]; this result is Lean-verified there. The heads of the wrapper move along rational near-identity Möbius maps (see [18, Section 13]). So Theorem Cq is a constraint on how a linear-time transposer moves its heads, not a lower bound by itself.

Remark 3.19 (How a machine can move along curves)**.** Proof sketch only: we indicate why curved macroscopic motion can be produced at amortised constant cost per step; this is a design sketch, not a result. A bounded number of constant-rate runs per leg gives piecewise affine limits, so a curved macroscopic leg needs unboundedly many runs whose speed, or duty cycle, varies along the leg. One way to produce this is to hold two numbers a, b of order L^δ, for some $0 < \delta < 1$, in unary on counter tapes and to alternate b moves of one head with a moves of another, each timed by a sweep of a counter head; a and b change by single-cell edits scheduled by a slower accumulator. The two heads then move at average speeds $b/(a+b)$ and $a/(a+b)$, with position error $o(L)$. For a Möbius profile f the function $1/f'$ is a quadratic polynomial, so a two-level hierarchy of unary accumulators, a discrete second difference, suffices. This realizes curved macroscopic motion only; the cell-exact pairings that data movement needs are a separate matter.

3.7 Consequences

The consequences concern the number of tapes that must interact, changes of clock, and hierarchical schedules; the first two, with Theorem Cq, give the design constraint of Corollary 3.24.

Corollary 3.20 (One work tape never expands)**.** Not formalized; a complete proof is given below.

(a) Let P be 1-Lipschitz on $[0, c_0]$. For all $\theta > 0$ and $\varepsilon > 0$ there is a Borel labelling β of $[0, c_0]$ whose classes have Lebesgue measure at most θ and whose crossing set satisfies $\nu(A(\beta)) \le \varepsilon$.

(b) Let $(P_t)_{t<k}$ be 1-Lipschitz functions on $[0, c_0]$ that satisfy $(\star)_\emptyset$. Then at least two of the P_t are not monotone, and the non-monotone ones are not all equal.

(c) Consequently, if M has one work tape and transposes, then $T_M(d)/L(d) \to \infty$. In every limit of a machine that transposes within $cL(d)$ steps on all inputs, at least two work-head limits are non-monotone, and they are not all equal.

Proof (not formalized). (a) For $m \geq 0$ let $N_m(p)$ be the number of intervals I of the partition of $[0, c_0]$ into 2^m intervals of equal length (half-open, the last one closed) with $p \in P(I)$. Each $P(I)$ is an interval, so N_m is Borel. We have $N_m \leq N_{m+1}$, since $P(I)$ is the union of the images of the two halves of I, and $N_m(p) \to \#\Lambda(p)$, since distinct intervals contain distinct points of $\Lambda(p)$ and any finitely many points of $\Lambda(p)$ lie in distinct intervals once m is large. So $Z_\infty = \{p : \#\Lambda(p) = \infty\}$ is Borel, and it is λ-null by Lemma 3.3(b). Let $Y = [0, c_0] \setminus P^{-1}(Z_\infty)$. By Lemma 3.3(a), $\nu([0, c_0] \setminus Y) = \int_{Z_\infty} \#\Lambda(p)\, \mathrm{d}p = 0$. The relation $F = \{(u, v) \in Y \times Y : P(u) = P(v)\}$ is a finite Borel equivalence relation on Y, since its classes are finite level sets. Put $F_j = F$ for all j and $B = D^+ \cap Y$. For $u \in B$ the point $\psi(u)$ lies in the level set of u, hence in Y, and $(u, \psi(u)) \in F$. Since $D^+ \setminus B \subseteq [0, c_0] \setminus Y$ is ν-null, Lemma 3.7 with $k = 1$ gives the claim.

(b) If P_t is monotone and $u \in D_t^+$, then P_t is constant on $[u, \psi_t(u)]$, so u lies in one of the countably many maximal nondegenerate intervals on which P_t is constant. On each of them $P_t' = 0$ except at the endpoints, so $\nu_t(D_t^+) = 0$. Now suppose that the non-monotone P_t, if there are any, are all equal to one path P. Take $\theta = \frac{1}{4}$, and let β be the labelling of (a) for P with $\varepsilon = 1/(4k)$, or any Borel labelling with classes of measure at most $\frac{1}{4}$ if all P_t are monotone. The heads with $P_t = P$ have the same returns and the same speed measure as P, so $\sum_t \nu_t(A_t(\beta)) \leq k \cdot \frac{1}{4k} = \frac{1}{4} < 1 - 2\theta$, which contradicts Theorem 3.6.

(c) If M has one work tape and transposes, and $T_M(d_i) \leq cL(d_i)$ along some $d_i \to \infty$, then by Theorem 3.2(b) M has a limit, which satisfies $(\star)_\emptyset$ and consists of a single path, against (b). The second statement is (b) for limits, which satisfy $(\star)_\emptyset$ by Theorem 3.2(a). □

Why it matters. For one work tape the statement is much weaker than the classical superlinear lower bounds in neighbouring models: $\Omega(L^{3/2}/(\log L)^{1/2})$ steps with one work tape and no output tape [31], and $\Theta(L^{5/4})$ for Boolean matrices with one work tape and a write-only output tape [30]. Its interest is the reason: the first returns of a single head stay in its level sets, which are finite for almost every level, so a single head can be labelled at no cost, however its motion is curved. The expansion that $(\star)$ demands must come from the interplay of the returns of two or more heads moving along different non-monotone paths. Heads in lockstep, stationary heads and monotone heads contribute nothing.

Corollary 3.21 (A common clock change does not help)**.** Not formalized; a complete proof is given below.

Let $(P_t)_{t<k}$ be 1-Lipschitz functions on $[0, c_0]$ that satisfy $(\star)_\emptyset$. Let φ be a homeomorphism of $[0, c_0]$ onto an interval J such that φ^{-1} maps λ-null sets to λ-null sets, for instance an increasing homeomorphism with absolutely continuous inverse. For each t let $E_{t,a}$ ($a \in \mathbb{N}$) be Borel subsets of J on each of which $P_t \circ \varphi^{-1}$ agrees with an affine function, and let $N_t = J \setminus \bigcup_a E_{t,a}$. Then $\sum_t \nu_t(\varphi^{-1}(N_t)) \geq \frac{1}{2}$.

Proof (not formalized). Choose affine maps $f_{t,a}$ with $P_t \circ \varphi^{-1} = f_{t,a}$ on $E_{t,a}$. Then $P_t = f_{t,a} \circ \varphi$ on the Borel set $\varphi^{-1}(E_{t,a})$, and these sets cover $[0, c_0]$ except $\varphi^{-1}(N_t)$. Lemma 3.9 gives Borel sets $B_t \subseteq D_t^+$ with $\nu_t(D_t^+ \setminus B_t) \leq 2\nu_t(\varphi^{-1}(N_t))$ and $\varphi(\psi_t(u)) \in \Gamma\varphi(u)$ for $u \in B_t$, where $\Gamma \leq \mathrm{Aff}(\mathbb{R})$ is the countable group generated by all the maps $f_{t,b}^{-1} \circ f_{t,a}$ with $f_{t,a}, f_{t,b}$ not constant. Take Y_0 and F_j from Lemma 3.8, let $Y = \varphi^{-1}(Y_0)$ and $F_j' = \{(u, v) \in Y \times Y : (\varphi(u), \varphi(v)) \in F_j\}$. These are increasing finite Borel equivalence relations on Y, with finite classes because φ is injective, and their union contains $(u, \psi_t(u))$ for every $u \in B_t' = B_t \cap Y \cap \psi_t^{-1}(Y)$. By hypothesis $[0, c_0] \setminus Y = \varphi^{-1}(J \setminus Y_0)$ is λ-null, so, as in the proof of Theorem 3.14, $\nu_t(D_t^+ \setminus B_t') \leq 2\nu_t(\varphi^{-1}(N_t))$. Lemma 3.7 and Theorem 3.6 then give $1 - 2\theta \leq 2\sum_t \nu_t(\varphi^{-1}(N_t)) + \varepsilon$ for all $\theta \in (0, \frac{1}{2})$ and $\varepsilon > 0$. □

With φ the identity this is Theorem Cq. The corollary covers, for example, a global time profile that slows all heads down and speeds them up together, or a shared duty cycle.

Why it matters. Curvature that is common to all heads does not create expansion: if one change of clock makes every head countably-piecewise affine, up to sets carrying little speed, then ($\star$) fails. Non-affine returns are therefore necessary but not sufficient. What is required is that no single such change of clock conjugates the transitions of all heads into a countable group of affine maps. Head-dependent clocks, $P_t = f_t \circ \varphi_t$ with affine f_t and different φ_t, are not covered; a Möbius clock on one head and affine motion on another produce Möbius transitions, which is the regime of the monotone expanders of Bourgain and Yehudayoff [17].

Next we consider hierarchical schedules in which every move is a sweep at constant speed and every excursion returns exactly to where it started. They may be arbitrarily deep and interleaved.

Definition 3.22 (Stroke trees)**.** Let $K, K' \geq 1$ be integers, $\gamma \in (0,1)$, $\eta_0 > 0$, $\rho \geq 0$ and $R > 0$, and let $x\colon \{0,\dots,T\} \to \mathbb{Z}$ be a walk of length $T+1$. For integers $0 \leq a \leq b \leq T$ the *time interval* $[a,b)$ is $\{a,\dots,b-1\}$; a finite family of time intervals *tiles* $[a,b)$ if its members are nonempty and pairwise disjoint and their union is $[a,b)$. A time interval $[a,b)$ is a *pause* if x is constant on $\{a,\dots,b\}$. A *stroke tree* for x with parameters (K, K', γ, η_0), tolerance ρ and scale R is a finite rooted forest whose nodes are *strokes* and *V's*, each with a time interval, such that:

(i) the intervals of the roots tile $[0,T)$, and every root is a stroke;

(ii) every stroke S has a direction $\sigma_S \in \{-1,1\}$, and its interval is tiled by at most K *runs*, any number of pauses, and the intervals of its children, which are V's; a run is a time interval $[a,b)$ such that $\sigma_S x$ is nondecreasing on $\{a,\dots,b\}$ and, for some $q \in [0,1]$, $|x(s)-x(a)-\sigma_S q(s-a)| \leq \rho$ for $a \leq s \leq b$;

(iii) every V, with interval $[a,b)$, satisfies $x(a) = x(b)$, and its interval is tiled by any number of pauses and the intervals of its children, which are at most K' strokes;

(iv) every root either has extent at least $\eta_0 R$ or has no runs, in which case it is a *home stroke*; and every stroke C that is a child of a V of a stroke S other than a home stroke satisfies $\operatorname{ext}(C) \leq \gamma \operatorname{ext}(S)$.

Here the *extent* $\operatorname{ext}(S)$ of a stroke S is the sum of $|x(b) - x(a)|$ over its runs $[a,b)$.

Every stroke moves monotonically through at most K runs, with any number of excursions (V's) inserted between runs and at its ends; every excursion returns exactly to its starting cell and consists of at most K' smaller strokes. A home stroke models a fixed home from which excursions of any size start.

Proposition 3.23 (Stroke trees are affine or stationary)**.** Not formalized; a complete proof is given below.

Let $K, K' \geq 1$, $\gamma \in (0,1)$, $\eta_0 > 0$ and $\rho(d) = o(L(d))$. Let M be a machine and t one of its work heads, and suppose that for all large d, on every matrix input of size d, the trajectory of head t up to the halting time has a stroke tree with parameters (K, K', γ, η_0), tolerance $\rho(d)$ and scale R_d. Let $(P_s)_{s<k}$ be a limit of M at some time constant, and let U be the union of the open intervals in $(0, c_0)$ on which P_t is affine. Then P_t is affine on each component of U, and $\nu_t([0,c_0]\setminus U) = 0$. Consequently, if every work head of M satisfies the hypothesis and M transposes, then $T_M(d)/L(d) \to \infty$.

Proof (not formalized). Write $P = P_t$, let d_i and w_i be as in Definition 3.1, let $x_i = x_{t,w_i}$, frozen after its halting time T_i, let $R_i = R_{d_i}$, and let $Q_i(u) = X_i(uR_i)/R_i$, where X_i interpolates x_i linearly between integers. As in the proof of Lemma 3.15, $Q_i \to P$ uniformly on $[0, c_0]$. The set U is open, P is affine on each of its components because it is locally affine there, and $Z = [0, c_0] \setminus U$ is closed.

Claim: $|P(v) - P(w)| \le \lambda([w, v] \cap U)$ *for all* $0 \le w < v \le c_0$. Fix $w < v$ and $\eta \in (0, \min(\eta_0, c_0))$, and fix a large i; we drop the index i where this causes no ambiguity. Call a stroke *big* if its extent is at least ηR and *small* otherwise; home strokes have no runs and play no role. Let $w'' = \min(\lceil wR \rceil, T)$ and $v'' = \min(\lfloor vR \rfloor, T)$. Since X is 1-Lipschitz and constant after T, $|Q(v) - Q(w)| \le (|x(v'') - x(w'')| + 2)/R$. For large i either $w'' < v''$, or $w'' = v'' = T$ and $x(v'') = x(w'')$; so assume $w'' < v''$, and let $I = [w'', v'')$.

Every time $s \in I$ has a *chain*: the root containing s; then the run, pause or V of that root containing s; then, if this is a V, the pause or stroke of that V containing s; and so on, down to a run or a pause. Call a V *inside* if its interval is contained in I. The *tile* of s is the interval of the first inside V on its chain, or, if there is none, the run or pause at the end of its chain. Two tiles are disjoint or equal: nested intervals of the forest belong to a node and a descendant, and a tile that is an inside V contains the tiles of none of its descendants, since their chains pass through it. So the sets $H \cap I$, for the tiles H, tile I into intervals, and $x(v'') - x(w'')$ is the sum of the displacements of x over them. Inside V's and pauses have displacement 0. A tile that is a run belongs to a stroke S none of whose ancestor V's is inside; call such strokes *exposed*. Since x is monotone with unit steps on a run, its displacement over a subinterval $[a', b')$ of a run $[a, b)$ is at most $\min(|x(b) - x(a)|,\, b' - a')$. Hence

$$|x(v'') - x(w'')| \le \sum_{S \text{ exposed}} c(S), \qquad c(S) \le \min\big(\operatorname{ext}(S),\ \#(I \cap \operatorname{runs}(S))\big),$$

where $\operatorname{runs}(S)$ is the union of the runs of S.

Small exposed strokes. An exposed stroke that is not a root has a parent V that meets I but is not inside; this V contains $w'' - 1$ and w'', or it contains $v'' - 1$ and v''. The V's of the first kind contain w'', so they lie on the chain of w'', and they are its first J V's for some J, say $V_1 \supset V_2 \supset \cdots \supset V_J$, with V_j a child of the stroke S_{j-1} of that chain and S_j a child of V_j. For $j \ge 2$ the stroke S_{j-1} is not a root, hence not a home stroke, and repeated use of (iv) gives $\operatorname{ext}(S_{j-1}) \le \gamma^{j-2} \operatorname{ext}(S_1) \le \gamma^{j-2} T$. So every child of V_j has extent at most $\gamma^{j-1} T \le \gamma^{j-1} c_0 R$, because $T \le cL \le c_0 R$. Each V_j has at most K' children, so the small exposed children of $V_1, \dots, V_J$ contribute at most

$$K'\eta R + K'R \sum_{j \ge 1} \min(\eta, \gamma^j c_0) \ \le\ K'\eta R\Big(1 + \log_{1/\gamma} \frac{c_0}{\eta} + \frac{1}{1-\gamma}\Big) :$$

at most $\log_{1/\gamma}(c_0/\eta)$ terms of the sum equal η, and the others form a geometric series with first term below η. The V's of the second kind lie on the chain of $v'' - 1$, and their small children contribute at most the same amount. Every small exposed stroke is a child of one of these V's, because the roots are big or home strokes. So the small exposed strokes contribute at most $\delta(\eta) R$, where $\delta(\eta) = 2K'\eta(1 + \log_{1/\gamma}(c_0/\eta) + 1/(1-\gamma))$ tends to 0 with η.

Big strokes. Runs of different strokes are disjoint, and a big stroke spends at least ηR steps in its runs. Since $T \le c_0 R$, there are at most c_0/η big strokes and at most $m = \lfloor K c_0/\eta \rfloor$ runs of big strokes. List these runs as $[a_{i,l}, b_{i,l})$, $l \le m$, padding with empty intervals, and let $J_{i,l} = [a_{i,l}/R_i,\ b_{i,l}/R_i]$. Since $[w''/R, v''/R] \subseteq [w, v]$, the big exposed strokes contribute at most $\sum_l \#(I \cap [a_{i,l}, b_{i,l})) \le R_i \sum_l \lambda([w, v] \cap J_{i,l})$. Altogether

$$|Q_i(v) - Q_i(w)| \ \le\ \delta(\eta) + \sum_{l \le m} \lambda([w, v] \cap J_{i,l}) + \frac{2}{R_i}.$$

Limit. By (ii), as in the proof of Lemma 3.15, $|Q_i(u) - Q_i(u') - q_{i,l}(u-u')| \le 2\rho(d_i)/R_i$ for $u, u' \in J_{i,l}$, with slopes $q_{i,l} \in [-1,1]$. Along a subsequence the endpoints of the $J_{i,l}$ and the slopes $q_{i,l}$ converge, to intervals $J_l \subseteq [0,c_0]$ and slopes q_l. The $J_{i,l}$ have pairwise disjoint interiors, hence so do the J_l. On the interior of each J_l the function P is affine with slope q_l, as in the proof of Lemma 3.15, so this interior lies in U. Moreover $\lambda([w,v] \cap J_{i,l}) \to \lambda([w,v] \cap J_l)$, and $\sum_l \lambda([w,v] \cap J_l) \le \lambda([w,v] \cap U)$ because the interiors of the J_l are disjoint and lie in U. Letting $i \to \infty$ gives $|P(v) - P(w)| \le \delta(\eta) + \lambda([w,v] \cap U)$, and letting $\eta \to 0$ proves the claim.

Conclusion. At almost every point u of Z the function P is differentiable and u is a Lebesgue density point of Z. For such u and small $\tau > 0$ the claim gives $2\tau|P'(u)| + o(\tau) = |P(u+\tau) - P(u-\tau)| \le \lambda([u-\tau, u+\tau] \setminus Z) = o(\tau)$, so $P'(u) = 0$. Hence $\nu_t(Z) = 0$. The last sentence of the proposition follows from Theorem 3.13, with U_s the set U of head s. □

Why it matters. Stroke trees model hierarchical copy-and-return schedules of unbounded depth, with roots of macroscopic size or a fixed home. The proposition says that, however deep and however interleaved, such motion is affine or stationary in the limit, and hence excluded when all work heads move this way. The non-affine sets $[0,c_0] \setminus U$ are nowhere dense, since $P_t' = 0$ almost everywhere on them. The Lebesgue form already excludes stroke trees whose non-affine sets have total Lebesgue measure less than $\frac{1}{2}$; the proposition is needed when that measure is at least $\frac{1}{2}$, as for Cantor-type arrangements of V's. A construction must break one of the hypotheses, for instance with strokes made of unboundedly many runs of varying density, which are curved; V's that drift; V's with unboundedly many strokes, or nested strokes whose extents do not contract geometrically; or roots that are not macroscopic, concatenated with varying density.

Corollary 3.24 (The motion a linear-time transposer needs)**.** Not formalized; a complete proof is given below.

Let M transpose within $cL(d)$ steps on every matrix input of every size d, for instance a machine that witnesses (T) with constant $c/2$, and let $(P_t)_{t<k}$ be a limit of M at time constant c; such limits exist along a subsequence of every sequence of sizes. Then:

- *(a) (curved) for all Borel sets $E_{t,a}$ on each of which P_t is affine, the uncovered sets N_t satisfy $\sum_t \nu_t(N_t) \ge \frac{1}{2}$;*
- *(b) (head-dependent) this survives every common change of clock: for every homeomorphism φ of $[0,c_0]$ onto an interval J such that φ^{-1} maps null sets to null sets, and all Borel sets $E_{t,a} \subseteq J$ on each of which $P_t \circ \varphi^{-1}$ is affine, the uncovered sets $N_t \subseteq J$ satisfy $\sum_t \nu_t(\varphi^{-1}(N_t)) \ge \frac{1}{2}$;*
- *(c) (interacting tapes) at least two of the P_t are not monotone, and these are not all equal; in particular $k \ge 2$.*

Proof (not formalized). Limits exist by Theorem 3.2(b), and they satisfy $(\star)_\emptyset$ by Theorem 3.2(a). Then (a) is Theorem 3.14, (b) is Corollary 3.21 and (c) is Corollary 3.20(b). □

Why it matters. This is the motion a construction must use: curved, with curvature that differs between heads, on at least two work tapes whose returns interact. The monotone expanders of Bourgain and Yehudayoff [17], driven along near-identity Möbius maps as in the architecture of [18, Section 13], are consistent with all three conditions. The corollary constrains only the macroscopic limit of the motion. Structure at mesoscopic scales, such as nested excursions much shorter than L, is invisible in the limit, and a curved macroscopic host can carry any mesoscopic structure; so the motion constraints say nothing about depth reduction (Section 5), which concerns all scales at

once. In this sense the results of this section are complementary to the decomposition of Section 4: they exclude classes of macroscopic motion without any coding hypothesis, while the decomposition applies to every motion of an oblivious machine but needs depth reduction and the crux.

4 The decomposition: depth reduction plus an index-coding crux

This section reduces the oblivious case of (T) to two statements of different kinds. For an oblivious machine and a fixed input length, the run is a circuit: the value computed at each step is a fixed function of the input bit under the input head, of the previous state, and of the symbols under the work heads, and each of these symbols was written at the previous visit of its head to its cell. Revealing the values at a set Γ of steps cuts this circuit. If every path of the circuit that avoids Γ is short, each output bit depends only on the revealed values and on few input bits, so the revealed values form an index code for the transpose demand with small side information.

Section 4.1 sets up the circuit and proves the cone lemma. Section 4.2 states the two hypotheses, depth reduction at linear cost (depth-TCP) and a lower bound for index codes on the resulting side information (the crux), and shows that together they exclude oblivious linear-time transposers (Theorem D). Section 4.3 records what the crux must deliver. Section 4.4 proves depth reduction for walks of linear leg area, in particular for home-based walks, for every coupling of the heads (Theorem B), and Section 4.5 turns its cuts into genies, which reduces home-based oblivious machines to the crux (Corollary B). None of the results of this section is formalized; each comes with a complete proof.

4.1 The dependency DAG of an oblivious run

Let M be a machine as in Section 2.1, with state set $\{0,\dots,q\}$ and $k \geq 1$ work tapes over the alphabet $\{0,\dots,g\}$, and let $L \geq 1$ be an input length at which M is oblivious, with halting time T. On every input $w \in \{0,1\}^L$ the machine executes steps $0,\dots,T-1$. Write $x_0(t)$ for the position of the input head and $x_j(t)$ for the position of work head j ($1 \leq j \leq k$) at time t; by obliviousness these positions, and the set of steps that append an output bit, do not depend on w. Step t reads the symbols under the heads at time t, writes the cells $x_1(t),\dots,x_k(t)$, moves the heads, and possibly appends an output bit. As in Section 2.2, $\mathrm{prev}_j(t) = \max\{s < t : x_j(s) = x_j(t)\}$ when this set is nonempty.

The *value* $v_t(w)$ of step t on input w consists of the state after step t, the k symbols written at step t, and the output bit appended at step t (or a symbol for "none"). It takes at most $2^{\beta(M)}$ values, where

$$\beta(M) = \lceil \log_2(q+1) \rceil + k \lceil \log_2(g+1) \rceil + 2$$

is the *value width* of M; the last two bits encode the output (none, 0 or 1).

Definition 4.1 (Dependency DAG)**.** The *dependency DAG* H_M of M at length L has vertex set $[0,T) = \{0,\dots,T-1\}$ and the following edges, all pointing backwards in time: a *state edge* $t \to t-1$ for every $1 \leq t < T$, and a *cell edge* $t \to \mathrm{prev}_j(t)$, labelled j, for every work tape j and every t for which $\mathrm{prev}_j(t)$ is defined. The *length* of an edge $t \to s$ is $t - s$.

Every vertex has out-degree at most $k+1$. For each j the map $t \mapsto \mathrm{prev}_j(t)$ is injective: if $\mathrm{prev}_j(t) = \mathrm{prev}_j(t') = s$ with $t < t'$, then $x_j(t') = x_j(s) = x_j(t)$ with $s < t < t'$, so $\mathrm{prev}_j(t') \geq t > s$. Hence the symbol written on tape j at step s is read along at most one edge, the state after step s only along $s+1 \to s$, and the output bit along none. The DAG depends only on the head schedule of M at length L, not on the input or on the transition table.

Lemma 4.2 (Local recursion)**.** Not formalized; a complete proof is given below.
For every $t < T$ there is a function Δ_t such that for every input $w \in \{0,1\}^L$,

$$v_t(w) = \Delta_t(v_{t-1}(w),\, w_{x_0(t)},\, v_{\mathrm{prev}_1(t)}(w), \ldots, v_{\mathrm{prev}_k(t)}(w)),$$

where an undefined argument is replaced by a constant: by the initial state if $t = 0$, by the end marker if $x_0(t) \notin [0, L)$, and by the blank symbol if $\mathrm{prev}_j(t)$ is undefined.

Proof (not formalized). Step t applies the transition function to the state at time t, the input symbol under the input head and the k symbols under the work heads, and $v_t(w)$ is read off from its result. The state at time t is the initial state if $t = 0$ and the state component of $v_{t-1}(w)$ otherwise. The input symbol is $w_{x_0(t)}$, or the end marker if $x_0(t) \notin [0, L)$. Every step writes the cell under each work head, so at time t the cell under work head j holds the symbol written on tape j at step $\mathrm{prev}_j(t)$, a component of $v_{\mathrm{prev}_j(t)}(w)$, and it is blank if the head has not visited it before time t; this is the cell-contents lemma of the first paper, Lean-verified there [18, Lemma 2.3]. The positions $x_0(t)$ and the steps $\mathrm{prev}_j(t)$ do not depend on w, so neither does Δ_t. □

Definition 4.3 (Genies, cones and depth)**.** A *genie* is a set $\Gamma \subseteq [0, T)$ of revealed steps. For an integer $\beta \geq \beta(M)$, its *revealed word* $G_\Gamma(w) = (v_s(w))_{s\in\Gamma}$ is written with β bits per value; this defines a map $G_\Gamma \colon \{0,1\}^L \to \{0,1\}^{\beta|\Gamma|}$, and $\beta|\Gamma|$ is the *cost* of Γ. A path $t_0 \to t_1 \to \cdots \to t_m$ in H_M has *length* m, and it is Γ*-free* if $t_1, \ldots, t_m \notin \Gamma$ (its first vertex may lie in Γ). The *cone* $C_\Gamma(t)$ is the set of endpoints of Γ-free paths starting at t, so $t \in C_\Gamma(t)$. Its *boundary* $\partial_\Gamma(t)$ is the set of $s \in \Gamma$ such that some edge of H_M leads from a vertex of $C_\Gamma(t)$ to s, and $J_\Gamma(t) = x_0(C_\Gamma(t)) \cap [0, L)$ is the set of input positions read in the cone. The genie has *depth at most* D if every Γ-free path has length at most D.

So revealing the value v_s blocks every edge into s.

Lemma 4.4 (Cone lemma)**.** Not formalized; a complete proof is given below.
Let Γ be a genie and $t < T$.

(a) If two inputs $w, w' \in \{0,1\}^L$ agree on $J_\Gamma(t)$ and satisfy $v_s(w) = v_s(w')$ for every $s \in \partial_\Gamma(t)$, then $v_t(w) = v_t(w')$. Thus v_t is a function of the values at the boundary and of the input bits read in the cone.

(b) If Γ has depth at most D, then $|J_\Gamma(t)| \leq |C_\Gamma(t)| \leq \sum_{i=0}^{D} (k+1)^i \leq (k+1)^{D+1}$.

Proof (not formalized). (a) By strong induction on t. Let w, w' be as stated; we show that the arguments of Δ_t in Lemma 4.2 agree on w and w'. Constant arguments agree trivially. If $x_0(t) \in [0, L)$, then $x_0(t) \in J_\Gamma(t)$ because $t \in C_\Gamma(t)$, so $w_{x_0(t)} = w'_{x_0(t)}$. Now let $t \to t'$ be an edge, with $t' = t - 1$ or $t' = \mathrm{prev}_j(t)$. If $t' \in \Gamma$, then $t' \in \partial_\Gamma(t)$, since the edge starts at $t \in C_\Gamma(t)$, and $v_{t'}(w) = v_{t'}(w')$ by hypothesis. If $t' \notin \Gamma$, then every Γ-free path from t', preceded by the edge $t \to t'$, is a Γ-free path from t. Hence $C_\Gamma(t') \subseteq C_\Gamma(t)$, so $J_\Gamma(t') \subseteq J_\Gamma(t)$ and $\partial_\Gamma(t') \subseteq \partial_\Gamma(t)$, and the induction hypothesis for $t' < t$ gives $v_{t'}(w) = v_{t'}(w')$. So all arguments agree, and $v_t(w) = v_t(w')$.

(b) Every vertex of $C_\Gamma(t)$ is the endpoint of a Γ-free path from t of length at most D, and since out-degrees are at most $k+1$, at most $(k+1)^i$ paths of length i start at t. The last inequality holds because $k \geq 1$. □

Corollary 4.5 (Output bits)**.** Not formalized; a complete proof is given below.
Suppose that M appends exactly L output bits at length L, the ℓ-th one ($0 \leq \ell < L$) at step t_ℓ, and let Γ be a genie. Put $I_\Gamma(\ell) = J_\Gamma(t_\ell)$. If $G_\Gamma(w) = G_\Gamma(w')$ and w, w' agree on $I_\Gamma(\ell)$, then output bit ℓ is the same on w and on w'. If Γ has depth at most D, then $|I_\Gamma(\ell)| \leq (k+1)^{D+1}$.

Proof (not formalized). Output bit ℓ is a component of v_{t_ℓ}, and $\partial_\Gamma(t_\ell) \subseteq \Gamma$, so $G_\Gamma(w) = G_\Gamma(w')$ gives $v_s(w) = v_s(w')$ for every $s \in \partial_\Gamma(t_\ell)$. Apply Lemma 4.4. □

Remark 4.6 (Field-level genies)**.** A genie may also reveal single *fields*: the state after a step, or the symbol written on one tape. Each field is read along at most one edge, so revealing it blocks exactly that edge, and the proof of Lemma 4.4 applies verbatim to paths that avoid the blocked edges. We do not need this refinement. Below, an edge $t \to s$ is blocked by putting s into Γ, at a cost of β bits, which also bounds the width of a single field.

4.2 Two hypotheses and Theorem D

The first hypothesis, depth-TCP (TCP stands for *trajectory cut problem*), says that a genie of linear cost reduces the depth of every dependency DAG of linear size to a constant.

Definition 4.7 (depth-TCP)**.** Let $k \geq 1$, $D \geq 0$ and $\beta \geq 2$ be integers and $C, \varepsilon > 0$ reals. The statement depth-TCP$(k, C, \beta, \varepsilon, D)$ says: for every machine M with k work tapes and value width $\beta(M) \leq \beta$ there is d_0 such that, for every $d \geq d_0$ at which M is oblivious at length $L = d^2 b(d)$ with halting time $T \leq C(L+1)$, the dependency DAG H_M at length L has a genie Γ of cost $\beta|\Gamma| \leq \varepsilon L$ and depth at most D.

This is a statement about head schedules: whether a genie of a given size and depth exists depends only on the DAG. By the trade-off of Proposition 4.13 below, applied to a machine that copies its input to its output, depth-TCP with $C \geq 1$ can hold only if $\varepsilon \geq \beta/(D+1)$.

For the second hypothesis recall the transpose map. Bit s of entry (i, j) of a $d \times d$ matrix with b-bit entries sits at input position $(di + j)b + s$, and in the encoding of the transpose at position $\mathrm{tpos}((di + j)b + s) = (dj + i)b + s$ [18, Section 2.2 and Lemma 2.5]. The map tpos is an involution of $[0, L)$, so on input w the output bit of a transposer at position ℓ is $w_{\mathrm{tpos}(\ell)}$.

Definition 4.8 (Index codes and the crux)**.** Let $d \geq 1$ and $L = d^2 b(d)$. A *side-information system* at size d is a family $I = (I(\ell))_{\ell \in [0,L)}$ of subsets of $[0, L)$: receiver ℓ knows the input bits at the positions in $I(\ell)$ and demands $w_{\mathrm{tpos}(\ell)}$. A map $G\colon \{0,1\}^L \to \{0,1\}^m$ is an *index code* for I with m bits if for every ℓ and all $w, w' \in \{0,1\}^L$ with $G(w) = G(w')$ that agree on $I(\ell)$ we have $w_{\mathrm{tpos}(\ell)} = w'_{\mathrm{tpos}(\ell)}$.

For a class $\mathcal{S}$ of side-information systems and a real $\rho \geq 0$, the statement $\mathrm{Crux}(\mathcal{S}, \rho)$ says: for every $\eta > 0$ there is d_0 such that every index code for every system of $\mathcal{S}$ at a size $d \geq d_0$ has at least $(\rho - \eta)L$ bits. In other words, index codes for systems of $\mathcal{S}$ have at least $\rho L - o(L)$ bits, with the $o(L)$ uniform over $\mathcal{S}$.

The class $\mathcal{S}_{k,C,\beta,D}$ consists of the systems $I_\Gamma = (I_\Gamma(\ell))_{\ell \in [0,L)}$ of Corollary 4.5, where $d \geq 1$; M is a machine with k work tapes and $\beta(M) \leq \beta$ that is oblivious at length $L = d^2 b(d)$ with halting time $T \leq C(L+1)$ and appends exactly L output bits; and Γ is a genie of depth at most D in H_M.

The class $\mathcal{S}_{k,C,\beta,D}$ depends on each machine only through its head schedule and its output times; no transition table enters. Its sets have at most $(k+1)^{D+1}$ elements, and it grows with D. The crux is a statement about *all* index codes, linear or not, and it must be: the revealed word of a genie is an arbitrary function of the input, determined by the transition table. A lower bound for linear index codes alone would cover only machines whose revealed words are linear functions of the input, such as machines with an $\mathbb{F}_2$-linear data path.

Remark 4.9 (Two readings of the crux)**.** (a) *The crux restricts head schedules.* Call receiver ℓ of a system I *trivial* if $\mathrm{tpos}(\ell) \in I(\ell)$, and let $\mathrm{triv}(I)$ be the number of trivial receivers. Broadcasting the demanded bits of the non-trivial receivers is an index code with $L - \mathrm{triv}(I)$ bits, so $\mathrm{Crux}(\mathcal{S}, \rho)$

implies $\mathrm{triv}(I) \le (1-\rho)L + o(L)$ for the systems $I \in \mathcal{S}$. For $\mathcal{S}_{k,C,\beta,D}$ this says that at most $(1-\rho)L + o(L)$ output positions ℓ have the property that the input head reads position $\mathrm{tpos}(\ell)$ at the end of a Γ-free path that starts at the step t_ℓ appending output bit ℓ.

(b) *Storage codes.* For a system I let $N_I(p) = I(\mathrm{tpos}(p))$, the side information of the receiver that demands w_p. If G is an index code for I with m bits, then every fibre $G^{-1}(z)$ is a binary storage code for N_I (Definition 2.1): two words of the fibre that agree on $N_I(p)$ agree at p. Some fibre has at least 2^{L-m} words. Hence, if every binary storage code for every N_I, $I \in \mathcal{S}$, has at most $2^{(1-\rho)L+o(L)}$ words, uniformly, then $\mathrm{Crux}(\mathcal{S}, \rho)$ holds; in particular, if every N_I satisfies the rate bound c (Definition 2.2), then $\mathrm{Crux}(\mathcal{S}, 1-c)$ holds by Theorem 6.3. Conversely, a linear storage code $U \subseteq \mathbb{F}_2^L$ for N_I of dimension λL gives the linear index code $w \mapsto w + U \in \mathbb{F}_2^L/U$ with $(1-\lambda)L$ bits: if $w - w' \in U$ vanishes on $N_I(p)$, it agrees there with $0 \in U$, so it vanishes at p. Restricted to linear codes, the crux at rate ρ is therefore equivalent to the statement that binary linear storage codes for the N_I, $I \in \mathcal{S}$, have dimension at most $(1-\rho)L + o(L)$, uniformly (the kernel of a linear index code is a linear storage code). For general codes the two notions are related by the duality recalled in Section 2.3.

Theorem 4.10 (Theorem D: decomposition)**.** Not formalized; a complete proof is given below.

Let $k \ge 1$, $D \ge 0$ and $\beta \ge 2$ be integers, $C > 0$, and $0 < \varepsilon < \rho$. Suppose that the statements depth-TCP$(k, C, \beta, \varepsilon, D)$ *and* $\mathrm{Crux}(\mathcal{S}_{k,C,\beta,D}, \rho)$ *hold. Then for every machine M with k work tapes and value width at most β there is d_0 such that for no $d \ge d_0$ is M oblivious at length $L = d^2 b(d)$ with halting time at most $C(L+1)$ and at the same time a transposer of all $d \times d$ matrices with $b(d)$-bit entries. In particular, no oblivious machine with k work tapes and value width at most β transposes within $C(L+1)$ steps for all d, or even for infinitely many d.*

Proof (not formalized). Fix M. Let d_1 be given by depth-TCP$(k, C, \beta, \varepsilon, D)$ for M, let d_2 be given by $\mathrm{Crux}(\mathcal{S}_{k,C,\beta,D}, \rho)$ for $\eta = (\rho - \varepsilon)/2$, and put $d_0 = \max(d_1, d_2)$. Suppose that for some $d \ge d_0$ the machine is oblivious at length $L = d^2 b(d)$ with halting time $T \le C(L+1)$ and transposes all $d \times d$ matrices. Every word of length L encodes such a matrix ([18, Lemma 2.5], Lean-verified there), so on every input w the output has length L and its bit ℓ is $w_{\mathrm{tpos}(\ell)}$; in particular, M appends exactly L output bits. By depth-TCP there is a genie Γ with $\beta|\Gamma| \le \varepsilon L$ and depth at most D, so $I_\Gamma \in \mathcal{S}_{k,C,\beta,D}$. By Corollary 4.5, if $G_\Gamma(w) = G_\Gamma(w')$ and w, w' agree on $I_\Gamma(\ell)$, then $w_{\mathrm{tpos}(\ell)} = w'_{\mathrm{tpos}(\ell)}$: the revealed word G_Γ is an index code for I_Γ with $\beta|\Gamma|$ bits. The crux gives $\beta|\Gamma| \ge (\rho - \eta)L = \frac{1}{2}(\rho + \varepsilon)L > \varepsilon L$, a contradiction. □

Corollary 4.11 (The two hypotheses refute oblivious (T))**.** Not formalized; a complete proof is given below.

Suppose that for all integers $k \ge 1$ and $\beta \ge 2$ and every real $C > 0$ there are reals $0 < \varepsilon < \rho$ and an integer $D \ge 0$ such that depth-TCP$(k, C, \beta, \varepsilon, D)$ *and* $\mathrm{Crux}(\mathcal{S}_{k,C,\beta,D}, \rho)$ *hold. Then for every oblivious machine M and every constant c there is d_0 such that for no $d \ge d_0$ does M transpose all $d \times d$ matrices within $c(L+1)$ steps. In particular, no oblivious machine witnesses (T).*

Proof (not formalized). If $c = 0$ there is nothing to prove, since a machine that halts at time 0 writes no output. A machine without work tapes first receives one idle work tape, which keeps it oblivious and changes neither its running time nor its output. So let M have $k \ge 1$ work tapes. It is oblivious at every input length, and if it transposes all $d \times d$ matrices within $c(L+1)$ steps, its halting time at length $L = d^2 b(d)$ is at most $c(L+1)$. Apply Theorem 4.10 with $C = c$ and $\beta = \beta(M) \ge 2$. □

Why it matters. Theorem D isolates exactly two statements that together refute the oblivious case of (T): a structural statement about tape dependency DAGs, which mentions neither the data nor the transition table, and an information-theoretic statement about index codes, which sees a machine only through its head schedule and its output times. An impossibility proof for oblivious machines can therefore be split into a depth-reduction part and a coding part; for instance, a crux at rate $\frac{1}{2}$ for every depth D would combine with depth reduction at a single $\varepsilon < \frac{1}{2}$. Theorem B and Corollary B below settle the depth-reduction part for home-based heads; in general both parts are open (Sections 5 and 6). On the construction side, Theorem D says what an oblivious transposer M with k work tapes and running time $C(L+1)$ must do; let $\beta = \beta(M)$. At every size at which it transposes and for all $\varepsilon > 0$ and $D \geq 0$, either H_M has no genie of cost at most εL and depth at most D, or the system $I_\Gamma \in \mathcal{S}_{k,C,\beta,D}$ of such a genie has an index code with at most εL bits, namely G_Γ; by Remark 4.9(b), some fibre of G_Γ is a storage code of rate at least $1-\varepsilon$ for N_{I_Γ}.

Theorem D is an elementary composition of the cone lemma with definitions tailored to it, the multitape analogue of the route from small-depth computation to index coding by depth reduction recalled in the introduction. Its content is the formulation of the two hypotheses and the bookkeeping of genie cost against depth.

Remark 4.12 (Relation to the first paper)**.** The first paper proves, Lean-verified, that the plain form of its Tape Coding Conjecture, which forbids causal codes for complete exchange on the block networks of tape diagrams of linear length, at the profile capacities, refutes the oblivious case of (T) [18, Theorem 9.9]. Theorem D is not a refinement of that reduction, and we know no implication, in either direction, between the plain Tape Coding Conjecture and the conjunction of depth-TCP and the crux: depth-TCP concerns the step-level dependency DAG and says nothing about codes on block networks, and the crux concerns index codes for cone side information rather than complete exchange.

4.3 What the crux must deliver

Theorem D needs the crux at a rate ρ above the relative cost ε of the genie. The next proposition bounds that cost from below in terms of the depth.

Proposition 4.13 (Forced trade-off)**.** Not formalized; a complete proof is given below.

Let M be oblivious at length L with halting time T, and let Γ be a genie of depth at most D. Then $|\Gamma| \geq \lfloor (T-1)/(D+1) \rfloor$. If moreover $T \geq L$, as for every machine that appends L output bits and in particular for every transposer, and $\beta|\Gamma| \leq \varepsilon L$, then

$$\varepsilon \;\geq\; \frac{\beta}{D+1} - \frac{\beta}{L}, \qquad \textit{equivalently} \qquad D+1 \;\geq\; \frac{\beta L}{\varepsilon L + \beta}.$$

Proof (not formalized). Let $n = \lfloor (T-1)/(D+1) \rfloor$ and $a_i = i(D+1)$ for $0 \leq i < n$. Then $a_i + D + 1 \leq n(D+1) \leq T-1$, and the path $a_i + D + 1 \to a_i + D \to \cdots \to a_i$ of state edges has length $D+1$, so it is not Γ-free: one of $a_i, \ldots, a_i + D$ lies in Γ. These n windows are disjoint, so $|\Gamma| \geq n \geq (T-1-D)/(D+1)$. With $T \geq L$ and $\beta|\Gamma| \leq \varepsilon L$ this gives $\varepsilon L \geq \beta(L-D-1)/(D+1) = \beta L/(D+1) - \beta$, which is either form of the claim. □

Why it matters. In Theorem D the crux is used at a rate $\rho > \varepsilon$, and for a transposer Proposition 4.13 gives $\varepsilon \geq \beta/(D+1) - \beta/L$. So at depth D the crux must deliver a rate of at least about $\beta/(D+1)$, and since $\beta \geq 2$, genies of cost εL have depth $D \geq 2/\varepsilon - 1 - o(1)$: the cheaper the genie, the deeper the side information that the crux has to handle. The next proposition shows that the size of the cones alone never suffices.

Proposition 4.14 (The counting bound cannot replace the crux)**.** Not formalized; a complete proof is given below.

Let I be a side-information system at size d with $|I(\ell)| \le B$ for all ℓ. Call a set W of non-trivial receivers (Remark 4.9(a)) acyclic *if there are no distinct $\ell_1, \dots, \ell_r \in W$, $r \ge 2$, with $\mathrm{tpos}(\ell_{i+1}) \in I(\ell_i)$ for all i (indices modulo r), and let $\mathrm{MAIS}(I)$ be the largest size of an acyclic set.*

(a) Every index code for I has at least $\mathrm{MAIS}(I)$ bits.

(b) $\mathrm{MAIS}(I) \ge (L - \mathrm{triv}(I))/(B+1)$.

(c) Suppose that a machine M with k work tapes, oblivious at length L with halting time $T \ge L$, has a genie Γ of depth at most D and cost $\beta|\Gamma| \le \varepsilon L$, where $\beta \ge 2$, $0 < \varepsilon \le 1$ and $L \ge \beta/\varepsilon$. Then $(k+1)^{D+1} > 1/\varepsilon$. Consequently, for every system I at this size whose sets have at most $B = (k+1)^{D+1}$ elements, in particular for every system of $\mathcal{S}_{k,C,\beta,D}$ (Corollary 4.5), the lower bound $(L - \mathrm{triv}(I))/(B+1)$ given by (a) and (b) is smaller than εL; it never contradicts the existence of an index code with εL bits, such as the revealed word G_Γ when M is a transposer.

Proof (not formalized). (a) This is the acyclic-subgraph bound of Bar-Yossef et al. [6, Theorem 3]; we include the short proof in our notation. Let W be acyclic, and order it as $\ell_1, \dots, \ell_r$ so that $\mathrm{tpos}(\ell_j) \in I(\ell_i)$ with $j \ne i$ implies $j < i$; this is possible because the digraph on W with an arc from ℓ to ℓ' when $\mathrm{tpos}(\ell') \in I(\ell)$ has no cycle. Let Z be the set of inputs that vanish outside $\mathrm{tpos}(W)$; it has $2^{|W|}$ elements because tpos is injective. We show that an index code G is injective on Z. Let $w, w' \in Z$ with $G(w) = G(w')$, and suppose w and w' agree at $\mathrm{tpos}(\ell_j)$ for all $j < i$. A position in $I(\ell_i)$ lies outside $\mathrm{tpos}(W)$, where both inputs vanish, or equals $\mathrm{tpos}(\ell_j)$ for some $j < i$, because $\mathrm{tpos}(\ell_i) \notin I(\ell_i)$ (receiver ℓ_i is non-trivial). So w and w' agree on $I(\ell_i)$, hence at $\mathrm{tpos}(\ell_i)$. By induction $w = w'$, so $2^m \ge 2^{|W|}$.

(b) Order the non-trivial receivers uniformly at random, and let W consist of those receivers ℓ that precede every non-trivial $\ell' \ne \ell$ with $\mathrm{tpos}(\ell') \in I(\ell)$. Then W is acyclic, since along a cycle in W each receiver would precede the next. As tpos is injective, a receiver ℓ has at most $|I(\ell)| \le B$ such receivers ℓ', so it lies in W with probability at least $1/(B+1)$, and $\mathbb{E}|W| \ge (L - \mathrm{triv}(I))/(B+1)$.

(c) By Proposition 4.13, $D + 1 \ge \beta L/(\varepsilon L + \beta) \ge \beta L/(2\varepsilon L) = \beta/(2\varepsilon) \ge 1/\varepsilon$, using $\beta \le \varepsilon L$ and $\beta \ge 2$. Hence $(k+1)^{D+1} \ge 2^{D+1} \ge 2^{1/\varepsilon} > 1/\varepsilon$, and $(L - \mathrm{triv}(I))/(B+1) \le L/(B+1) < \varepsilon L$. □

Why it matters. The combined bound $(L - \mathrm{triv}(I))/(B+1)$ of (a) and (b) uses only the sizes of the sets of the side information and the number of trivial receivers. Part (c) shows that it is always below εL: the depth that a genie of cost εL must reach is so large that the size bound $(k+1)^{D+1}$ on the cones is far bigger than $1/\varepsilon$. No argument that uses only the sizes can do better, because splitting the receivers into groups of $B+1$, each receiver knowing the bits demanded by the others in its group, gives a system with sets of size B and an index code of $\lceil L/(B+1) \rceil$ bits, one XOR per group. A proof of the crux must use the structure of the side information produced by tape schedules.

Remark 4.15 (The rate needed)**.** For each depth D, let $\rho^*(D)$ be the supremum of all rates ρ such that the statement $\mathrm{Crux}(\mathcal{S}_{k,C,\beta,D}, \rho)$ holds. It is non-increasing in D, because the classes grow with D. Suppose that depth-TCP holds with parameters $(k, C, \beta, \varepsilon, D(\varepsilon))$ for all small $\varepsilon > 0$. By Theorem D, oblivious machines with these parameters are then excluded as soon as $\rho^*(D(\varepsilon)) > \varepsilon$ for a single ε. If $D(\varepsilon) \le c\,\varepsilon^{-a}$ for constants c and a (by Proposition 4.13, necessarily $a \ge 1$ if depth-TCP is to apply to transposers), a crux with $\rho^*(D) \ge D^{-1/a+\delta}$ for some $0 < \delta < 1/a$ suffices, because then

$$\rho^*(D(\varepsilon)) \;\ge\; \left(c\,\varepsilon^{-a}\right)^{-1/a+\delta} \;=\; c^{-1/a+\delta}\,\varepsilon^{1-a\delta} \;>\; \varepsilon$$

for all small ε. For heads of leg area at most AT (home-based heads have $A = 1$), Corollary B below gives depth $D(\varepsilon) = 2^{O(k^3(1+A)\beta C/\varepsilon)}$; there a crux at a single depth D with a rate exceeding about $456k^3(1 + A)\beta C/\log_2 D$ suffices. Both requirements are far above the rate, at most $(k + 1)^{-D-1}$, that the counting bound of Proposition 4.14 gives.

4.4 Depth reduction for home-based walks: Theorem B

Theorem B is a statement about walks; machines reappear in Section 4.5. Throughout this subsection $x_1, \dots, x_k$ are walks of length T (Section 2.2), indexed by *heads* $h \in \{1, \dots, k\}$. They may be coupled arbitrarily: no relation between them is assumed.

The walk DAG and cuts. The *walk DAG* of $x_1, \dots, x_k$ has vertex set $[0, T)$ and two kinds of edges: *state edges* $t \to t - 1$ for $1 \le t < T$, and *long hops* $u \to \mathrm{prev}_h(u)$, one for every head h and every time u with $u - \mathrm{prev}_h(u) \ge 2$ (the hop $(\mathrm{prev}_h(u), u)$ of Section 2.2, read backwards). A pause $u \to u - 1$ of a head joins the same two vertices as the state edge $u \to u - 1$ and is identified with it; in general an edge is determined by its two endpoints, so a pair that is a long hop of several heads is a single edge. A *cut* is a set F of edges, and it has *depth at most* D if every path that uses no edge of F contains at most D long hops. For the work-head walks of an oblivious machine M at length L, the walk DAG has the vertices of H_M (Definition 4.1), and every edge of H_M joins the endpoints of an edge of the walk DAG: a cell edge of length 1 is a pause, and a cell edge of length at least 2 is a long hop.

Bands, legs and leg area. For $z \in \mathbb{Z}$ let $\lambda(z) = \lfloor \log_2 |z| \rfloor$ if $|z| \ge 2$ and $\lambda(z) = 0$ otherwise. The *band* of head h at time t is $\lambda_h(t) = \lambda(x_h(t))$, a function of the cell. Bands, scales, floors and the exponents θ of radii 2^θ are integers throughout. For integers $\theta \ge 1$, $\lambda(z) \ge \theta$ if and only if $|z| \ge 2^\theta$; and since $|x_h|$ changes by at most 1 per step, so does λ_h. The *scale* of a leg (Section 2.2) is the largest band over it. For $\theta \ge 0$ let $N_h(\theta, t)$ be the number of legs of head h of scale at least θ that begin at or before time t, and let $N_h(\theta) = N_h(\theta, T - 1)$. The *leg area* of head h is

$$S_h = \sum_{\text{legs } \ell \text{ of } h} 2^{\mathrm{scale}(\ell)},$$

and the *area hypothesis* with constant $A > 0$ is

$$\text{(A)} \qquad S_h \le AT \quad \text{for every head } h.$$

Every leg contributes at least 1, so head h has at most S_h legs. The statements below use only that the legs of each walk partition $[0, T)$ into intervals on which the walk is weakly monotone.

Lemma 4.16 (Leg area of home-based walks)**.** Not formalized; a complete proof is given below.
If every leg of a walk x of length T begins or ends at a cell z with $|z| \le 1$, then every leg ℓ satisfies $2^{\mathrm{scale}(\ell)} \le |\ell|$, and the leg area of x is at most T. In particular, home-based walks satisfy (A) with $A = 1$.

Proof (not formalized). If the scale s of ℓ is 0, then $2^s = 1 \le |\ell|$. Otherwise ℓ contains a time t with $|x(t)| \ge 2^s$ and has an endpoint e with $|x(e)| \le 1$. Since x moves by at most 1 per step, $|t - e| \ge 2^s - 1$, and the interval ℓ contains the $|t - e| + 1 \ge 2^s$ times between e and t. Summing over the legs, which partition $[0, T)$, bounds the leg area by T. □

The walk DAG does not change when a single walk is translated, so everything below applies to heads with individual fixed home cells, the bands of each head being measured from its home.

Lemma 4.17 (Crossings and spanning hops)**.** Not formalized; a complete proof is given below.

Let x be a walk of length T with leg area S, let $\theta \geq 1$, and let $N(\theta)$ be the number of legs of x of scale at least θ. A crossing *of radius 2^θ at time t ($1 \leq t < T$) is* outward *if $|x(t-1)| < 2^\theta \leq |x(t)|$ and* inward *if $|x(t-1)| \geq 2^\theta > |x(t)|$.*

(F1) *The walk has at most $2N(\theta)$ outward crossings, and at most $5N(\theta)$ crossings in total, of radius 2^θ. Moreover $\sum_{\theta \geq 1} 2^\theta N(\theta) \leq 2S$.*

(F2) *For every time τ and every integer $R \geq 1$, at most $2R-1$ hops $u \to \mathrm{prev}(u)$ of x span τ, that is, satisfy $\mathrm{prev}(u) < \tau \leq u$, and have $|x(u)| < R$.*

Proof (not formalized). (F1) Assign each outward crossing t to the leg ℓ that contains t; the scale of ℓ is at least θ because $|x(t)| \geq 2^\theta$. Either t is the first time of ℓ, which happens for at most one crossing per leg, or $t-1 \in \ell$ as well. In the second case suppose that x is non-decreasing on ℓ; the non-increasing case follows by the symmetry $z \mapsto -z$. If $x(t) \leq 0$, then $x(t-1) \leq x(t) \leq 0$ and $|x(t-1)| \geq |x(t)|$, which is impossible; so $x(t) \geq 2^\theta$, and $x \geq 2^\theta$ at all later times of ℓ, so no later outward crossing t' with $t'-1 \in \ell$ is assigned to ℓ. Thus each leg of scale at least θ receives at most two outward crossings and every other leg none, which gives the first bound. Crossings of radius 2^θ alternate between outward and inward, as each changes whether $|x| \geq 2^\theta$. So the inward crossings number at most the outward ones plus one, and the extra one occurs only if $|x(0)| \geq 2^\theta$, in which case the first leg has scale at least θ and $N(\theta) \geq 1$. Altogether there are at most $2N(\theta) + 3N(\theta)$ crossings. Finally, $\sum_{\theta \geq 1} 2^\theta N(\theta) = \sum_\ell \sum_{1 \leq \theta \leq \mathrm{scale}(\ell)} 2^\theta \leq \sum_\ell 2^{\mathrm{scale}(\ell)+1} = 2S$.

(F2) For such a hop, x visits the cell $x(u)$ at time $\mathrm{prev}(u) < \tau$ and at no time strictly between $\mathrm{prev}(u)$ and u, so u is the first visit to $x(u)$ at or after τ. Hence each of the $2R-1$ cells z with $|z| < R$ is the cell of at most one such hop. □

Lemma 4.18 (Legs and long hops)**.** Not formalized; a complete proof is given below.

Let $u \to p$ be a long hop of head h, so $p = \mathrm{prev}_h(u) \leq u-2$. The leg of h that contains u begins after p and has scale at least $\lambda_h(u)$. Hence $N_h(\theta, u) \geq N_h(\theta, p) + 1$ for every $\theta \leq \lambda_h(u)$.

Proof (not formalized). Pick t with $p < t < u$. By the definition of prev_h, head h is not at the cell $x_h(u) = x_h(p)$ at time t. If p and u lay in one leg, x_h would be weakly monotone on an interval containing $p < t < u$, so $x_h(t)$ would lie between the equal values $x_h(p)$ and $x_h(u)$, a contradiction. So the leg containing u is a later interval than the leg containing p, and it begins after p; its scale is at least $\lambda_h(u)$ because it contains u. For $\theta \leq \lambda_h(u)$ this leg is counted in $N_h(\theta, u)$ but not in $N_h(\theta, p)$, and every leg counted in $N_h(\theta, p)$ is counted in $N_h(\theta, u)$. □

Lemma 4.18 fails for pauses, whose two endpoints may lie in one leg; this is why pauses are treated as state edges.

Definition 4.19 (The band label and its cut)**.** Fix integers $a \geq 1$ (the *group width*), $r \in \{0, \ldots, a-1\}$ (the *offset*), $K \geq 2$ (the *modulus*) and $\omega_1, \ldots, \omega_k \in \{0, \ldots, K-1\}$.

- The *group* of head h at time t is $g_h(t) = \lfloor (\lambda_h(t) + r)/a \rfloor$, and the *floor* of a group $G \geq 0$ is $\theta(G) = \max(aG - r, 0)$, the smallest band in group G. The floors of the groups $G \geq 1$ are the integers $\theta \geq 1$ with $\theta \equiv -r \pmod a$.
- The *occupied groups* at time t are the distinct values among $g_1(t), \ldots, g_k(t)$, listed from the top as $G_1(t) > G_2(t) > \cdots > G_{m(t)}(t)$, with floors $\theta_i(t) = \theta(G_i(t))$; here $m(t) \leq k$.
- The *shared counts* are $n_i(t) = \sum_{h=1}^k N_h(\theta_i(t), t)$ for $i \leq m(t)$: the number of legs, of all heads, of scale at least $\theta_i(t)$ begun by time t.

- The *label* is
$$\Psi(t) = \sum_{i=1}^{m(t)} \big((n_i(t) + \omega_i) \bmod K\big)\, K^{k-i} \ \in [0, K^k).$$
- The *cut* $F = F(a, r, K, \omega)$ consists of the long hops $u \to p$ with $\Psi(p) \geq \Psi(u)$ and the state edges $t \to t-1$ with $\Psi(t-1) > \Psi(t)$.

So $\Psi(t)$ is the integer whose base-K digits, most significant first, are $\psi_i(t) = (n_i(t)+\omega_i) \bmod K$ for $i \leq m(t)$ and 0 for $m(t) < i \leq k$, and labels compare as these digit strings compare lexicographically. Whether an edge is cut depends only on its endpoints. For a fixed θ, the shared count $\sum_h N_h(\theta, t)$ is non-decreasing in t; this monotonicity and Lemma 4.18 are what the label uses.

Theorem 4.20 (Theorem B: depth reduction for home-based walks)**.** Not formalized; a complete proof is given below.

Let $x_1, \dots, x_k$ be walks of length T, coupled arbitrarily, that satisfy the area hypothesis (A) with constant A. Let $a \geq 1$ and $K \geq 2$ be integers, and let $r \in \{0, \dots, a-1\}$ and $\omega \in \{0, \dots, K-1\}^k$ be uniformly random and independent. Then the cut $F = F(a, r, K, \omega)$ has depth at most $K^k - 1$, and

$$\mathbb{E}_{r,\omega}|F| \ \leq \ \frac{(20k^2A + k)\,T}{a} + \Big(9 + \frac{2^{a+2}}{a}\Big)\frac{k^2AT}{K} \ \leq \ 20k^2(1+A)\,T\Big(\frac{1}{a} + \frac{2^a}{aK}\Big) + \frac{20k^2AT}{K}. \quad (2)$$

Consequently:

(a) for every $D \geq 4^k$, the choice $K = \lfloor D^{1/k} \rfloor$ and $a = \lfloor \log_2 D/(2k) \rfloor$, with suitable offsets, gives a cut of depth at most $D-1$ with $|F| \leq 114k^3(1+A)T/\log_2 D$;

(b) for every $0 < \varepsilon \leq 1$, some cut of size at most εT has depth less than $2^{\lceil 114k^3(1+A)/\varepsilon \rceil}$;

(c) in particular, for every coupling of k home-based walks, εT edges suffice to reach depth less than $2^{\lceil 228k^3/\varepsilon \rceil} = 2^{O(k^3/\varepsilon)}$.

Proof (not formalized). Depth. Along a path that avoids F, Ψ does not increase along state edges and strictly decreases along long hops. Since Ψ takes values in $[0, K^k)$, the path contains at most $K^k - 1$ long hops.

Notation. Fix the walks, a and K; probabilities and expectations are over r and ω. We bound the number of cut long hops counted with their head, as pairs (h, u), which is at least the number of cut long-hop edges, and the number of cut state edges. Consider a long hop $u \to p$ of head h, with cell $z = x_h(u) = x_h(p)$. Since bands are functions of the cell, h has the same band $\lambda = \lambda(z)$ and the same group $G = g_h(u) = g_h(p)$ at p and at u. Let $\theta^+ = \theta(G+1) = a(G+1) - r \geq 1$ be the floor of the next group: a head is in a group above G at time t exactly when its cell has absolute value at least 2^{θ^+}, and $|z| < 2^{\theta^+}$. Let $U(t)$ be the set of occupied groups above G at time t. Whether $U(p) = U(u)$ holds, and all counts n_i, depend on r only.

Step 1: $U(p) = U(u)$. Then G has the same position $j = |U(p)| + 1$ in the lists at p and at u, and $G_i(p) = G_i(u)$, with a common floor θ_i, for every $i \leq j$. Let $\Delta_i = n_i(u) - n_i(p)$ for $i \leq j$. For $i < j$, $\Delta_i \geq 0$ by monotonicity at a fixed floor. For $i = j$ we have $\theta_j = \theta(G) \leq \lambda = \lambda_h(u)$, so Lemma 4.18 gives $N_h(\theta_j, u) \geq N_h(\theta_j, p) + 1$, and with monotonicity for the other heads, $\Delta_j \geq 1$. Say that digit i *wraps* if $\lfloor (n_i(p) + \omega_i)/K \rfloor \neq \lfloor (n_i(u) + \omega_i)/K \rfloor$. If no digit $i \leq j$ wraps, then $\psi_i(p) = \psi_i(u) - \Delta_i$ for every $i \leq j$, so the first position at which the digit strings of p and u differ is at most j and carries the smaller digit at p. Thus $\Psi(p) < \Psi(u)$, and the hop is not cut; the digits below position j play no role. Given r, since ω_i is uniform on $\{0, \dots, K-1\}$, digit i wraps with probability at most Δ_i/K. So, given r, a hop of Step 1 is cut with probability at most $\sum_{i \leq j} \Delta_i/K$.

Step 2: $U(p) \neq U(u)$. We count every such hop as cut and charge it to a crossing. Let $G' > G$ be a group that lies in exactly one of $U(p)$ and $U(u)$. Since h is in group G at both times, at one of the two times G' is occupied by a head $h' \neq h$ that is not in G' at the other time. Groups change by at most one per step, and a head moves from a group G'' to $G''+1$ exactly when its band reaches $\theta(G''+1)$, that is, at an outward crossing of radius $2^{\theta(G''+1)}$; it moves back at an inward crossing of the same radius. If h' is in G' at time p, consider its first move out of G' after p; if h' is in G' at time u, consider its last move into G' at or before u. In either case the move happens at a step $\tau - 1 \to \tau$ with $p < \tau \leq u$, and it is a crossing of radius $2^{\theta(G'+1)}$ or $2^{\theta(G')}$. So h' has a crossing of radius 2^{θ^*} at a time $\tau \in (p, u]$, where θ^* is a floor with $\theta^* \geq \theta(G') \geq \theta^+$. Charge the hop to one such crossing (h', θ^*, τ), chosen by a fixed rule. Every hop (h, u) charged to (h', θ^*, τ) satisfies $\mathrm{prev}_h(u) < \tau \leq u$ and $|x_h(u)| = |z| < 2^{\theta^+} \leq 2^{\theta^*}$, so by (F2) with $R = 2^{\theta^*}$, applied to each of the k heads, at most $k\,2^{\theta^*+1}$ hops are charged to each crossing. By (F1), head h' has at most $5N_{h'}(\theta)$ crossings of radius 2^θ. Hence, for fixed r,

$$\#\{\text{hops of Step 2}\} \;\leq\; \sum_{h'=1}^{k} \sum_{\theta \geq 1 \text{ a floor}} 5N_{h'}(\theta) \cdot k\,2^{\theta+1}.$$

Each $\theta \geq 1$ is a floor for exactly one value of r, hence with probability $1/a$, and by (F1) and (A),

$$\mathbb{E}\,\#\{\text{hops of Step 2}\} \;\leq\; \frac{10k}{a} \sum_{h'} \sum_{\theta \geq 1} 2^\theta N_{h'}(\theta) \;\leq\; \frac{10k}{a} \sum_{h'} 2S_{h'} \;\leq\; \frac{20k^2AT}{a}.$$

Step 3: wraps. By Step 1, the expected number of cut hops of Step 1 is at most $\frac{1}{K}\mathbb{E}\sum\sum_{i\leq j}\Delta_i$, where the outer sum runs over the hops of Step 1. For such a hop and $i \leq j$, Δ_i is the number of legs, of all heads, that have scale at least θ_i and begin in $(p, u]$. We exchange the order of summation: for each leg ℓ, of any head, beginning at time τ_ℓ with scale s_ℓ, we count the hops that span τ_ℓ in the sense of (F2), that is, satisfy $p < \tau_\ell \leq u$. By the argument of (F2), for each head and each cell at most one hop of that head spans τ_ℓ and ends at that cell.

Own digit ($i = j$). A hop spanning τ_ℓ contributes ℓ to its Δ_j only if $\theta(G) \leq s_\ell$, where G is its group. Since $\theta(G)$ is the largest floor at most λ (or 0 if there is none), this holds if $\lambda \leq s_\ell$, and for $\lambda > s_\ell$ it holds exactly when no floor lies in $(s_\ell, \lambda]$. The floors at least 1 form an arithmetic progression of difference a with a uniformly random offset, so the latter has probability $\max(0, 1 - (\lambda - s_\ell)/a)$. There are $2^{s+2} - 1$ cells of band at most s, and $2^{\lambda+1}$ cells of band $\lambda \geq 1$. So the expected number of hops counted for ℓ is at most

$$k\Big(2^{s_\ell+2} + \sum_{i=1}^{a-1} 2^{s_\ell+i+1}\Big(1 - \frac{i}{a}\Big)\Big) \;\leq\; k\Big(2^{s_\ell+2} + \frac{2^{s_\ell+a+2}}{a}\Big),$$

because $\sum_{i=1}^{a-1} 2^i(a-i) = 2^a \sum_{j=1}^{a-1} j2^{-j} < 2^{a+1}$. Summing over the legs of all heads, with $\sum_\ell 2^{s_\ell} = \sum_{h'} S_{h'} \leq kAT$, gives $\mathbb{E}\sum\Delta_j \leq 4k^2AT(1 + 2^a/a)$.

Higher digits ($i < j$). Here $G_i > G$, so $\theta_i \geq \theta^+$ is a floor at least 1 that exceeds the band of the hop, and distinct i give distinct floors. So, for each leg ℓ, the pairs consisting of a hop spanning τ_ℓ and an index $i < j$ with $\theta_i \leq s_\ell$ number at most the sum, over the floors $1 \leq \theta \leq s_\ell$, of the number of hops spanning τ_ℓ whose cell has absolute value less than 2^θ. By (F2) this number is at most $k(2^{\theta+1} - 1)$, and since consecutive floors differ by $a \geq 1$, the sum is at most $k\sum_{\theta=1}^{s_\ell} 2^{\theta+1} < k\,2^{s_\ell+2}$. Summing over the legs gives $\sum\sum_{i<j}\Delta_i \leq 4k^2AT$ for every r.

Together, the expected number of cut hops of Step 1 is at most $(4k^2AT(1+2^a/a) + 4k^2AT)/K = (8 + 2^{a+2}/a)\,k^2AT/K$.

Step 4: state edges. Consider the state edge $t \to t-1$. If no head changes group at step $t-1 \to t$, the occupied groups and their floors are the same at $t-1$ and t, and $n_i(t-1) \le n_i(t)$ for every i; as in Step 1, the edge is cut only if some digit wraps, which, given r, has probability at most $\sum_i (n_i(t) - n_i(t-1))/K$. Here $n_i(t) - n_i(t-1)$ is the number of heads that begin, at time t, a leg of scale at least θ_i. A head begins at most one leg at a time and there are at most k digits, so the sum over i and t is at most k times the number of legs, which is at most $\sum_{h'} S_{h'} \le kAT$; so these cuts number at most k^2AT/K in expectation. Otherwise some head changes group at step $t-1 \to t$, and we count the edge as cut. A head changes group at a step only if its band moves between two values λ and $\lambda+1$ and $\lambda+1$ is a floor, which has probability $1/a$; so the expected number of such steps is at most kT/a.

Total. Adding Steps 2, 3 and 4,

$$\mathbb{E}|F| \;\le\; \frac{20k^2AT}{a} + \Big(8 + \frac{2^{a+2}}{a}\Big)\frac{k^2AT}{K} + \frac{k^2AT}{K} + \frac{kT}{a},$$

which is the first bound of (2); the second follows from $k \le 20k^2$, $9 \le 20$ and $2^{a+2}A \le 20(1+A)2^a$. Some offsets r, ω achieve at most the expectation.

Parameters. For (a) put $y = \log_2 D/(2k) \ge 1$. Then $a = \lfloor y \rfloor \ge y/2$, and $K = \lfloor 2^{2y} \rfloor \ge 2^{2y-1}$ because $2^{2y} \ge 4$; moreover $K^k \le D$. Using $2^{-y} \le 1/y$ for $y > 0$,

$$\frac{1}{a} \le \frac{2}{y} = \frac{4k}{\log_2 D}, \qquad \frac{2^a}{K} \le 2^{1-y} \le \frac{2}{y} = \frac{4k}{\log_2 D}, \qquad \frac{1}{K} \le 2^{1-2y} \le 2^{-y} \le \frac{1}{y} = \frac{2k}{\log_2 D}.$$

Inserting these into the first bound of (2), with $2^{a+2}/(aK) \le 4 \cdot 2^a/K$, gives

$$\mathbb{E}|F| \;\le\; (80k^3A + 4k^2 + 18k^3A + 16k^3A)\frac{T}{\log_2 D} \;\le\; \frac{114k^3(1+A)\,T}{\log_2 D},$$

and the depth is at most $K^k - 1 \le D - 1$. For (b) take $\log_2 D = \lceil 114k^3(1+A)/\varepsilon \rceil$, which is at least $2k$. Part (c) is (b) with $A = 1$, by Lemma 4.16. □

Why it matters. Theorem B proves the walk form of the first hypothesis of Theorem D for all heads of linear leg area, in particular for home-based heads, and for every coupling: it makes no assumption on hop lengths, on independence, or on how the heads move relative to each other. So, for an oblivious machine running in time at most $C(L+1)$, a genie of cost εL and depth at most $D(\varepsilon)$, depending on k, A, β and C, exists for every $\varepsilon \in (0,1]$ (see the paragraph after the proof of Corollary 4.23) unless the leg area of some work head is not $O(T)$ for any choice of home cell. That requires many legs whose duration is much shorter than 2^{scale}; by Lemma 4.16 such legs neither begin nor end near the home cell, so the head turns around far from every fixed home. Section 5 studies walks of this kind.

4.5 Home-based oblivious machines reduce to the crux

Theorem B cuts edges, counts only long hops, and pays one unit per cut edge. Two conversions turn its cut into a genie in the sense of Definition 4.3: an edge $t \to s$ is blocked by revealing s, at a cost of β bits, and the runs of short edges are cut by revealing periodic steps.

Lemma 4.21 (Long-edge conversion)**.** Not formalized; a complete proof is given below.

Let M be oblivious at length L with halting time T, let $\mu \ge 1$ and $P > \mu$ be integers, and let Γ_0 be a genie such that every Γ_0-free path has at most D' edges of length greater than μ. Then the genie

$$\Gamma = \Gamma_0 \cup \{t \in [0,T) : t \bmod P \ge P - \mu\}$$

has depth at most $D' + (D'+1)(P-\mu) \le (D'+1)P - 1$, *and* $|\Gamma| \le |\Gamma_0| + \mu\lceil T/P\rceil$; *for* $\mu = 1$, $|\Gamma| \le |\Gamma_0| + \lfloor T/P\rfloor$.

Proof (not formalized). Let $t_0 \to t_1 \to \cdots \to t_m$ be Γ-free. It is Γ_0-free, so at most D' of its edges have length greater than μ, and removing them leaves at most $D'+1$ maximal runs of consecutive edges of length at most μ. Let $t_a \to \cdots \to t_b$ be such a run. Its vertices $t_{a+1} > \cdots > t_b$ are not in Γ, so each lies in an interval $[nP, nP + P - \mu)$ with $n \ge 0$. If t_i lies in such an interval and $t_{i+1} < nP$ for consecutive vertices of the run, then $n \ge 1$ and $t_{i+1} \ge t_i - \mu \ge nP - \mu$, so $t_{i+1} \bmod P \ge P - \mu$ and $t_{i+1} \in \Gamma$, which is impossible. Hence $t_{a+1}, \ldots, t_b$ are distinct elements of one interval of $P - \mu$ consecutive integers, and the run has $b - a \le P - \mu$ edges. So $m \le D' + (D'+1)(P-\mu)$, and this is at most $(D'+1)P - 1$ as $\mu \ge 1$. Each of the $\lceil T/P\rceil$ periods $[nP, (n+1)P)$ that meet $[0, T)$ contains μ revealed residues; for $\mu = 1$ the revealed steps are $P-1, 2P-1, \ldots$, of which $\lfloor T/P\rfloor$ are less than T. □

Remark 4.22 (Why blocks of μ steps)**.** For $\mu = 1$ one revealed step per period suffices, because a run of edges of length 1 visits consecutive times. Single revealed steps do not cut runs of edges of length at most μ when $\mu \ge 2$: a work head that oscillates between two adjacent cells has $\mathrm{prev}_j(t) = t - 2$ for all $t \ge 2$, and if P is even, the path $t \to t-2 \to t-4 \to \cdots$ through even times avoids every step $t \equiv P - 1 \pmod P$. A depth-reduction statement that treats such short revisits (jitter, or revisits within a window of μ steps) as free must therefore be combined with Lemma 4.21 for that μ, which reveals blocks of μ consecutive steps at μ times the cost. Theorem B needs only $\mu = 1$, because it counts every hop of length at least 2 as long.

Corollary 4.23 (Corollary B: home-based oblivious machines reduce to the crux)**.** Not formalized; a complete proof is given below.

Let $k \ge 1$ *and* $\beta \ge 2$ *be integers,* $C \ge 1$, $A > 0$ *and* $0 < \varepsilon \le 1$, *and put*

$$D = D(\varepsilon) = 2^{\lceil 456k^3(1+A)\beta C/\varepsilon\rceil} \cdot \lceil 4\beta C/\varepsilon\rceil = 2^{O(k^3(1+A)\beta C/\varepsilon)}.$$

Suppose that $\mathrm{Crux}(\mathcal{S}_{k,C,\beta,D}, \rho)$ *holds for some* $\rho > \varepsilon$. *Then for every machine* M *with* k *work tapes and value width at most* β *there is* d_0 *such that for no* $d \ge d_0$ *do the following three conditions hold together at* $L = d^2 b(d)$: M *is oblivious at length* L *with halting time* $T \le C(L+1)$; *each work-head walk* x_j *of* M *on* $[0, T)$ *has leg area at most* AT *measured from some cell* c_j *(that is, the walk* $x_j - c_j$ *satisfies (A)); and* M *transposes all* $d \times d$ *matrices with* $b(d)$*-bit entries. In particular, if an oblivious machine with* k *work tapes and value width at most* β, *whose work heads are home-based at every input length, transposes within* $C(L+1)$ *steps for infinitely many* d, *then for every* $\varepsilon \in (0, 1]$ *the crux* $\mathrm{Crux}(\mathcal{S}_{k,C,\beta,D(\varepsilon)}, \rho)$, *with* $A = 1$ *in* $D(\varepsilon)$, *fails for every rate* $\rho > \varepsilon$.

Proof (not formalized). Put $D_1 = 2^{\lceil 456k^3(1+A)\beta C/\varepsilon\rceil}$ and $P = \lceil 4\beta C/\varepsilon\rceil$, so that $D = D_1 P$; note that $D_1 \ge 4^k$, because $456k^3(1+A)\beta C/\varepsilon \ge 2k$. Fix M, let d_0 be given by $\mathrm{Crux}(\mathcal{S}_{k,C,\beta,D}, \rho)$ for $\eta = (\rho - \varepsilon)/2$, and suppose that the three conditions hold at some $d \ge d_0$. As in the proof of Theorem 4.10, M appends exactly L output bits, and on every input w its output bit ℓ is $w_{\mathrm{tpos}(\ell)}$.

Apply Theorem 4.20(a) with depth parameter D_1 to the translated work-head walks $x_1 - c_1, \ldots, x_k - c_k$ on $[0, T)$, whose walk DAG is that of $x_1, \ldots, x_k$: some cut F of depth at most $D_1 - 1$ has $|F| \le 114k^3(1+A)T/\log_2 D_1 \le \varepsilon T/(4\beta C)$. Let Γ_0 be the set of steps s such that some edge of F ends at s; so $|\Gamma_0| \le |F|$. Every edge of a Γ_0-free path in H_M ends outside Γ_0, so its endpoints are not those of an edge of F. Read in the walk DAG, the path therefore avoids F, and its edges of length at least 2 are long hops; so it has at most $D_1 - 1$ edges of length greater than 1. By

Lemma 4.21 with $\mu = 1$, the genie $\Gamma = \Gamma_0 \cup \{t < T : t \equiv P - 1 \pmod P\}$ has depth at most $D_1 P - 1 < D$, and

$$\beta|\Gamma| \;\le\; \beta|F| + \beta\lfloor T/P \rfloor \;\le\; \frac{\varepsilon T}{4C} + \frac{\varepsilon T}{4C} \;\le\; \frac{\varepsilon(L+1)}{2} \;\le\; \varepsilon L,$$

using $T \le C(L+1)$ and $L \ge 1$. So $I_\Gamma \in \mathcal{S}_{k,C,\beta,D}$, and by Corollary 4.5 the revealed word G_Γ is an index code for I_Γ with at most εL bits, while the crux requires at least $(\rho - \eta)L > \varepsilon L$ bits. This contradiction proves the first statement. For the second, a work head that is home-based with home cell c_j satisfies (A) with $A = 1$ when measured from c_j, by Lemma 4.16 applied to $x_j - c_j$; a machine that transposes within $C(L+1)$ steps at length L has halting time at most $C(L+1)$ there; so a machine as described would violate the first statement at infinitely many d if the crux held at some rate $\rho > \varepsilon$. □

The genie Γ built in this proof does not use that M transposes. It shows that the conclusion of depth-TCP$(k, C, \beta, \varepsilon, D(\varepsilon))$ holds for every machine M with k work tapes and value width at most β, at every length $L = d^2 b(d)$ at which M is oblivious with halting time at most $C(L+1)$ and its work-head walks satisfy (A), measured from some home cells.

Why it matters. Theorem D needs two hypotheses. For home-based heads Theorem B supplies the first, so Corollary B reduces home-based oblivious machines to the crux alone, whatever the coupling of their heads. Read as a requirement on the crux, it suffices to prove $\mathrm{Crux}(\mathcal{S}_{k,C,\beta,D}, \rho)$ at a single depth $D = D(\varepsilon)$ with a rate $\rho > \varepsilon$, that is, a rate exceeding about $456k^3(1+A)\beta C/\log_2 D$ at depth D (Remark 4.15); this is far beyond the rate, at most $(k+1)^{-D-1}$, that the counting bound of Proposition 4.14 gives, but, for large D, far below rate $\frac{1}{2}$. Read as a design constraint, an oblivious linear-time transposer must either have a work head whose leg area, measured from any home cell, is not $O(T)$, or produce, for every $\varepsilon \in (0, 1]$ and at every size at which it transposes, an index code with at most εL bits for its own cone side information at depth $D(\varepsilon)$, and hence a storage code of rate at least $1 - \varepsilon$ (Remark 4.9(b)).

Remark 4.24 (Adaptive machines)**.** The argument uses obliviousness twice: the dependency DAG and the times of the output bits must be the same for all inputs. We do not treat adaptive machines here. We expect that an extension of Corollary B to adaptive machines whose heads are home-based on every input would need additional hypotheses on head motion far from home and a stronger form of the crux, for index codes that need to be correct only on a large set of inputs. No result of this paper depends on such an extension.

5 Depth reduction for interacting walks

Theorem 4.10 reduces the oblivious case of (T) to two hypotheses. This section is about the structural one: deleting a small fraction of the edges of the walk DAG of a head schedule so that every remaining path contains few long hops. Theorem 4.20 does this for home-based walks, for every coupling. Here we state the general problem at a fixed fraction ε (Problem 5.1), rule out packing certificates for lower bounds (Proposition 5.14), give evidence against per-head band bookkeeping (Remarks 5.6 and 5.7, proof sketches only), and study what we regard as the sharpest open instance: two deeply nested walks, shifted against each other in time (Definition 5.9). One such walk alone is easy (Lemma 5.8), and the pair with m levels of nesting is depth-reducible: clock labels cut it at a depth exponential in m (Proposition 5.13). On a cyclic cover of the pair the cost of every label obeys exact identities (Lemma Zq, Lemmas 5.17 and 5.18); with them we prove, for the pair

with one level of nesting, that the labels that count exactly along one head pay a price per hop on the cover that does not depend on the depth (Theorems 5.21 and 5.22). Whether the pairs are depth-robust as m grows remains open (Conjecture 5.25).

5.1 The problem

We use the walk DAG of Section 4.4: for walks $x_1, \dots, x_k$ of length T, its vertices are the times $[0, T)$, its edges are the state edges $t \to t-1$ and the long hops $u \to \mathrm{prev}_h(u)$, a pause is identified with the state edge joining the same two times, and an edge is determined by its endpoints. A *cut* is a set of edges; it has *depth at most* D if every path that uses no edge of the cut contains at most D long hops.

Problem 5.1 (TCP(k) at fixed ε)**.** *Let $k \geq 1$. Is there, for every $\varepsilon > 0$, an integer $D = D(k, \varepsilon)$ such that for every T and all walks $x_1, \dots, x_k$ of length T, coupled arbitrarily, the walk DAG has a cut with at most εT edges and depth at most D?*

A family of k-tuples of walks is *depth-robust* if for some $\varepsilon > 0$ no single D serves all its members. Cuts are conveniently described by labels. For an integer $D \geq 0$ and a map $\Psi\colon [0, T) \to \{0, \dots, D\}$, call a long hop $u \to p$ *violated* if $\Psi(u) \leq \Psi(p)$, call a time $t \geq 1$ a *descent* if $\Psi(t) < \Psi(t-1)$, and let $\mathrm{cost}(\Psi)$ be the number of violated long hops plus the number of descents.

Lemma 5.2 (Cuts are labels)**.** Not formalized; a complete proof is given below.

For all walks $x_1, \dots, x_k$ of length T and every $D \geq 0$, the smallest size of a cut of depth at most D equals the smallest value of $\mathrm{cost}(\Psi)$ over all maps $\Psi\colon [0, T) \to \{0, \dots, D\}$.

Proof (not formalized). Given Ψ, let F consist of the violated long hops and of the state edges $t \to t-1$ at descents t; so $|F| = \mathrm{cost}(\Psi)$. Along every edge $a \to b$ outside F we have $\Psi(b) \leq \Psi(a)$, with strict inequality for long hops. Along a path avoiding F, Ψ therefore never increases and drops at every long hop; as it takes values in $\{0, \dots, D\}$, the path contains at most D long hops. Conversely, let F be a cut of depth at most D, and let $\Psi(t)$ be the largest number of long hops on a path that starts at t and avoids F; so $0 \leq \Psi(t) \leq D$. For an edge $a \to b$ not in F, prepending it to a path from b gives $\Psi(a) \geq \Psi(b) + 1$ if the edge is a long hop and $\Psi(a) \geq \Psi(b)$ otherwise. Hence every violated long hop and every state edge at a descent of Ψ belongs to F, and $\mathrm{cost}(\Psi) \leq |F|$. □

Why the problem matters. Problem 5.1 is the walk-level form of the structural hypothesis of Theorem 4.10. If it has a positive answer for k, the proof of Corollary 4.23, with the cut provided by Problem 5.1 in place of the one provided by Theorem 4.20, shows that depth-TCP$(k, C, \beta, \varepsilon, D)$ holds for all $C \geq 1$, $\beta \geq 2$ and $0 < \varepsilon \leq 1$, with $D = (D(k, \varepsilon/(4\beta C)) + 1)\lceil 4\beta C/\varepsilon \rceil$; Theorem 4.10 then reduces the oblivious case of (T) to the crux alone, as Corollary 4.23 does for home-based heads. The problem is posed for all walks, and not only for the head schedules of transposers, because a machine can run any schedule that its finite control generates on extra work tapes without changing its output; such dummy traffic enters the dependency DAG. A family that oblivious machines can generate in linear time, and that is depth-robust at every ε below the crux rates in question, would therefore make the structural hypothesis fail for some machines, and the route of Theorem 4.10 would need a version of it that sees only the part of the DAG able to carry the transposition. On the construction side, Theorem 4.10 says that an oblivious linear-time transposer either beats the crux or has a head schedule without genies of small cost and bounded depth; up to constants this means that its walk DAG has no cut of size εT and bounded depth, since a genie Γ yields a cut of at most $(k+1)|\Gamma|$ edges and the conversions of Section 4.5 go the other way. The walks of this section show what the second alternative could look like.

For $k = 1$ the problem is easy. Call a long hop (p, u) of a single walk x an upper hop if $x > x(p)$ on (p, u) and a lower hop if $x < x(p)$ there. Two upper hops never cross as intervals of time: if $p < p' < u < u'$, then $x(p') > x(p)$ because $p' \in (p, u)$, while $u \in (p', u')$ and $x(u) = x(p) < x(p')$, contradicting that $x > x(p')$ on (p', u'); the same holds for lower hops. Every long hop is an upper or a lower hop, since $x \neq x(p)$ on (p, u) and x moves by at most 1 per step. Drawing the upper hops on one side of the time axis and the lower hops on the other shows that the walk DAG of one walk is planar; its maximum degree is at most 4. By the planar separator theorem of Lipton and Tarjan [47], applied recursively as in [48], some set of $O(T/\sqrt{M})$ vertices leaves components of at most M vertices. Cutting the at most four edges at each of these vertices cuts $O(T/\sqrt{M})$ edges, and a path that avoids the cut stays inside one component, so it has at most $M - 1$ long hops; this gives $D(1, \varepsilon) = O(\varepsilon^{-2})$. For $k \geq 2$ the union of the walks' hop families is in general not planar, and only special classes are solved, chief among them the home-based walks of Theorem 4.20.

Relation to segregators. Paul, Pippenger, Szemerédi and Trotter [57] showed that the computation graphs of multitape machines have small segregators: deleting $O(n/\log^* n)$ vertices leaves every vertex with at most $O(n/\log^* n)$ predecessors. This is the combinatorial core of their separation of deterministic from nondeterministic linear time. Problem 5.1 is a depth version of the same problem with constant parameters. Through the conversions of Section 4.5, a positive answer gives genies of cost εL and bounded depth D', and by the cone lemma (Lemma 4.4) every step then has at most $(k+1)^{D'+1}$ predecessors reached without passing through the genie: the deleted set is a fixed fraction of the steps, far more than $n/\log^* n$, but the remaining cones have bounded size instead of size $n/\log^* n$. Santhanam stated that if, for every r, the multi-pushdown graphs $H_r(n)$, a class containing the computation graphs of multitape machines, had segregators deleting $o(n/\log n)$ vertices and leaving every vertex at most $o(n)$ predecessors, then deterministic and nondeterministic time $n \log n$ would differ [60, Thm. 2.7], and indicated that its proof is analogous to that of his Theorem 2.5. Since the deleted part in Problem 5.1 is linear, a positive answer would not by itself give a new separation of complexity classes.

5.2 Uniform recursive walks

The instances of this section are built from one family of walks in which every level of a recursive nesting receives the same total time. Since Problem 5.1 concerns all schedules, these walks are relevant even though the results of Section 3 constrain the macroscopic motion of transposers.

Definition 5.3 (Uniform recursive walks)**.** Let $g \geq 1$ and $m \geq 0$ be integers, and let s_0 be an integer with $s_i := s_0 - ig \geq 1$ for $0 \leq i \leq m$. A *level-m stroke* of direction $\sigma \in \{-1, 1\}$ is the sequence of 2^{s_m} steps equal to σ. For $i < m$, a *level-i stroke* of direction σ is the concatenation, for $j = 0, \ldots, 2^{g-1} - 1$, of a *segment* of $2^{s_{i+1}+1}$ steps equal to σ followed by a V: a level-$(i+1)$ stroke of direction $-\sigma$ (the *back stroke*) followed by a level-$(i+1)$ stroke of direction σ (the *forward stroke*). The walk $U(s_0, m, g)$ is the walk $x\colon \mathbb{Z} \to \mathbb{Z}$ with $x(0) = 0$, periodic with period $P := 4(m+1)2^{s_0}$, whose steps $x(t+1) - x(t)$, $t \in [0, P)$, are those of four level-0 strokes of directions $1, -1, -1, 1$, in this order.

By induction on $m - i$, a level-i stroke has $(m - i + 1)2^{s_i}$ steps and moves the head by $\sigma 2^{s_i}$: its segments contribute $2^{g-1} \cdot 2^{s_{i+1}+1} = 2^{s_i}$, and each V returns the head to where it started. So every period has $4 \cdot 2^{s_0} = P/(m+1)$ steps belonging to segments of level-i strokes for each $i < m$, and as many belonging to level-m strokes, and the walk never pauses. Each step belongs to exactly one segment or level-m stroke; we say that the stroke emits it, and a time t belongs to the stroke that

emits the step from t to $t+1$. The *span* of a stroke or of a V is the set of times belonging to it, and its *end time* is the first time after the span. The terms *during*, *first time* and *last time* refer to spans.

If a level-i stroke of direction σ, $i < m$, starts at cell a, its segment j ends at the *grid point* $b_j = a + \sigma(j+1)2^{s_{i+1}+1}$, and the V that follows is *attached* at b_j. By Lemma 5.4(a) below, this V stays in its *range*, the interval of cells between its *base* b_j and its *valley cell* $b_j - \sigma 2^{s_{i+1}}$; the *level* of the V is the level of its strokes, its *apex time* is the first time of its back stroke and its *valley time* the first time of its forward stroke. The *open half-segments* of the stroke are the open intervals of cells between consecutive points of $a + \sigma 2^{s_{i+1}}\mathbb{Z}$ that lie between a and $a + \sigma 2^{s_i}$; the *open range* of a level-i stroke is the set of cells strictly between a and $a + \sigma 2^{s_i}$, and for a level-m stroke we use it in place of the open half-segments; the open range of a V is its range without the base and the valley cell. A *turn* is a time t with $x(t+1) - x(t) \neq x(t) - x(t-1)$, and its *level* is the level of the stroke to which t belongs. Intervals of cells are written without regard to orientation, and "beyond b in direction σ" means $\sigma(c - b) > 0$.

Lemma 5.4 (Structure of U)**.** Not formalized; a complete proof is given below.
Let $x = U(s_0, m, g)$.

(a) During a level-i stroke S of direction σ starting at cell a, including its end time, the head stays between a and $a + \sigma 2^{s_i}$, and it is at $a + \sigma 2^{s_i}$ at the end time. The first and the last step of S equal σ. If $i < m$, the part of S up to the end time of the V at b_j stays between a and b_j, and the part after it stays between b_j and $a + \sigma 2^{s_i}$. During a V, including its end time, the head stays in the range of the V.

(b) The turns are exactly the apex and valley times of all V's, each of the level of its V, and, in every period, the first times of the second and of the fourth level-0 stroke, which have level 0; at these two times the head is at 2^{s_0} and at -2^{s_0}.

(c) During a V of level i, every cell strictly inside its range is visited exactly $2(m-i+1)$ times and each of its two endpoints exactly $m - i + 1$ times. In every period, every cell c with $|c| < 2^{s_0}$ is visited exactly $2(m+1)$ times and each of $\pm 2^{s_0}$ exactly $m+1$ times.

(d) For every stroke S whose first time t_S is nonnegative, the number of times $t \in [0, t_S)$ with $x(t) = c$ is the same for all cells c of any one open half-segment of S (of the open range of S if S has level m).

(e) Let $1 \le j \le m$. If no turn of level less than j lies strictly inside a hop (p, u), then $|x(t) - x(p)| \le 2^{s_j}$ for all $t \in [p, u]$.

Proof (not formalized). (a) Induction on $m - i$. A level-m stroke is monotone. For $i < m$, the segments move the head monotonically from a to $a + \sigma 2^{s_i}$, reaching b_j at the end of segment j. By the induction hypothesis, the back stroke of the V at b_j stays between b_j and $b_j - \sigma 2^{s_{i+1}}$ and ends at the valley cell, and the forward stroke stays in the same range and ends at b_j. This range lies between $b_j - \sigma 2^{s_{i+1}+1}$, the start of segment j, and b_j, which gives all the claims about S. The first step of S is a segment step, and its last step is the last step of its last forward stroke, which equals σ by induction.

(b) Inside a segment and inside a level-m stroke all steps are equal. Let t be a time at which the steps from $t-1$ to t and from t to $t+1$ are emitted by different strokes. Let Y be the largest segment or stroke whose span ends at $t-1$, and Z the largest segment or stroke whose span begins at t. They are consecutive parts of one stroke or of the sequence of level-0 strokes, so the pair (Y, Z) is of one of four kinds, and by (a) the last step of Y and the first step of Z are their directions.

(1) Y is a segment of a stroke of direction σ and Z the back stroke of the V attached at its end: the steps are σ and $-\sigma$, and t is the apex time, a turn of the level of the back stroke. (2) Y is a back stroke and Z the forward stroke of the same V: the steps are $-\sigma$ and σ, and t is the valley time, a turn of the level of the forward stroke. (3) Y is a forward stroke and Z the next segment of the stroke to which its V is attached: both steps are σ, and t is not a turn. (4) Y and Z are consecutive level-0 strokes: their directions $1, -1, -1, 1$, followed by 1 in the next period, give a turn exactly when Z is the second or the fourth stroke of a period, and then the head is at 2^{s_0} or -2^{s_0} by (a).

(c) By induction on $m-i$ we prove the claim about V's. A V of level m consists of two monotone runs of 2^{s_m} steps in opposite directions: the back stroke visits every cell of the range except the valley cell, the forward stroke every cell except the base. Let $i < m$, and suppose by symmetry that the back stroke goes down, so that the range is $[v, b]$ with $v = b - 2^{s_i}$. Put $s = s_{i+1}$ and $K = 2^{g-1}$, and let $v = \gamma_0 < \gamma_1 < \cdots < \gamma_K = b$ be the points of $v + 2^{s+1}\mathbb{Z}$ in $[v, b]$; since $2^{s_i} = K2^{s+1}$, these are also the points of $b - 2^{s+1}\mathbb{Z}$ in $[v, b]$. The segments of the back stroke visit $b, b-1, \ldots, v+1$ once each, and its V's have level $i+1$, are attached at $\gamma_{K-1}, \ldots, \gamma_0$ and have ranges $[\gamma_k, \gamma_k + 2^s]$. The segments of the forward stroke visit $v, v+1, \ldots, b-1$ once each, and its V's are attached at $\gamma_1, \ldots, \gamma_K$ and have ranges $[\gamma_k - 2^s, \gamma_k]$. By the induction hypothesis each of these V's visits the cells strictly inside its range $2(m-i)$ times and its endpoints $m-i$ times, and the spans of the segments and of these V's partition the span of the V. A cell strictly between γ_k and $\gamma_k + 2^s$, or strictly between $\gamma_k + 2^s$ and γ_{k+1}, lies strictly inside exactly one of the attached V's and is visited $1 + 1 + 2(m-i)$ times. The cell $\gamma_k + 2^s$ $(0 \le k < K)$ is an endpoint of two attached V's, the back stroke's at γ_k and the forward stroke's at γ_{k+1}, and so is every γ_k with $0 < k < K$; these cells are visited $1 + 1 + 2(m-i)$ times. The base b is visited once by the back stroke's segments and $m-i$ times by the forward stroke's V at $\gamma_K = b$, and the valley cell v once by the forward stroke's segments and $m-i$ times by the back stroke's V at $\gamma_0 = v$. This proves the claim for V's. In a period, the first two level-0 strokes have, over $[0, 2^{s_0}]$, the same segments and attached V's as the forward and back strokes of a V over this range (in the other order), so the count for V's applies verbatim and shows that they visit every c with $0 < c < 2^{s_0}$ exactly $2(m+1)$ times and 0 and 2^{s_0} exactly $m+1$ times; the last two strokes do the same over $[-2^{s_0}, 0]$.

(d) For a stroke S with $t_S \ge 0$ write $\mathrm{pre}_S(c)$ for the number of times $t \in [0, t_S)$ with $x(t) = c$. By (a) and (c), during a level-i stroke with $i < m$ the number of visits to c is constant on each open half-segment: the segments visit each cell between the first cell (included) and the last cell (excluded) once, and the V at b_j adds $2(m-i)$ visits on the open half-segment of segment j next to b_j and none elsewhere inside open half-segments. During a level-m stroke every cell of the open range is visited once. We now argue down the tree of strokes; since time 0 starts a level-0 stroke, the parent of every stroke S with $t_S \ge 0$ also starts at a nonnegative time. Let S be a level-0 stroke of period n. If it is the first or the third stroke of the period, $\mathrm{pre}_S(c) = 2(m+1)n$ on its open range by (c). If it is the second (fourth) stroke, pre_S is the sum of $\mathrm{pre}_{S'}$ and of the visits during S', where S' is the first (third) stroke of the period, and S and S' have the same open half-segments because 2^{s_0} is a multiple of 2^{s_1} (for $m = 0$, the same open range). Next let S be the back stroke of a V attached at b_j to a stroke S'. The open range of the V is an open half-segment of S', and on it $\mathrm{pre}_S = \mathrm{pre}_{S'} + 1$: by (a) the part of S' before segment j stays between the first cell of S' and the start of segment j, and segment j visits each cell of the open range once. So pre_S is constant on the open range of S, which contains all its open half-segments. If S is the forward stroke of that V and B its back stroke, pre_S is the sum of pre_B and of the visits during B, and B and S have the same open half-segments, or the same open range at level m, because their common displacement is a multiple of the spacing of their half-segments. Hence pre_S is constant on the open half-segments, or the open range, of every stroke.

(e) Let $c = x(p)$. As $u - p \ge 2$ and the walk moves by ± 1, the first time $\tau^* \in (p, u)$ at which

$|x(t) - c|$ is largest is a turn, and it suffices to show $|x(\tau^*) - c| \le 2^{s_j}$. The level of τ^* is at least $j \ge 1$, so by (b) τ^* is the apex or valley time of a V of level at least j, and therefore belongs to the span of a V of level exactly j, say $\mathcal{V}$: a V of level greater than j lies inside a stroke of level at least j, and such a stroke lies inside a V of level j. Let $\mathcal{V}$ be attached to the stroke S', of level $j-1$ and direction σ, at its base b, so that its range is $[v, b]$ with $v = b - \sigma 2^{s_j}$. By (a), $x(\tau^*) \in [v, b]$, so it suffices to show $c \in [v, b]$. Suppose not. By (a) the head is in $[v, b]$ during $\mathcal{V}$ and at its end time t_E, so p precedes the apex time t_A of $\mathcal{V}$ and u follows t_E.

Case 1: c lies beyond b in direction σ. By (a) the part of S' before $\mathcal{V}$ stays between the first cell of S' and b, so p precedes the first time of S'. If $j \ge 2$, S' is a back or forward stroke of a V of level $j-1$, whose first time is an apex or valley time, a turn of level $j-1$ in (p, t_A). If $j = 1$ and S' is the second or fourth level-0 stroke of its period, its first time is a turn of level 0 in (p, t_A). If S' is the first (third) level-0 stroke, then $\sigma = 1$ and $c > b > 0$ ($\sigma = -1$ and $c < b < 0$); the two preceding level-0 strokes stay in $[-2^{s_0}, 0]$ (in $[0, 2^{s_0}]$), so p precedes both of them, and the first time of the later one, the fourth (second) level-0 stroke, is a turn of level 0 in (p, t_A).

Case 2: c lies beyond v in direction $-\sigma$. Let $A_{j-1} = S'$, and for $1 \le l \le j-1$ let A_{l-1} be the stroke to which the V containing A_l is attached; A_l has level l. From t_E on, by (a), the head stays beyond b in direction σ, or at b, until the end time of A_{j-1}. If A_{j-1} is a back stroke, its end time is the valley time of its V, a turn of level $j-1$. If A_{j-1} is a forward stroke, it has the direction σ of A_{j-2}, its end time is the end time of a V attached to A_{j-2} at the last cell of A_{j-1}, and by (a) the head stays beyond that cell, hence beyond b, until the end time of A_{j-2}. Repeating this, if some A_l with $1 \le l \le j-1$ is a back stroke and l^* is the largest such l, the head stays beyond b (or at b) from t_E until the end time of A_{l^*}, which is a valley time, a turn of level $l^* \le j-1$. Otherwise all A_l have direction σ, and the head stays beyond b (or at b) until the end time of the level-0 stroke A_0. If A_0 is the first (third) stroke of its period, that end time is a turn of level 0. If it is the second (fourth) stroke, the next level-0 stroke continues in direction σ and stays beyond b, and its end time is a turn of level 0. In every case the turn follows t_E, and the head is not at c between t_E and the turn, so the turn precedes u.

In both cases a turn of level less than j lies strictly inside (p, u), contrary to the hypothesis. □

5.3 Union labels and the nesting barrier

The labels of Theorem 4.20 are lexicographic: they carry counts of events at a few scales and compare them from the top. A natural way to extend this to arbitrary walks is to let each head decide, at each cell and time, which bands of scales (groups of consecutive scales; in Section 4.4 a band was a single scale) it currently needs, and to carry, for every band needed by some head, a count of the events of that band (a *union label*). Two requirements make such a label work. A hop must find its own band among the tracked bands at both of its ends, so that it is decided there; and a change of the bands tracked by one head breaks the comparison along the hops of the other heads that span it, of which there can be about 2^θ when θ is the bottom scale of the changed band, so changes at high bands must be rare. We make the per-head part precise and sketch, without a full proof, why it cannot be met with bounded load on deeply nested walks. We do not claim that such rules would suffice for Problem 5.1; the point is that they appear not to exist.

Definition 5.5 (Band rules)**.** Let x be a walk of length T without pauses. For a turn τ at which x has a local maximum, a time t is *higher* than τ if $x(t) > x(\tau)$, or $x(t) = x(\tau)$ and $t < \tau$; the *prominence* $\mathrm{pr}(\tau)$ is the least $d \ge 1$ such that on one side of τ the walk reaches a time higher than τ without going below $x(\tau) - d$ on the way, and $x(\tau) - \min x$ if no time is higher than τ. For local minima exchange above and below, an earlier time at the same height counting as lower. Fix integers

$h_0, W \geq 1$ and offsets $r \in [0, h_0)$, $r' \in [0, W)$. A scale $\theta \geq 0$ lies in *band* $\lfloor(\lfloor(\theta + r)/h_0\rfloor + r')/W\rfloor$, and $\theta(b)$ is the least scale of band b. The band of a turn is the band of $\lfloor \log_2 \mathrm{pr}(\tau)\rfloor$. The *owner* of a hop (p, u) is the first time in (p, u) at which $|x(t) - x(p)|$ is largest, a turn, and the *own band* of the hop is the band of its owner. For a band $b \geq 1$, the *b-gaps* are the intervals $[\tau, \tau')$ between consecutive turns of band at least b (and the interval before the first such turn). A *band rule* assigns to every cell c and time t a set $\mathcal{M}(c, t)$ of bands containing 0. It is *consistent* if for every $b \geq 1$ the membership $b \in \mathcal{M}(c, t)$ depends only on c and on the b-gap containing t; its *load* is $\max_t |\mathcal{M}(x(t), t)|$; Viol is the number of hops whose own band is missing from $\mathcal{M}(x(p), p)$ or from $\mathcal{M}(x(u), u)$; and

$$D_{\mathrm{sum}} = \sum_t \sum_{b \geq 1,\ b \in \mathcal{M}(x(t),t) \,\triangle\, \mathcal{M}(x(t+1),t+1)} 2^{\theta(b)},$$

while $D_{\max}$ counts, for each t, only the largest band of the symmetric difference.

A union label for k heads tracks, at time t, the union of the sets $\mathcal{M}_h(x_h(t), t)$ of the heads; with load P_0 per head it has at most kP_0 digits, so its depth is bounded in terms of k, P_0 and the moduli. Its cost is bounded by the heads' violations, by their changes weighted as above (a change at band b is charged to the hops of other heads that span it), and by wrap terms. This is the setting of the following two statements.

Remark 5.6 (The nesting barrier for consistent band rules)**.** Proof sketch only: we state the result precisely, but its proof exists only at the level of the following sketch; exact linear programs support it numerically. Let $g \geq \max(Wh_0, 2)$ and $s_m \geq Wh_0 + 1$, let $x = U(s_0, m, g)$ on a whole number of periods, T steps in all, and put $M = T/(m + 1)$. Then for every choice of the offsets, every consistent band rule of load at most P_0 satisfies

$$\mathrm{Viol} + \gamma D_{\mathrm{sum}} \ \geq\ (m + 2 - P_0)M - 2^{1-g}T - 2^{1-s_m}M - O(m2^{s_0}), \qquad \gamma = 2^{Wh_0 - g};$$

if the rule need not contain band 0, replace $m + 2 - P_0$ by $m + 1 - P_0$. *Sketch.* Up to boundary effects the prominences of the turns of U are 2^{s_0+1} and $2^{s_1}, \ldots, 2^{s_m}$, one value per level of nesting (at a double extremum the later turn carries the prominence of the deeper level), and for every offset their bands $\beta_0 > \beta_1 > \cdots > \beta_m$ are distinct, with $\theta(\beta_i) \geq s_i - Wh_0 + 1$. For a level-$i$ stroke S let $\mathrm{gap}(S)$ be the β_i-gap containing its first segment step, and let $y(c, S) \in \{0, 1\}$ be the membership of β_i at $(c, \mathrm{gap}(S))$, well defined by consistency. (1) For every cell c that the segments of S visit inside $\mathrm{gap}(S)$, the first visit of c in $\mathrm{gap}(S)$ ends a hop of own band β_i, which $y(c, S) = 0$ violates; distinct pairs (c, S) give distinct hops, except for a fraction at most $q_0 2^{-g}$ of each level, $q_0 = 1/(2(1 - 2^{-g}))$, which comes from the last V's of strokes whose gap ends at their own arrival, and except for the first cell of each back stroke. (2) Every switch of $y(\cdot, S)$ between adjacent cells is a change of β_i at a segment step inside $\mathrm{gap}(S)$, of weight $2^{\theta(\beta_i)} \geq 2^{1-Wh_0} \cdot 2^{s_i}$, and distinct pairs (stroke, switch) give distinct pairs (step, band). (3) A cell c of a level-m stroke with ancestors $S_0 \supset S_1 \supset \cdots \supset S_m$ is visited by the segments of S_m at a time that lies in $\mathrm{gap}(S_i)$ for every i, except at exceptional levels, where S_m lies in the last V of an ancestor whose gap ends early; so the load gives $\sum_i y(c, S_i) \leq P_0 - 1 + \#\{\text{exceptional levels of } c\}$, and the expected number of exceptional levels of a uniform such cell is at most $2q_0 2^{-g} m$. (4) Averaging (3) over a uniform level-m stroke and a uniform cell of it, which gives every block (a segment with its V) of every ancestor the same mass, and comparing with (1) and (2) yields the bound; the exceptional terms add up to at most $2^{1-g}T$ because $g \geq 2$.

Per head, a union label would need, heuristically, $\mathrm{Viol} \leq \varepsilon(N + T)$ and $D_{\max} \leq \varepsilon(N + T)$, where $N \leq T$ is the number of long hops. As $D_{\mathrm{sum}} \leq D_{\max}/(1 - 2^{-Wh_0})$, the bound excludes this for all P_0, W and h_0 once $\varepsilon < 0.37$ (take $g = Wh_0 + 4$, $m = 10P_0$ and s_m large), and once $\varepsilon < 0.21$ at

$g = Wh_0 + 2$. Numerically, the exact optimum of $\mathrm{Viol} + \gamma D_{\mathrm{sum}}$ over all consistent rules, computed by linear programming on seventeen instances with $g \in \{1, 2, 3, 4\}$ and $m \le 5$, was $(m + 2 - P_0)M' - 1$, where M' is the mass per level in the window used; the bound is loose only by its correction terms.

Remark 5.7 (The nesting barrier without consistency)**.** Proof sketch only: under the hypotheses of Remark 5.6, let $t \mapsto \mathcal{M}(t)$ be any choice of sets of bands containing 0, with $|\mathcal{M}(t)| \le P_0$ and no consistency requirement; it may depend on the whole walk, and it may be randomised or fractional. Let Viol_u be the number of hops whose own band is missing from $\mathcal{M}(u)$ at the later endpoint u, and let D_{sum} count the changes of non-zero bands with weights $2^{\theta(b)}$. Then $\mathrm{Viol}_u + \gamma D_{\mathrm{sum}}$ satisfies the bound of Remark 5.6. Every band rule, read along the walk as $\mathcal{M}(t) = \mathcal{M}(x(t), t)$, is such a choice, with the same load and changes and with $\mathrm{Viol}_u \le \mathrm{Viol}$; so relaxing consistency does not help, and the thresholds 0.37 and 0.21 remain. *Sketch.* Steps (2) and (3) of the sketch of Remark 5.6 are the only uses of consistency, and both are replaced by a trichotomy in time. For a level-i stroke S and its block j (segment j with its V), let J be the block's time window, from just after the start of segment j to the start of segment $j + 1$ (the first and last windows extend to the ends of $\mathrm{gap}(S)$); the windows of one level are disjoint. Exactly one of the following holds: β_i changes at a step inside J, at cost $\gamma 2^{\theta(\beta_i)} \ge 2^{s_{i+1}+1}$, the number of cells of segment j; β_i is absent throughout J, so all the hops entering the cells of segment j, whose later endpoints lie in J, are violated; or β_i is present throughout J, in particular at the deepest visits inside the V of block j, where it uses load. The load at the deepest visit of a cell then bounds the number of levels i of the third kind by $P_0 - 1$, up to exceptional levels, and the averaging of step (4) concludes. A layer-cake decomposition of $1 - \mathcal{M}$ gives a second proof, as a max-flow certificate. Numerically, when the own band is required at both endpoints of a hop, the optimum over such choices of sets was lower than over consistent rules only by a boundary term on the instances computed, and a bound with only boundary losses is conjectured; when it is required at the later endpoint alone, the optimum was lower by a small multiple of M.

Why they matter. Remarks 5.6 and 5.7 indicate, with proof sketches only, that the union-label route to Problem 5.1 fails in its natural form: per-head band bookkeeping with load independent of the depth of nesting, own-band decisions and 2^{θ}-weighted changes would fail already on a single walk, offline and for every offset. The obstruction does not concern TCP(1), which holds by planarity; what fails is a per-head format that would have handled all couplings at once. A repair must decide some hops at a band other than their own, as the labels of Theorem 4.20 do, or weigh changes differently, which is legitimate only against other heads that are not adversarial, or use a different mechanism altogether.

One nested walk is nevertheless easy. A single digit that ignores bands handles all levels of one walk at once.

Lemma 5.8 (One nested walk is easy)**.** Not formalized; a complete proof is given below.
Let $x = U(s_0, m, g)$ on $[0, T)$, where T is a multiple of P, let $q \ge 1$, and let $\kappa(t)$ be the number of times $s \in [0, t)$ with $x(s) = x(t)$. For $\omega \in \{0, \dots, q-1\}$ put $\Psi_\omega(t) = (\kappa(t) + \omega) \bmod q$. Then

$$\frac{1}{q} \sum_{\omega=0}^{q-1} \mathrm{cost}(\Psi_\omega) \;\le\; \frac{\#\{\textit{long hops}\}}{q} + \frac{6T}{(m+1)2^{s_m}}.$$

In particular, for every $\varepsilon > 0$ the walk DAG has a cut of at most εT edges and depth at most $\lceil 2/\varepsilon \rceil - 1$ as soon as $(m+1)2^{s_m} \ge 12/\varepsilon$.

Proof (not formalized). Hops. For a hop (p, u) we have $\kappa(u) = \kappa(p) + 1$, so $\Psi_\omega(u) \le \Psi_\omega(p)$ if and only if $\Psi_\omega(p) = q - 1$, which holds for exactly one ω. Averaged over ω, the number of violated hops is $\#\{\text{long hops}\}/q$ (all hops of U are long).

Descents. A descent at t requires $\kappa(t) \ne \kappa(t-1)$; we count such t. Let the step from $t-1$ to t be emitted by a segment of a level-i stroke S, $i < m$, running from a_j to the grid point b_j in $2^{s_{i+1}+1}$ steps, so that its middle cell is $a_j + \sigma 2^{s_{i+1}}$. If neither $x(t-1)$ nor $x(t)$ is one of a_j, $a_j + \sigma 2^{s_{i+1}}$ and b_j, both cells lie in one open half-segment of S, both times belong to the segment, and $\kappa(t-1) = \kappa(t)$: indeed, for a segment time s at such a cell c, the visits to c before s are those before the first time of S, because by Lemma 5.4(a) the earlier parts of S stay between the first cell of S and a_j and the segment reaches c only at s; and the number of earlier visits is the same for all cells of the open half-segment by Lemma 5.4(d). So at most four steps emitted by the segment change κ: those leaving a_j, entering and leaving the middle cell, and entering b_j. In the same way a level-m stroke from a to $a + \sigma 2^{s_m}$ emits at most two such steps, those leaving a and entering $a + \sigma 2^{s_m}$. A period contains $4 \cdot 2^{gi}$ strokes of level i, so the number of changes of κ per period is at most

$$4 \sum_{i<m} 4 \cdot 2^{gi} \cdot 2^{g-1} + 2 \cdot 4 \cdot 2^{gm} \;\le\; \frac{8 \cdot 2^g}{2^g - 1} 2^{gm} + 8 \cdot 2^{gm} \;\le\; 24 \cdot 2^{gm} = \frac{6P}{(m+1)2^{s_m}},$$

using $2^{gm} = 2^{s_0 - s_m}$. Each change of κ is at most one descent of Ψ_ω, for every ω, which gives the second term.

For the last claim take $q = \lceil 2/\varepsilon \rceil$: the first term is at most $\varepsilon T/2$, the second at most $\varepsilon T/2$, and some ω does at least as well as the average; Lemma 5.2 turns Ψ_ω into a cut. □

Why it matters. One deeply nested walk costs essentially nothing beyond the floor $1/q$ per hop, at every depth of nesting, and larger m even helps. The label is a single anchored digit, since κ rises by exactly 1 along every hop whatever happens in between, and it is not a band rule, so it escapes Remarks 5.6 and 5.7. The difficulty of the instance studied next lies entirely in the interaction of two heads.

5.4 The deep-nesting pair

Theorem 4.20 covers walks of linear leg area, in particular walks whose legs all begin or end near a home. In $U(s_0, m, g)$ most legs do neither: the nesting happens far from home, at every level. Two copies of U shifted against each other in time give the instance that we regard as the sharpest open case of Problem 5.1 for $k = 2$.

Definition 5.9 (The deep-nesting pair)**.** Let s_0, m, g be as in Definition 5.3 and $P = 4(m+1)2^{s_0}$. The *golden shift* ξ is the least odd integer at least $\lfloor 0.381966\, P \rfloor$. The *deep-nesting pair* $\mathrm{DNP}(s_0, m, g)$ consists of the walks $x_1 = U(s_0, m, g)$ and $x_2(t) = x_1(t + \xi)$, $t \in \mathbb{Z}$. For $r \ge 1$, its *r-fold cover* is the pair on the cyclic time axis $\mathbb{Z}_T$, $T = rP$. For $t \in \mathbb{Z}_T$ and a head h, $\mathrm{prev}_h(t)$ is the residue modulo T of the previous visit, on $\mathbb{Z}$, of head h to $x_h(t)$; this defines bijections $\mathrm{prev}_1, \mathrm{prev}_2$ of $\mathbb{Z}_T$, and the *hops* of head h on the cover are the pairs $(\mathrm{prev}_h(t), t)$, $t \in \mathbb{Z}_T$, so there are $n_\mathrm{h} = 2T$ hops in all. A *label* on the cover is a map $G\colon \mathbb{Z}_T \to \{0, \dots, q-1\}$ with $q \ge 1$. A hop (p, u) is *violated* if $G(u) \le G(p)$, a time t is a *descent* if $G(t) < G(t-1)$, and $\mathrm{cost}(G)$ is the number of violated hops of both heads plus the number $\mathrm{desc}(G)$ of descents. Hops are indexed by head and later endpoint, so a pair that is a hop of both heads would count twice: $\mathrm{cost}(G)$ counts violations per head, unlike $\mathrm{cost}(\Psi)$ on the walk DAG; by Remark 5.23 no such pair exists when $s_m \ge 2$.

The constant 0.381966 is $2 - \phi$ to six decimals, ϕ the golden ratio, and we also call $\mathrm{DNP}(s_0, m, g)$ the *golden pair*. It is meant to keep the two hierarchies misaligned, but because of the truncation it

fails to do so when 15625 divides $m+1$: then $\xi \equiv 1 \pmod{2^{s_0-3}}$, and the hierarchies are aligned at every scale up to 2^{s_0-3}; for some other m they nearly align at several scales (Remark 5.24). Only the arithmetic of Theorem 5.22, where $m = 1$, depends on its exact value.

The cover is a model of the walk DAG of many periods in which the identities below are exact; Remark 5.23 transfers the main bound back to the walk DAG. A label on the cover plays the role of a label of depth $q-1$; by Lemma 5.2, labels lose nothing on the walk DAG. A cell c is *interior* if $|c| < 2^{s_0}$; by Lemma 5.4(c), on the r-fold cover each head visits every interior cell $r(2m+2)$ times and each of the two tips $\pm 2^{s_0}$ $r(m+1)$ times.

For $t \in \mathbb{Z}_T$, identified with its representative in $[0, T)$, the *visit index* $\kappa_h(t)$ is the number of times $s \in [0, t)$ with $x_h(s) = x_h(t)$. The *error* of a hop (p, u) is $e(p, u) = (G(u) - G(p) - 1) \bmod q \in \{0, \dots, q-1\}$, and G is *exact* on the hop if the error is 0, that is, $G(u) \equiv G(p) + 1 \pmod{q}$. The labels $(\kappa_1 + \omega) \bmod q$ of Lemma 5.8 are exact on every head-1 hop at an interior cell when q divides $r(2m+2)$. The *level* of a hop (p, u) of head h is the least level of a turn of x_h strictly inside (p, u) (on the cover, inside any lift of the hop to $\mathbb{Z}$); every hop contains a turn, its owner, so levels are defined.

Anchored counts. The visit index is anchored: it rises by exactly one along every hop of its head. A count restricted to a band of levels rises by exactly one along the hops of that band and is unchanged along the others.

Lemma 5.10 (Level-band anchored counts)**.** Not formalized; a complete proof is given below.

For a head h, a set B of levels and a time $t \geq 0$, let $U_h^B(t)$ be the number of times $s \in [0, t)$ with $x_h(s) = x_h(t)$ such that the hop of head h that begins at s has its level in B. Then for every hop (p, u) of head h with $p \geq 0$, $U_h^B(u) - U_h^B(p)$ is 1 if the level of (p, u) lies in B and 0 otherwise.

Proof (not formalized). The visits to $x_h(u)$ in $[0, u)$ are those in $[0, p)$ and p itself, because p and u are consecutive visits. The visit p is counted in $U_h^B(u)$ exactly when the hop that begins at p, which is (p, u), has its level in B. □

With B all levels, U_h^B is the visit index. Along the *chain* of a cell, the visits of one head (here head 1) to it in time order, a label exact on the head-1 hops whose level lies in B rises by one modulo q wherever U_1^B rises, so modulo q it differs from U_1^B by a function of the cell and of the number of earlier head-1 hops at that cell with level outside B; this is the class used in Remark 5.24.

Remark 5.11 (Numerical: how often band counts change)**.** Numerically, on all instances checked, $U_1^{\{0,\dots,j\}}$ is constant during every V of level greater than j at all times at which the head is not at the base of the V; at the base it may change, so the plain statement "constant during the V" is false. The total number of its changes was of the order of $T/((m+1)2^{s_{j+1}}) + T(m-j)/((m+1)2^{s_j})$, so the gain $1/(m+1)$ of Lemma 5.8 is lost for $j < m$. Nothing below uses this remark.

Lemma 5.12 (No monotone anchoring)**.** Not formalized; a complete proof is given below.

Let x be a time shift of $U(s_0, m, g)$, let $1 \leq i < m$, and let $\mathcal{V}$ be a V of level i with valley cell v whose forward stroke has direction σ. For $1 \leq \rho < 2^{s_i}$ let p_ρ be the last visit to $v + \sigma\rho$ during the back stroke and u_ρ its first visit during the forward stroke; (p_ρ, u_ρ) is a hop. Let K be an integer function on the span of $\mathcal{V}$ with $K(t) \geq K(t-1)$ whenever $t-1$ and t belong to the span, and with $K(u') \geq K(p') + 1$ for every hop (p', u') of level $i+1$ with both endpoints in the span. Then

$$K(u_\rho) - K(p_\rho) \;\geq\; \rho/2^{s_{i+1}} - 3/2 \qquad (1 \leq \rho < 2^{s_i});$$

in particular $K(u_\rho) - K(p_\rho) \geq 2^g - 2$ for $\rho = 2^{s_i} - 1$.

Proof (not formalized). After its last visit to $v+\sigma\rho$ the back stroke ends at v, so by continuity it stays on the side of v; the forward stroke starts at v and first reaches $v+\sigma\rho$ at u_ρ. So p_ρ, u_ρ are consecutive visits. Put $s = s_{i+1}$ and let $\gamma_k = v + \sigma k 2^{s+1}$. By Lemma 5.4(a), the back stroke has V's of level $i+1$ attached at γ_k with ranges $[\gamma_k, \gamma_k + \sigma 2^s]$, $0 \le k < 2^{g-1}$, and the forward stroke has V's attached at γ_{k+1} with ranges $[\gamma_{k+1} - \sigma 2^s, \gamma_{k+1}]$. If $(2k+1)2^s < \rho$, the back stroke's V at γ_k lies strictly on the side of v of $v+\sigma\rho$; it is attached after the back stroke has passed $v+\sigma\rho$, and nothing after it in the back stroke returns to $v+\sigma\rho$, so it lies inside (p_ρ, u_ρ). There are at least $(\rho/2^s - 1)/2$ such k. If $(k+1)2^{s+1} < \rho$, the forward stroke reaches γ_{k+1} and completes its V there before it reaches $v+\sigma\rho$, so this V lies inside (p_ρ, u_ρ) as well; there are at least $\rho/2^{s+1} - 1$ such k. In all, at least $\rho/2^s - 3/2$ V's of level $i+1$ lie inside (p_ρ, u_ρ), and their spans are disjoint.

In each of them, with valley cell v' and forward direction σ', take the hop between the last visit of the back stroke and the first visit of the forward stroke to $v' + \sigma'$. As $2^s \ge 2$, this cell is not the base, so the hop lies after the apex time; by Lemma 5.4(b) the only turns strictly inside it are the valley time of this V and turns of V's nested in it, so the hop has level $i+1$. Listing the chosen hops in time order and joining them by the stretches of time between them, K gains at least 1 along each chosen hop and does not decrease along the stretches. Hence $K(u_\rho) - K(p_\rho)$ is at least the number of chosen hops. For $\rho = 2^{s_i} - 1$ the bound is $2^g - 2^{-s} - 3/2 \ge 2^g - 2$. □

Clock labels. Counting turns instead of visits gives digits that never decrease. One digit per scale suffices for both heads, because the two walks of a deep-nesting pair have the same scales.

Proposition 5.13 (Clock labels)**.** Not formalized; a complete proof is given below.

Let x_1, x_2 be time shifts of $U(s_0, m, g)$, for instance a deep-nesting pair, on $[0, T)$, where T is a multiple of P. For $0 \le i \le m$ let $N^{(i)}(t)$ be the number of turns of level i of either head at times in $[0, t]$. For an integer $B \ge 2$ and $\omega \in \{0, \dots, B-1\}^{m+1}$ let

$$\Psi_\omega(t) = \sum_{i=0}^{m} B^{m-i}\big((N^{(i)}(t) + \omega_i) \bmod B\big) \in \{0, \dots, B^{m+1} - 1\}.$$

Then the average of $\mathrm{cost}(\Psi_\omega)$ over all ω is at most $12T/B$. Consequently, for every $\varepsilon > 0$ the walk DAG of the pair has a cut of at most εT edges and depth at most $\lceil 12/\varepsilon \rceil^{m+1} - 1$.

Proof (not formalized). The vector $(N^{(0)}(t), \dots, N^{(m)}(t))$ is coordinatewise non-decreasing in t, and $\Psi_\omega(t)$ is the integer whose base-B digits, most significant first, are $(N^{(i)}(t) + \omega_i) \bmod B$.

Hops. Let (p, u) be a hop of head $h' \in \{1, 2\}$ inside $[0, T)$. Its owner is a turn strictly inside (p, u), so some $N^{(i)}$ increases from p to u; let j be the least such i and $\Delta = N^{(j)}(u) - N^{(j)}(p) \ge 1$. The digits of index less than j agree at p and u. If $(N^{(j)}(p) + \omega_j) \bmod B < B - \Delta$, digit j increases and $\Psi_\omega(u) > \Psi_\omega(p)$ whatever the later digits are; for uniform ω_j the residue $(N^{(j)}(p) + \omega_j) \bmod B$ is uniform, so the hop is violated with probability at most Δ/B. Now Δ is the number of turns of level j of either head at times in $(p, u]$, and no turn of level less than j of either head lies in $(p, u]$. Fix a turn τ of level j and consider the hops (p, u) of head h' with $\tau \in (p, u]$ and without turns of head h' of level less than j strictly inside. By Lemma 5.4(e) if $j \ge 1$, and because $|x_{h'}| \le 2^{s_0}$ if $j = 0$, the cell of every such hop lies within distance 2^{s_j} (within 2^{s_0+1} if $j = 0$) of $x_{h'}(\tau)$; and at each cell at most one hop of head h' has $\tau \in (p, u]$, because the hops of one head at one cell are disjoint half-open intervals of time. So the turn τ is counted, over both heads, at most $2(2^{s_j+1} + 1)$ times if $j \ge 1$ and at most $2(2^{s_0+1} + 1)$ times if $j = 0$. By Lemma 5.4(b), each head has, per period, two turns of level 0 and $4 \cdot 2^{gj}$ turns of level j for $1 \le j \le m$ (a period contains $2 \cdot 2^{gj}$ V's of level j).

Summing over the turns of both heads in one period gives at most

$$4 \cdot 2(2^{s_0+1}+1) + \sum_{j=1}^{m} 8 \cdot 2^{gj} \cdot 2(2^{s_j+1}+1) = 32(m+1)2^{s_0} - 16 \cdot 2^{s_0} + 8 + 16 \sum_{j=1}^{m} 2^{gj} \;\le\; 8P + 8 \;\le\; 9P,$$

because $16 \sum_{j \le m} 2^{gj} \le 32 \cdot 2^{gm} \le 16 \cdot 2^{s_0}$ (as $s_m \ge 1$) and $P \ge 8$. So the expected number of violated hops is at most $9T/B$.

Descents. If no turn occurs at t, then $\Psi_\omega(t) = \Psi_\omega(t-1)$. Otherwise the coordinates increase by one for each head that turns at t, so by at most two in all; the first changed digit increases by 1 or 2, and Ψ_ω descends only if this digit wraps, which has probability at most (number of turns at t)$/B$. The two heads have at most $4 + 8 \sum_{j \le m} 2^{gj} \le 4 + 8 \cdot 2^{s_0} \le 3P$ turns per period, so the expected number of descents is at most $3T/B$.

For the last claim we may assume $\varepsilon < 12$, since cutting all of the at most $3T$ edges gives depth 0; take $B = \lceil 12/\varepsilon \rceil \ge 2$, choose ω with cost at most the average, and apply Lemma 5.2. □

Why they matter. By Proposition 5.13 every deep-nesting pair of fixed nesting depth m is depth-reducible, with $D(\varepsilon)$ of the form $(c/\varepsilon)^{m+1}$; so depth-robustness can only appear as m grows. The two lemmas show why the obvious ways of beating the clock fail. For $g \ge 2$, a digit that decides one head's hops at several levels exactly, with slack below $2^g - 2$, must descend inside every V of that head (Lemma 5.12; for $g = 1$ the lemma is empty), and the other head's long hops that span such a descent see it. A digit that decides the other head's hops must instead be non-decreasing across them. Anchored digits (Lemmas 5.8 and 5.10) and clock digits are the two ends of this spectrum, and every explicit label we know is built from them.

Packing certificates. A proof that the pairs are depth-robust as m grows must show that every cut of depth D is large. The standard way to certify such lower bounds, linear programming duality or packings of local obstructions, cannot do it.

Proposition 5.14 (No packing certificate)**.** Not formalized; a complete proof is given below.

*Let $x_1, \dots, x_k$ be walks of length T, let H be the set of long hops of their walk DAG, and let $D \ge 0$. Call a set F of edges D-*forcing *if no map $\Psi\colon [0,T) \to \{0,\dots,D\}$ satisfies $\Psi(a) \ge \Psi(b)+1$ for every long hop $a \to b$ in F and $\Psi(a) \ge \Psi(b)$ for every state edge $a \to b$ in F.*

(a) Every D-forcing set contains a path with at least $D+1$ long hops.

(b) Let $F_1, \dots, F_n$ be D-forcing and $y_1, \dots, y_n \ge 0$ with $\sum_{j : e \in F_j} y_j \le 1$ for every edge e. Then every cut of depth at most D has at least $\sum_j y_j$ edges, and $\sum_j y_j \le |H|/(D+1) \le kT/(D+1)$.

(c) The linear program that minimises $\sum_e z_e$ over $z \ge 0$ subject to $\sum_{e \in \pi} z_e \ge 1$ for every path π with at least $D+1$ long hops has value at most $|H|/(D+1)$.

Hence a lower bound εT on the size of all cuts of depth at most D can be certified in the form (b) or (c) only if $D < k/\varepsilon$; such certificates never show depth-robustness.

Proof (not formalized). (a) If F contains no path with $D+1$ long hops, let $\Psi(a)$ be the largest number of long hops on a path in F that starts at a. Then $0 \le \Psi \le D$, and for every edge $a \to b$ of F, $\Psi(a) \ge \Psi(b) + 1$ if it is a long hop and $\Psi(a) \ge \Psi(b)$ otherwise; so F is not forcing.

(b) Let F be a cut of depth at most D, and let Ψ be the map of the proof of Lemma 5.2; it satisfies the constraints of all edges outside F. Since each F_j is forcing, it contains an edge of F, and $|F| \ge \sum_{e \in F} \sum_{j : e \in F_j} y_j \ge \sum_j y_j$. By (a), $|F_j \cap H| \ge D+1$, so $(D+1)\sum_j y_j \le \sum_j y_j |F_j \cap H| =$

$\sum_{e\in H}\sum_{j:e\in F_j} y_j \le |H|$. Finally $|H| \le kT$, because each time is the later endpoint of at most one hop of each head.

(c) Put $z_e = 1/(D+1)$ on long hops and $z_e = 0$ on state edges.

For the last sentence, $|H|/(D+1) \ge \varepsilon T$ forces $D+1 \le k/\varepsilon$. □

On a cyclic time axis, such as the covers of Definition 5.9, every cyclic hop chain is forcing: a chain of n hops always contains at least $\lceil n/q \rceil$ violated hops, more than the average n/q unless q divides n. This is why we use covers on which q divides the visit numbers of the cells that matter; the effect does not arise on the walk DAG, which is acyclic.

Why it matters. Proposition 5.14 rules out the natural strategy for proving depth-robustness by duality: packings of forcing structures, of any shape, certify at most what the trivial assignment of weight $1/(D+1)$ to every long hop certifies. Lemma 5.20(c) below records the same barrier for cycle bounds in the cost identity. A robustness proof cannot be a fractional packing; it has to exploit integrality, and we expect it to be multiplicative: it would show that a label with q values cannot serve many levels of both heads at once.

5.5 Exact identities for labels

On the cover every label obeys an exact cost identity, which separates the unavoidable floor $1/q$ per hop from the errors of the hops.

Lemma 5.15 (Lemma Zq: the cost identity)**.** `Lean`: `lemma_Zq`, `card_violated_eq`, `card_descents_eq` [`DepthLabels.lean`]

Let τ and H be finite sets, $q \ge 1$ an integer, $G\colon \tau \to \{0,\dots,q-1\}$, and $t \mapsto t^-$ a bijection of τ. Let $p, u\colon H \to \tau$ be maps and succ *a bijection of H with $p(\mathrm{succ}(h)) = u(h)$ for all $h \in H$. Put $e_h = (G(u(h)) - G(p(h)) - 1) \bmod q$ and $\chi_t = (G(t) - G(t^-)) \bmod q$, residues in $\{0,\dots,q-1\}$. Then*

$$q \cdot \#\{h : G(u(h)) \le G(p(h))\} = |H| + \sum_h e_h, \qquad q \cdot \#\{t : G(t) < G(t^-)\} = \sum_t \chi_t,$$

and hence $q\,(\#\{h : G(u(h)) \le G(p(h))\} + \#\{t : G(t) < G(t^-)\}) = |H| + \sum_h e_h + \sum_t \chi_t$.

Outline of the formal proof. For $-q \le y < q$ one has $y \bmod q = y + q\cdot[y<0]$ (`emod_of_bounds`). With $y = G(u(h)) - G(p(h)) - 1$, which lies in $[-q, q-2]$, this gives $e_h = G(u(h)) - G(p(h)) - 1 + q\cdot[G(u(h)) \le G(p(h))]$. Summing over H and using $\sum_h G(u(h)) = \sum_h G(p(h))$, obtained by reindexing along the bijection succ, gives the first identity. The second is the same computation with $y = G(t) - G(t^-) \in (-q, q)$ and $\sum_t G(t^-) = \sum_t G(t)$; the third is their sum. □

On the cover of Definition 5.9 take $\tau = \mathbb{Z}_T$, $t^- = t-1$, let $H = \{1,2\} \times \mathbb{Z}_T$ index the hops of the two heads, with p, u their endpoints, and let succ map a hop to the next hop of the same head at the same cell; the hypothesis only says that the hops of each head at each cell form a cycle. Since $\sum_t \chi_t = q\,\mathrm{desc}(G)$, the lemma reads

$$\mathrm{cost}(G) \;=\; \mathrm{desc}(G) + \frac{1}{q}\sum_{\text{hops } h} (1 + e_h). \tag{3}$$

Every hop costs at least $1/q$, and beyond that exactly its error divided by q; every descent costs 1. For $q = 2$, (3) says that twice the cost is n_{h} plus the number of hops with $G(u) = G(p)$ plus the number of times with $G(t) \ne G(t-1)$, so depth 1 is a problem of frustration in a signed graph. The identity also shows that $\mathrm{cost}(G)$ does not change when G is replaced by $(G + c) \bmod q$.

Corollary 5.16 (Per-class split)**.** Not formalized; a complete proof is given below.
On the cover of a deep-nesting pair, for every label G,

$$\operatorname{cost}(G) = \operatorname{desc}(G) + \sum_{h=1}^{2} \sum_{i=0}^{m} \frac{1}{q} \sum_{\substack{\text{hops of head } h \\ \text{of level } i}} (1 + e).$$

Proof (not formalized). Group the hops in (3) by head and level; each hop has exactly one of each. □

So the classes (head, level) contribute additively, and how well G serves a class is measured by the error mass of the class: a class on which G is exact costs its size over q.

Lifts. To state the next lemmas without boundary terms we lift labels, walks and hops to $\mathbb{Z}$: G and the walks are T-periodic, the hops of head h are the pairs $(\operatorname{prev}_h(u), u)$, $u \in \mathbb{Z}$, with prev_h the previous visit on $\mathbb{Z}$, and each hop of the cover is the residue of such a pair. For a head h, a cell c and a visit $s \in \mathbb{Z}$ of head h to c, let $\hat{\kappa}_h(s)$ be the number of visits of head h to c at times in $[0, s)$ if $s \geq 0$, and minus the number at times in $[s, 0)$ if $s < 0$. Thus $\hat{\kappa}_h(s') - \hat{\kappa}_h(s) = 1$ for consecutive visits $s < s'$, $\hat{\kappa}_h(s+T) = \hat{\kappa}_h(s) + n_h(c)$, where $n_h(c)$ is the number of visits of head h to c per cover, and $\hat{\kappa}_h = \kappa_h$ on $[0, T)$. Let $t_0(c)$ be the first visit of head h to c in $[0, T)$, and let $\hat{E}_h(s)$ be the total error of the hops of head h at c whose later endpoint lies in $(t_0(c), s]$ if $s \geq t_0(c)$, and minus the total error of those with later endpoint in $(s, t_0(c)]$ if $s < t_0(c)$. So $\hat{E}_h(s') - \hat{E}_h(s) \geq 0$ is the error mass of the hops of c ending in $(s, s']$ when $s < s'$.

Lemma 5.17 (Chain decomposition)**.** Lean (partial): `congr_of_exact` [`DepthLabels.lean`]
Not formalized: the general form (a), with error terms, and its reading (c) on the cover.

(a) For every head h, every cell c and every visit $s \in \mathbb{Z}$ of head h to c,

$$G(s) \equiv G(t_0(c)) + \hat{\kappa}_h(s) + \hat{E}_h(s) \pmod{q},$$

and the total error of the $n_h(c)$ hops of head h at c on the cover is congruent to $-n_h(c)$ modulo q.

(b) (Exact case, abstract form.) Let $\operatorname{prev}\colon \tau \to \tau$ *and* $\operatorname{cell}\colon \tau \to C$ *be maps on a set τ with* $\operatorname{cell}(\operatorname{prev}(t)) = \operatorname{cell}(t)$*, such that whenever* $\operatorname{cell}(s) = \operatorname{cell}(t)$ *some iterate of* prev *maps t to s. Let $S \subseteq C$, and let $G, \kappa\colon \tau \to \mathbb{Z}$ satisfy $G(t) - G(\operatorname{prev}(t)) \equiv 1$ and $\kappa(t) - \kappa(\operatorname{prev}(t)) \equiv 1 \pmod{q}$ whenever $\operatorname{cell}(t) \in S$. Then $G(s) - \kappa(s) \equiv G(t) - \kappa(t) \pmod{q}$ whenever $\operatorname{cell}(s) = \operatorname{cell}(t) \in S$.*

(c) If G is exact on all hops of head h at a cell c, then q divides $n_h(c)$ and $G \equiv \kappa_h + G(t_0(c)) \pmod{q}$ at the visits of c in $[0, T)$. Conversely, if q divides $n_h(c)$ and $G - \kappa_h$ is constant modulo q on the visits of c in $[0, T)$, then G is exact on all hops of head h at c.

Outline of the formal proof of (b). Induction on n shows $G(\operatorname{prev}^n(t)) - \kappa(\operatorname{prev}^n(t)) \equiv G(t) - \kappa(t)$: the cell stays in S because $\operatorname{cell} \circ \operatorname{prev} = \operatorname{cell}$, and at each step the two congruences cancel. The orbit hypothesis supplies n with $\operatorname{prev}^n(t) = s$. □

Proof of (a) and (c) (not formalized). (a) For consecutive visits $s < s'$ of head h to c, $G(s') - G(s) \equiv 1 + e(s, s')$ by the definition of the error, while $\hat{\kappa}_h$ rises by 1 and $\hat{E}_h$ by $e(s, s')$. So $G - \hat{\kappa}_h - \hat{E}_h$ is constant modulo q on the visits of c, and at $t_0(c)$ it equals $G(t_0(c))$. At $s = t_0(c) + T$ we have

$G(s) = G(t_0(c))$, $\hat{\kappa}_h(s) = n_h(c)$, and $\hat{E}_h(s)$ is the total error of the $n_h(c)$ hops of c on the cover; this gives the second claim.

(c) If all these errors vanish, (a) gives $0 \equiv -n_h(c)$ and $G(s) \equiv G(t_0(c)) + \kappa_h(s)$ for $s \in [0, T)$. Conversely, let $t_0 < \cdots < t_{n-1}$ be the visits of c in $[0, T)$. For $1 \le j < n$ the hop (t_{j-1}, t_j) satisfies $G(t_j) - G(t_{j-1}) \equiv \kappa_h(t_j) - \kappa_h(t_{j-1}) = 1$, and the remaining hop of c on the cover, from t_{n-1} to t_0, satisfies $G(t_0) - G(t_{n-1}) \equiv 0 - (n-1) \equiv 1$ because q divides n. □

The visit numbers of the tips are $r(m+1)$ and those of interior cells $r(2m+2)$. So if q divides $r(2m+2)$ but not $r(m+1)$, for instance $r = q/\gcd(q, 4)$ with $4 \mid q$ at $m = 1$, no label is exact on all head-1 hops; the natural class is that of labels exact on all head-1 hops *at interior cells*, which by (c) are the labels $G \equiv \kappa_1 + \Phi(x_1) \pmod q$ at interior cells, for arbitrary functions Φ of the cell. It contains the visit-index labels of Lemma 5.8.

Lemma 5.18 (Transfer identity)**.** Lean (partial): `cycleDefect`, `cycle_errors_congr` [`DepthLabels.lean`]
Not formalized: the general form (a), with head-1 error mass, and the absorption reading (c).

(a) Let $h_j = (p_j, u_j)$, $j \in \mathbb{Z}_\ell$, be hops of head 2 on $\mathbb{Z}$ with $x_1(u_j) = x_1(p_{j+1})$ for every j. Put $\Delta\hat{\kappa}_1(h_j) = \hat{\kappa}_1(u_j) - \hat{\kappa}_1(p_j)$ and define the defect $\delta = \sum_j (\Delta\hat{\kappa}_1(h_j) - 1)$*. Then*

$$\sum_j e(h_j) \;\equiv\; \delta - \sum_j (\hat{E}_1(p_{j+1}) - \hat{E}_1(u_j)) \pmod q.$$

(b) (Exact case, abstract form.) Let $\mathrm{prev}_2 \colon \tau \to \tau$, $\mathrm{cell} \colon \tau \to C$, $S \subseteq C$ and $G, \kappa \colon \tau \to \mathbb{Z}$ be such that $G - \kappa$ is constant modulo q on the times of each cell of S. Let $b \colon \mathbb{Z}_\ell \to \tau$ satisfy $\mathrm{cell}(b(j)) = \mathrm{cell}(\mathrm{prev}_2(b(j+1)))$ and $\mathrm{cell}(b(j)) \in S$ for all j. Then

$$\sum_j (G(b(j)) - G(\mathrm{prev}_2(b(j))) - 1) \;\equiv\; \sum_j (\kappa(b(j)) - \kappa(\mathrm{prev}_2(b(j))) - 1) \pmod q;$$

the right side is the defect `cycleDefect` *of κ, prev_2 and b.*

(c) (Absorption.) In (a), call the head-1 stretch from u_j to p_{j+1} forward *if $u_j < p_{j+1}$ and* backward *otherwise, and let $A_j = |\hat{E}_1(p_{j+1}) - \hat{E}_1(u_j)|$ be its error mass. If $-q < \delta < 0$ and $\sum_j e(h_j) < q - |\delta|$, then either the backward stretches carry error mass at least $|\delta| + \sum_j e(h_j)$, or the forward stretches carry error mass at least $q - |\delta| - \sum_j e(h_j) + \sum_{\text{backward}} A_j$.*

Outline of the formal proof of (b). After reindexing one of the sums by the rotation $j \mapsto j+1$ of $\mathbb{Z}_\ell$, the difference of the two sides becomes $\sum_j [(G-\kappa)(b(j)) - (G-\kappa)(\mathrm{prev}_2(b(j+1)))]$. Each term is divisible by q, because $b(j)$ and $\mathrm{prev}_2(b(j+1))$ are times of the same cell of S. □

Proof of (a) and (c) (not formalized). (a) By the definition of the error, $G(u_j) - G(p_j) \equiv 1 + e(h_j)$. By Lemma 5.17(a) for head 1 at the cell $x_1(u_j) = x_1(p_{j+1})$, whose base term cancels, $G(p_{j+1}) - G(u_j) \equiv \hat{\kappa}_1(p_{j+1}) - \hat{\kappa}_1(u_j) + \hat{E}_1(p_{j+1}) - \hat{E}_1(u_j)$. Summing both congruences over $j \in \mathbb{Z}_\ell$, the values of G cancel, and $\sum_j (\hat{\kappa}_1(p_{j+1}) - \hat{\kappa}_1(u_j)) = -\sum_j \Delta\hat{\kappa}_1(h_j)$. Rearranging gives (a).

(c) Write $\sigma_e = \sum_j e(h_j)$, $\delta' = -\delta \in (0, q)$, and let F_A and B_A be the error masses of the forward and of the backward stretches. The term $\hat{E}_1(p_{j+1}) - \hat{E}_1(u_j)$ is A_j for a forward and $-A_j$ for a backward stretch, so (a) says $B_A - F_A \equiv \sigma_e + \delta' \pmod q$, where $0 < \sigma_e + \delta' < q$. Write $B_A - F_A = \sigma_e + \delta' + qz$ with $z \in \mathbb{Z}$. If $z \ge 0$, then $B_A \ge \sigma_e + \delta'$. If $z \le -1$, then $F_A = B_A - \sigma_e - \delta' - qz \ge q - \delta' - \sigma_e + B_A$. □

For labels exact on head 1 at interior cells, all errors of head-1 hops at interior cells vanish, so by (a) or (b) every closed sequence of head-2 hops through interior head-1 cells has $\sum_j e(h_j) \equiv \delta$: a cell correction Φ telescopes and absorbs nothing. In general, by (c), a defect $-q < \delta < 0$ can be absorbed cheaply, at a cost of order $|\delta|/q$, only by errors of head-1 hops on backward stretches, which lie inside the overlaps of consecutive head-2 hops; errors on forward stretches enter with the wrong sign and help only after a wrap, at a cost of nearly a whole unit.

Remark 5.19 (Numerical: who absorbs in optimised labels)**.** For the deep-nesting pair with $m = 3$, $g = 1$ and $s_0 = 6$, we examined the best labels found by optimisation at $D = 7, 15, 31, 63$; they cost about four to five times less than clock labels with the same number of values. For the pairs of head-2 hops $a \to b$, $b \to a$ in head-1 cell space whose defect a label exact on head 1 could not absorb, head-1 errors on backward stretches did the absorbing in 55%, 83%, 88% and 96% of the cases, and head 2 paid (error sum at least $q/2$) in 28%, 9%, 9% and 4%. The transfer identity held on all 6925 pairs examined.

Lemma 5.20 (Euler: balanced degrees, chain defects, and the barrier in $\mathbb{Z}_q$ form)**.** Not formalized; a complete proof is given below.

On the r-fold cover of a deep-nesting pair:

(a) The multigraph M_2 on the cells of head 1 with one edge $x_1(p) \to x_1(u)$ for every hop (p, u) of head 2 is balanced: the in-degree and the out-degree of a cell c both equal the number of visits of head 1 to c.

(b) Let head 2 visit the cell c' at the times $t_0 < \cdots < t_{n-1}$ of $[0, T)$. Its hops at c', lifted to (t_{j-1}, t_j) for $1 \le j < n$ and $(t_{n-1}, t_0 + T)$, satisfy the hypothesis of Lemma 5.18(a), and their defect is $n_1(x_1(t_0)) - n_2(c')$. It is 0 if c' and $x_1(t_0)$ are both interior cells or both tips.

(c) In the setting of Lemma 5.18(a), with A_j as in Lemma 5.18(c), $\sum_j e(h_j) + \sum_j A_j \ge \|\delta\|_q$, where $\|\cdot\|_q$ is the distance to $q\mathbb{Z}$, and $|\delta|$ is at most the number of hops on the closed walk formed by the h_j and the head-1 stretches, counted with multiplicity. Consequently, if such closed walks $C_1, \ldots, C_n$ use every hop of either head at most once in all, the bound they yield through (3), $\mathrm{cost}(G) \ge (n_\mathrm{h} + \sum_i \|\delta(C_i)\|_q)/q$, is at most $2n_\mathrm{h}/q$.

Proof (not formalized). (a) Since prev_2 is a bijection of $\mathbb{Z}_T$, every time t is the later endpoint of exactly one hop of head 2 and the earlier endpoint of exactly one; the first enters $x_1(t)$ in M_2 and the second leaves it.

(b) Consecutive hops of the list share an endpoint in time, so the head-1 cells match, and the last hop ends at $t_0 + T$, where head 1 is at $x_1(t_0)$. The sum of the $\Delta\hat{\kappa}_1$ telescopes to $\hat{\kappa}_1(t_0 + T) - \hat{\kappa}_1(t_0) = n_1(x_1(t_0))$, and there are $n = n_2(c')$ hops. As x_2 is a time shift of x_1, $n_2(c') = n_1(c')$, which is $r(2m + 2)$ for interior cells and $r(m + 1)$ for tips.

(c) By Lemma 5.18(a), δ is congruent to $\sum_j e(h_j) \pm A_1 \pm \cdots \pm A_\ell$, whose absolute value is at most $\sum_j e(h_j) + \sum_j A_j$. Next, $\hat{\kappa}_1(p_{j+1}) - \hat{\kappa}_1(u_j)$ is plus or minus the number of head-1 hops on the j-th stretch, so $\delta = -\ell - \sum_j (\hat{\kappa}_1(p_{j+1}) - \hat{\kappa}_1(u_j))$ and $|\delta|$ is at most the number of hops on the closed walk. If the walks C_i use every hop at most once, the errors they involve are distinct terms of $\sum_h e_h$, so (3) gives $q\mathrm{cost}(G) \ge n_\mathrm{h} + \sum_h e_h \ge n_\mathrm{h} + \sum_i \|\delta(C_i)\|_q$; and $\sum_i \|\delta(C_i)\|_q \le \sum_i |\delta(C_i)| \le n_\mathrm{h}$. □

Why they matter. Lemma 5.15 makes the depth problem on the cover an exact accounting problem: a label pays $1/q$ per hop, its errors, and its descents, and the error of a hop is a residue modulo q. Lemma 5.17 says that a label is the visit index of either head plus a cell correction plus that head's integrated errors, and Lemma 5.18 says how the two descriptions interact: around a

closed walk of head-2 hops, the defect of head 1's visit index must be paid by head 2 or absorbed by head 1's errors on backward stretches. By Lemma 5.20(b), the chain of a head-2 cell has zero defect unless exactly one of that cell and the head-1 cell of its first visit is a tip, and by Lemma 5.20(c) lower bounds obtained from hop-disjoint cycles in this way are bounded by twice the floor: the identity alone does not give robustness, but it says exactly what a proof has to count.

5.6 Labels exact on one head

The one-head mechanism of Lemma 5.8 counts exactly along one head; adding an arbitrary correction per cell gives the natural class of labels exact on that head. For this class the transfer identity has no absorption, and negative cycles force a price that does not depend on q.

Theorem 5.21 (Theorem E: labels exact on one head, abstract form)**.** Lean: `cost_ge_of_negative_cycles`, `le_of_modEq_neg` [`DepthLabels.lean`]

(a) If $q \geq 1$, $E \geq 0$ and $\delta < 0$ are integers with $E \equiv \delta \pmod q$, then $E \geq q + \delta$.

(b) Let τ be a finite set, $q \geq 1$, $G\colon \tau \to \{0, \dots, q-1\}$, and let $t \mapsto t^-$, prev_1 and prev_2 be bijections of τ. Let $\mathrm{cell}\colon \tau \to C$, $S \subseteq C$ and $\kappa\colon \tau \to \mathbb{Z}$ be such that $G - \kappa$ is constant modulo q on the times of each cell of S. Let I be a finite set and, for $i \in I$, let $b_i\colon \mathbb{Z}_{\ell_i} \to \tau$ be maps such that $(i, j) \mapsto b_i(j)$ is injective, $\mathrm{cell}(b_i(j)) = \mathrm{cell}(\mathrm{prev}_2(b_i(j+1)))$ and $\mathrm{cell}(b_i(j)) \in S$ for all i, j, and

$$\delta_i := \sum_{j \in \mathbb{Z}_{\ell_i}} \big(\kappa(b_i(j)) - \kappa(\mathrm{prev}_2(b_i(j))) - 1\big) < 0 \qquad (i \in I).$$

Then

$$2|\tau| + \sum_{i \in I} (q + \delta_i) \\ \leq\ q\Big(\#\{t : G(t) \leq G(\mathrm{prev}_1 t)\} + \#\{t : G(t) \leq G(\mathrm{prev}_2 t)\} + \#\{t : G(t) < G(t^-)\}\Big).$$

Outline of the formal proof. (a) Write $E - \delta = qz$; then $z \geq 1$ because $E \geq 0 > \delta$, so $E \geq q + \delta$. (b) Apply `card_violated_eq` to each head, with the hops $(\mathrm{prev}_h t, t)$, $t \in \tau$, and the successor bijection prev_h^{-1}, and apply `card_descents_eq`; all reduced errors are nonnegative. For each i, the reduced errors of the hops $(\mathrm{prev}_2(b_i(j)), b_i(j))$ have a sum congruent to the unreduced sum, which is congruent to δ_i by `cycle_errors_congr`; so by (a) it is at least $q + \delta_i$. By injectivity these hops are distinct, so the sum over i is at most the total reduced error of head 2. Linear arithmetic combines the three identities. □

On the cover of a deep-nesting pair take $\tau = \mathbb{Z}_T$, $t^- = t - 1$, the hop maps $\mathrm{prev}_1, \mathrm{prev}_2$, $\mathrm{cell} = x_1$, S the set of interior cells and $\kappa = \kappa_1$; the right side is then $q\mathrm{cost}(G)$, and $2|\tau| = n_{\mathrm{h}}$. If q divides $r(2m+2)$, every label exact on all head-1 hops at interior cells satisfies the hypothesis on $G - \kappa_1$, by Lemma 5.17(c). A family (b_i) is a set of hop-disjoint closed walks of head-2 hops in M_2 through interior head-1 cells, each with negative defect. A closed walk with $-q < \delta_i < 0$ forces head-2 error at least $q + \delta_i$, nearly q, on its hops, which by (3) costs nearly one unit, whatever q is. The formal statement needs only $\delta_i < 0$; for $\delta_i \leq -q$ the contribution $q + \delta_i$ is not positive and lowers the bound, so such walks should be left out of the family. Here δ_i is computed from the residues κ on the cover, as in the Lean statement; it differs from the lifted defect of Lemma 5.18(a) by $n_1(\cdot)$ on every hop that wraps around the cover, and in Theorem E_1 no hop wraps.

Why it matters. Proposition 5.14 and Lemma 5.20(c) bound every packing certificate for all labels by about twice the floor. Theorem E escapes this because the class is restricted: exactness on one head removes all absorption (Lemma 5.18), so each negative cycle costs about one unit at every depth. A packing of such cycles with positive density therefore gives a price per hop independent of D, which rules out the whole class as a depth-reduction mechanism for the instance; the next theorem supplies such a packing in closed form.

Theorem 5.22 (Theorem E_1)**.** Not formalized; a complete proof is given below.

Let $s_0 \geq 4$, $\Lambda = 2^{s_0}$, let $x_1 = U(s_0, 1, 1)$, of period $P = 8\Lambda$, and let $x_2(t) = x_1(t+\xi)$ with $\xi = 3\Lambda + \eta$ for an integer η with $0 < 8\eta < \Lambda$. Let $r, q \geq 1$ with $q \mid 4r$. Then every label G on the r-fold cover (defined as in Definition 5.9 for this shift) that is exact on all hops of head 1 at interior cells satisfies

$$\mathrm{cost}(G) \;\geq\; \nu_1 n_{\mathrm{h}}, \qquad \nu_1 = \frac{\Lambda/2 - 4\eta - 1}{16\Lambda},$$

and $\mathrm{cost}(G) \geq \nu_1 n_{\mathrm{h}} + (1 - 4\nu_1) n_{\mathrm{h}}/q$ if $q > 4$. For the golden shift of Definition 5.9 the condition $0 < 8\eta < \Lambda$ holds for every $s_0 \geq 4$; for $s_0 \geq 5$, $\nu_1 \geq 67/4096 \approx 0.01636$, with equality exactly at $s_0 = 8$, and for $s_0 = 4$, $\nu_1 = 3/256$.

Proof (not formalized). The walk. By Definition 5.3 with $m = g = 1$, each level-0 stroke is one segment of Λ steps followed by a V of two plain strokes of $\Lambda/2$ steps. So on $[0, 4\Lambda]$

$$x_1(t) = \begin{cases} t, & 0 \leq t \leq \Lambda, \\ 2\Lambda - t, & \Lambda \leq t \leq 3\Lambda/2, \\ t - \Lambda, & 3\Lambda/2 \leq t \leq 2\Lambda, \\ 3\Lambda - t, & 2\Lambda \leq t \leq 3\Lambda, \\ t - 3\Lambda, & 3\Lambda \leq t \leq 7\Lambda/2, \\ 4\Lambda - t, & 7\Lambda/2 \leq t \leq 4\Lambda, \end{cases}$$

and $x_1(t + 4\Lambda) = -x_1(t)$ for all t. In $[0, 8\Lambda)$, a cell c with $0 < c < \Lambda/2$ is visited at the times c, $3\Lambda - c$, $3\Lambda + c$, $4\Lambda - c$, and a cell with $\Lambda/2 < c < \Lambda$ at c, $2\Lambda - c$, $\Lambda + c$, $3\Lambda - c$; we call $0, 1, 2, 3$ the *indices* of these visits. A cell c with $-\Lambda < c < 0$ is visited at the times $4\Lambda + t$, for the visits t of $-c$ just listed, with the same indices. Each of these cells is interior and has 4 visits per period; so for a time t of the k-th period copy $[8\Lambda k, 8\Lambda(k+1))$ of the cover at such a cell, $\kappa_1(t)$ is $4k$ plus the index of the visit. A pair (p, u) is a hop of head 2 if and only if $(p + \xi, u + \xi)$ is a hop of head 1.

The family. For every integer a with $1 \leq a \leq \Lambda/2 - 4\eta - 1$ consider the four head-2 hops of the following table, where the third column is the head-1 hop they come from, with its cell, and the last two columns give the cells of head 1 at the two endpoints and the difference of the indices of these visits.

	head-2 hop (p, u)	head-1 hop $(p+\xi, u+\xi)$; its cell	$x_1(p) \to x_1(u)$	$\Delta\kappa_1$
e_1	$(a, \Lambda - a - 2\eta)$	$(3\Lambda + a + \eta, 4\Lambda - a - \eta)$; $a + \eta$	$a \to \Lambda - a - 2\eta$	0
e_2	$(2\Lambda + a + 2\eta, 3\Lambda - a - 4\eta)$	$(5\Lambda + a + 3\eta, 6\Lambda - a - 3\eta)$; $a + 3\eta - \Lambda$	$\Lambda - a - 2\eta \to a + 4\eta$	-2
e_3	$(3\Lambda - a - 4\eta, 3\Lambda + a + 2\eta)$	$(6\Lambda - a - 3\eta, 6\Lambda + a + 3\eta)$; $a + 3\eta - \Lambda$	$a + 4\eta \to a + 2\eta$	1
e_4	$(3\Lambda - a - 2\eta, 3\Lambda + a)$	$(6\Lambda - a - \eta, 6\Lambda + a + \eta)$; $a + \eta - \Lambda$	$a + 2\eta \to a$	1

The bounds on a give $0 < a + \eta < a + 2\eta < a + 3\eta < a + 4\eta < \Lambda/2$, which places every cell below in the stated range. Row e_1: the cell $a + \eta$ lies in $(0, \Lambda/2)$ and is visited at $3\Lambda + (a + \eta)$ and $4\Lambda - (a + \eta)$ with indices 2 and 3, consecutively; at the times a and $\Lambda - a - 2\eta$ head 1 is on its first stroke, at cells $a \in (0, \Lambda/2)$ and $\Lambda - a - 2\eta \in (\Lambda/2, \Lambda)$, both with index 0. Row e_2: the cell $\Lambda - a - 3\eta$ lies in $(\Lambda/2, \Lambda)$

and is visited at $\Lambda+a+3\eta$ and $2\Lambda-a-3\eta$ with indices 1 and 2, so its negative $a+3\eta-\Lambda$ is visited at the times of the table with indices 1 and 2, consecutively; at $2\Lambda+a+2\eta=3\Lambda-(\Lambda-a-2\eta)$ head 1 visits $\Lambda-a-2\eta\in(\Lambda/2,\Lambda)$ with index 3, and at $3\Lambda-(a+4\eta)$ it visits $a+4\eta\in(0,\Lambda/2)$ with index 1. Row e_3: the visits of $a+3\eta-\Lambda$ at $6\Lambda-a-3\eta$ and $6\Lambda+a+3\eta=4\Lambda+3\Lambda-(\Lambda-a-3\eta)$ have indices 2 and 3; at $3\Lambda+(a+2\eta)$ head 1 visits $a+2\eta\in(0,\Lambda/2)$ with index 2. Row e_4: the cell $\Lambda-a-\eta$ lies in $(\Lambda/2,\Lambda)$ and is visited at $2\Lambda-a-\eta$ and $2\Lambda+a+\eta$ with indices 2 and 3, so its negative $a+\eta-\Lambda$ is visited at the times of the table with indices 2 and 3; at $3\Lambda-(a+2\eta)$ head 1 visits $a+2\eta$ with index 1, and at $3\Lambda+a$ it visits a with index 2. In each row $p<u$, and all eight times lie in $[0,4\Lambda)$, so the differences of κ_1 in any period copy are the differences of the indices.

Closure and defect. The head-1 cells run $a\to\Lambda-a-2\eta\to a+4\eta\to a+2\eta\to a$, so $x_1(u_j)=x_1(p_{j+1})$ for $j\in\mathbb{Z}_4$, identically in a, Λ and η, and all four cells are interior. The defect is $(0-1)+(-2-1)+(1-1)+(1-1)=-4$. (At $a=0$ and at $a=\Lambda/2-4\eta$ other negative cycles arise, with other index patterns; we do not use them.)

The mirror family. Since $x_1(t+4\Lambda)=-x_1(t)$ for all t, also $x_2(t+4\Lambda)=-x_2(t)$, so adding 4Λ to all times of a member gives four head-2 hops whose head-1 cells are negated. The cycle condition is preserved, the indices are the same by the description of the visits of negative cells, and the defect is again -4.

Disjointness. A hop of head 2 on the cover is determined by its later endpoint. In $[0,8\Lambda)$ the later endpoints of e_1, e_2, e_3, e_4 lie in $(\Lambda/2,\Lambda)$, $(5\Lambda/2,3\Lambda)$, $(3\Lambda,7\Lambda/2)$ and $(3\Lambda,7\Lambda/2)$ respectively, and those of the mirror family in the same intervals shifted by 4Λ; these sets are disjoint except for the last two. For different values of a the endpoints of e_1 differ, as do those of e_2, of e_3 and of e_4, and $3\Lambda+a+2\eta=3\Lambda+a'$ exactly when $a'=a+2\eta$. So two members share a hop only if their parameters differ by 2η, the shared hop being e_3 of the smaller and e_4 of the larger, and within a member the four hops are distinct. Let $n=\Lambda/2-4\eta-1$ be the number of members of each family. The members split into chains $a,a+2\eta,a+4\eta,\dots$; keeping every other member of each chain, starting with the first, keeps $\lceil\ell/2\rceil$ members of a chain of ℓ, hence at least $n/2$ members of each family, and the kept members are pairwise hop-disjoint. With the mirror family this gives at least n hop-disjoint cycles per period.

Counting. On the r-fold cover, take these cycles in each of the r period copies, translated by multiples of 8Λ; they remain hop-disjoint, each has defect -4 computed with κ_1, and their number n_{c} is at least rn. Interior cells have $4r$ visits per cover and $q\mid 4r$, so by Lemma 5.17(c) the label G satisfies the hypothesis of Theorem 5.21(b) with $\kappa=\kappa_1$. That theorem gives $q\mathrm{cost}(G)\ge 2T+n_{\mathrm{c}}(q-4)$, that is, $\mathrm{cost}(G)\ge n_{\mathrm{c}}+(2T-4n_{\mathrm{c}})/q\ge n_{\mathrm{c}}$, because the n_{c} cycles use $4n_{\mathrm{c}}$ distinct hops among the T hops of head 2. Since $n_{\mathrm{h}}=2T=16r\Lambda$ and $n_{\mathrm{c}}\ge rn$, this is $\mathrm{cost}(G)\ge\nu_1 n_{\mathrm{h}}$. If $q>4$, the right side $n_{\mathrm{c}}(1-4/q)+2T/q$ increases with n_{c}, and $n_{\mathrm{c}}\ge 2T\nu_1$ gives $\mathrm{cost}(G)\ge\nu_1 n_{\mathrm{h}}+(1-4\nu_1)n_{\mathrm{h}}/q$.

The golden shift. Here $P=8\Lambda$ and $0.381966\,P=3.055728\Lambda$. For $s_0=5,6,7,8$ one finds $\lfloor 3.055728\Lambda\rfloor=97,195,391,782$, so $\xi=97,195,391,783$, $\eta=1,3,7,15$, $8\eta<\Lambda$, and $\nu_1=11/512$, $19/1024$, $35/2048$, $67/4096$. For $s_0\ge 9$, $3.055728\Lambda-1<\xi\le 3.055728\Lambda+1$, so $0.055728\Lambda-1<\eta\le 0.055728\Lambda+1$; hence $0<8\eta<\Lambda$ and $\nu_1\ge(0.277088-5/\Lambda)/16\ge 0.0167>67/4096$, as $\Lambda\ge 512$. □

Why it matters. Theorem E_1 is the first price per hop that does not depend on the depth, on the sharpest open instance, with one level of nesting, for a natural class of labels: the class that extends the mechanism solving one walk by an arbitrary correction per cell. On the golden pair with $m=1$ and $s_0\ge 5$, such labels cost at least 0.016 per hop on the cover, about $0.033\,T$, at every depth D, so they cannot give cost $\le\varepsilon T$ for $\varepsilon<0.03$; for the aligned shift $\xi=1$ the analogous certificates are close to zero numerically. The theorem says nothing about labels with small but

nonzero errors on head 1, and these are exactly what optimised labels use (Remark 5.19).

Remark 5.23 (From the cover to the walk DAG)**.** The same argument works on the walk DAG of the pair of Theorem 5.22 on $[0, NP)$, $N \geq 1$, for labels $\Psi\colon [0, NP) \to \{0, \dots, q-1\}$ that are exact modulo q on all head-1 hops at interior cells. The chains of hops no longer close up, so summing $q[\Psi(u) \leq \Psi(p)] = 1 + e - (\Psi(u) - \Psi(p))$ over the hops of a chain gives $q \cdot \#\{\text{violated hops}\} \geq \#\{\text{hops}\} + \sum e - (q-1)\#\{\text{chains}\}$; there are at most $2(2\Lambda+1)$ chains. Along each interior head-1 chain $\Psi - \kappa_1$ is constant modulo q, with no divisibility condition, and the cycles of the family that lie in $[0, NP)$, at least Nn after selection, force head-2 error at least $q-4$ each. Hence, counting hops per head, $\mathrm{cost}(\Psi) \geq 2\nu_1 NP - O(\Lambda)$ for $q > 4$, and $\mathrm{cost}(\Psi) \geq NP/2 - O(\Lambda)$ for $q \leq 4$ from the floor. A pair (p, u) that is a hop of both heads would be a single edge of the walk DAG, but there is none. Indeed, all turn times and turn positions of $U(s_0, m, g)$ are multiples of 2^{s_m}, so on every maximal monotone run $x(t) = \pm t + c$ with $2^{s_m} \mid c$. A hop (p, u) leaves $x(p)$ at time p and returns at time u along runs of opposite slopes, so $p + u \equiv 0 \pmod{2^{s_m}}$, while a hop of x_2 satisfies $p + u \equiv -2\xi$; and $2\xi \not\equiv 0 \pmod{2^{s_m}}$ for odd ξ when $s_m \geq 2$; for the pair of Theorem 5.22, where $\xi = 3\Lambda + \eta$ need not be odd, $2^{s_1} = \Lambda/2$ and $2\xi \equiv 2\eta \not\equiv 0 \pmod{\Lambda/2}$ because $0 < 2\eta < \Lambda/4$.

Remark 5.24 (Numerical: nesting depths $m \leq 6$)**.** Theorem E_1 is a closed form of computer-found certificates. For the golden pair $\mathrm{DNP}(s_0, m, 1)$ with $1 \leq m \leq 6$, explicit one-parameter families of negative cycles in M_2, fitted as affine functions of a, Λ and ξ on ranges of s_0 between 9 and 14 and checked member by member on instances outside the fitting range (up to $s_0 = 16$), certify for labels exact on all head-1 hops at interior cells, at every q, prices per hop of 0.0509, 0.0554, 0.0527, 0.0519, 0.0526, 0.0480 for $m = 1, \dots, 6$ (from $12, 16, 95, 208, 245, 531$ families, with the fractional packing that gives each member C the weight $1/\max_{h \in C} \mathrm{load}(h)$, where the load of a hop is the number of members through it). These values were stable in s_0; they are per-instance certificates, not proofs for all s_0, and they have not been checked independently. For $m = 1$ a greedy selection of hop-disjoint members does better: it finds exactly $\Lambda - 4$ negative 4-cycles per period on all instances checked ($7 \leq s_0 \leq 14$ and $s_0 = 16$), that is, a price $1/16 - 1/(4\Lambda)$ per hop at every q; a proof would need the coincidences between the twelve families. For labels exact only on the head-1 hops of some levels, the certified price falls as levels are released: at $m = 1$ exactness on the finest level alone admits an explicit label of cost at most about $4/q$ per hop, while at $m = 3$ and $s_0 = 7$, exactness on the finest one, two or three levels or on all four carries certified prices of about 0.003, 0.026, 0.066 and 0.107. Why "every m" is open: the families show no recursion in m, and the phases of the shift relative to the level grids, $\psi_i = \mathrm{frac}(2^i \xi / 2^{s_0})$, which are the same for all strokes of level i, degenerate for special m; for instance $\psi_2, \dots, \psi_6$ lie between 0.003 and 0.06 at $m = 8$, and when 15625 divides $m + 1$ exact arithmetic gives $\psi_i = 2^{i - s_0}$ for every $i \geq 3$. A proof for all m therefore needs a rule generating level-i families from local configurations, a bound on their overlaps independent of m, and a positive share of good levels, not equidistribution of phases.

5.7 The remaining gap

Labels found by optimisation are not exact on either head. They trade errors of head 1 on backward stretches against errors of head 2 (Remark 5.19), and by Lemma 5.18(c) such an absorption costs only about $|\delta|/q$ per cycle. A lower bound for all labels must show that absorptions cannot be supplied consistently: an error on a head-1 hop lies on backward stretches of some head-2 cycles, where it helps, and on forward stretches of others, where it hurts, and symmetrically for head 2. We expect that this forces a multiplicative loss at every level that both heads use, and we state the expected outcome as a count of served levels.

Let the pair of Definition 5.9 be defined, verbatim, for any odd shift ξ. For a label G on its cover and $0 < \theta < 1$, say that level i is *θ-served for head h* if the hops of head h of level i have total error at most $\theta q n_{\mathrm{h}}/(2(m+1))$, and *doubly θ-served* if it is θ-served for both heads. When the $n_{\mathrm{h}}/2$ hops of each head are spread evenly over the $m+1$ levels, as numerical counts indicate, this says that the mean error per hop of the class is at most θq. Two kinds of labels serve every level for trivial reasons; the conjecture excludes the first by its bound on the cost and the second by its choice of the shift. Labels with few or random values serve every level at large θ, but cost at least n_{h}/q, more than $n_{\mathrm{h}}/8$ if $q < 8$, respectively about $n_{\mathrm{h}}/2$, by (3). And for a shift aligned with the dyadic scales, such as the golden shift when 15625 divides $m+1$, a label exact on one head is nearly exact on the other.

Conjecture 5.25 (Conjecture A′: few doubly served levels)**.** *There are $\theta_0 > 0$ and constants C, C', and for all s_0 and m an odd shift ξ, such that for all r and q every label G with $\mathrm{cost}(G) \le n_{\mathrm{h}}/8$ on the r-fold cover of the pair $x_1 = U(s_0, m, 1)$, $x_2 = x_1(\cdot + \xi)$ has at most $C \log_2 q + C'$ doubly θ_0-served levels.*

We expect shifts misaligned with the dyadic scales at every level to qualify. The golden shift is not always one of them: even when $15625 \nmid m+1$, its phases nearly align at several levels for some m, such as $m = 8$ (Remark 5.24).

If the conjecture holds, let G be a label, on a cover of the pair with the shift that it provides, of cost $\mathrm{cost}(G) \le \varepsilon n_{\mathrm{h}}$, where $\varepsilon \le 1/8$ and $\varepsilon < \theta_0/2$. By Corollary 5.16, every level that is not doubly θ_0-served contributes more than $\theta_0 n_{\mathrm{h}}/(2(m+1))$ to the cost, so fewer than $2\varepsilon(m+1)/\theta_0$ levels are not doubly served, and $(m+1)(1 - 2\varepsilon/\theta_0) < C \log_2 q + C'$. This gives

$$D = q - 1 \;\ge\; \exp\bigl(c(1 - 2\varepsilon/\theta_0)(m+1) - c'\bigr) - 1$$

for constants $c, c' > 0$, so the depth needed at a fixed cost per hop grows exponentially in the depth of nesting, matching the clock labels of Proposition 5.13 up to the constant in the exponent. The same statement for labels on the walk DAG of many periods would bound the depth of every cut of size εT in the same way, because (3) and Corollary 5.16 hold there up to boundary terms and, when $s_m \ge 2$, no pair is a hop of both heads (both as in Remark 5.23). It would make the family of these pairs with $s_m \ge 2$, $m \to \infty$, depth-robust, and Problem 5.1 for $k = 2$ would have a negative answer.

Remark 5.26 (Numerical: evidence for Conjecture A′)**.** The evidence is limited. In the best labels found by optimisation for $m = 3$ ($g = 1$, $s_0 = 6$), with mean error at most $q/10$ per hop counted as served, the number of doubly served levels was 1, 1, 2 and 2 or 3 at $D = 7, 15, 31, 63$, consistent with growth like $\log q$, and level 0 of head 1 was never served (mean error about $0.17q$ to $0.20q$ at every D). Labels exact on several levels of one head carry a price that grows with the number of exact levels (Remark 5.24), a restricted form of the conjecture. Small instances cannot decide asymptotic questions, and these labels are upper bounds of uncertain quality.

Status. We do not know whether the deep-nesting pairs are depth-robust as m grows. No mechanism we know cuts them at a depth independent of m: per-head band rules appear to fail (proof sketches of Remarks 5.6 and 5.7); for $g \ge 2$, anchored digits cannot be monotone (Lemma 5.12); and explicit combinations of visit indices and band counts kept a cost per hop bounded away from zero at every depth tried, in computations up to $m = 8$. Clock labels cut each pair, but only at depth $(c/\varepsilon)^{m+1}$ (Proposition 5.13). On the other side, the only lower bounds that hold at every depth are for restricted classes (Theorems 5.21 and 5.22), and packing certificates cannot do better (Proposition 5.14).

Why it matters. Problem 5.1 is the walk-level form of the structural hypothesis of Theorem 4.10, and Theorem 4.20 solves it for home-based walks. On the sharpest open instance, the evidence just listed leaves a robust multiplicative count of absorptions as the missing step, of which Conjecture 5.25 is one form. Either answer is informative. If the pairs are depth-robust as m grows, TCP(2) fails as stated; if moreover oblivious linear-time machines can generate them as dummy traffic and they are depth-robust at every ε below the crux rates in question, the structural hypothesis of Theorem 4.10 must be weakened as explained in Section 5.1, and these pairs, placed at intermediate scales inside curved motion, are candidates for the depth-robust schedule that an oblivious linear-time transposer obeying the crux would need. If they are depth-reducible at a depth independent of m, the mechanism must beat clock labels by absorbing defects consistently across levels, and it would be the core of a general proof of TCP(2).

6 Storage codes on sparse side information

The second hypothesis of Theorem D is an all-codes index-coding statement, the crux (Definition 4.8): on the side information produced by a bounded-depth genie, every index code for the transposition demand must broadcast a constant fraction of the input. An oblivious linear-time transposer whose dependency DAG has cheap genies of bounded depth needs the opposite, broadcasts of εL bits with ε as small as the trade-off of Proposition 4.13 allows. Rate $\frac{1}{2}$ is the natural threshold: when the receivers can be paired so that each knows its partner's demand, broadcasting the XOR of each pair serves everyone with $L/2$ bits, and no construction we know does better on bounded-degree side information of growing odd girth.

In the hardest instances we know, coming from schedules whose head motion expands (the ones Section 3 leaves open), the side information that matters is symmetric and sparse: two receivers that know each other's demands form an edge of a bounded-degree graph that looks locally like a tree. This is a heuristic description, not a theorem, but it singles out the following question, the undirected form of the crux. By the duality between index codes and storage codes (Section 2.3), it asks whether storage codes can beat rate $\frac{1}{2}$.

Question 6.1 (Undirected crux)**.** *Let (G_n) be graphs of maximum degree at most Δ that are* locally bipartite*: for each fixed ℓ, only $o(|V(G_n)|)$ vertices lie on odd cycles of length at most ℓ. Does every storage code on G_n have rate at most $\frac{1}{2} + o(1)$, over every alphabet?*

A negative answer is necessary for the kind of coding gain a linear-time transposer would exploit on such side information; a positive answer, in the directed and depth-bounded generality of Definition 4.8, would combine with depth reduction to refute oblivious (T). This section proves the positive answer for restricted classes of codes and of graphs, and shows where the remaining difficulty lies:

- every storage code, linear or not, on any graph that maps homomorphically to C_5, to the circular clique $K_{14/5}$ or to the Petersen graph has rate at most $\frac{1}{2}$, with no degree assumption (Corollary 6.9);
- on triangle-free graphs of maximum degree 3, no linear storage code over any field beats rate $\frac{1}{2}$ (Theorem S, Theorem 6.13);
- with bounded complex coefficients, no linear code beats rate $\frac{1}{2} + o(1)$ on locally bipartite graphs of bounded degree, and over finite fields, with a fixed number of symbols per vertex, a gain must come from p-torsion (Theorem A, Theorem 6.17 and Corollary 6.22);

- at degree 5 the threshold fails: the Clebsch graph carries a binary linear code of rate 5/8, and no storage code does better, so its guessing number is exactly 10 (Corollary 6.11);
- local arguments inside tree-like regions cannot give rate $\frac{1}{2} + o(1)$ at degree at least 4, real eigenvalue codes cannot reach rate $\frac{1}{2}$ at girth at least 7, and Mycielski's construction does not preserve goodness; whether goodness bounds the fractional chromatic number is open (Sections 6.9 and 6.10).

Every numbered result of Sections 6.1 to 6.6 is formalized; Sections 6.7 to 6.10 are not formalized.

6.1 Entropic polymatroids

Lemma 6.2 (Entropy of a recoverable word)**.** Lean: `codeEntropy`, `codeEntropy_isPolymatroid`, `codeEntropy_isValid` [`StorageCode.lean`]

Let $X : \Omega \to \alpha^V$ be a random word on a finite probability space. Then $S \mapsto H(X_S)$ is a polymatroid on V. If X is recoverable for N, this polymatroid is valid for N.

Outline of the formal proof. $H(X_S)$ is the entropy of the observation $\omega \mapsto X(\omega)|_S$, written as a word in $(\alpha \cup \{\bot\})^V$ that is $\bot$ outside S, in the second paper's entropy library [20]. The empty observation is constant, so it has entropy 0. Monotonicity holds because X_S is a function of X_T for $S \subseteq T$. Submodularity is that library's strong subadditivity for the observations X_S, X_T and $X_{S \cap T}$. Validity: recoverability makes $X_{N(v) \cup \{v\}}$ a function of $X_{N(v)}$, so the two entropies agree. □

Theorem 6.3 (Rate bounds bound codes)**.** Lean: `codeEntropy_univ_le_of_rateBound`, `codeEntropy_univ_uniform`, `log_card_le_of_rateBound` [`StorageCode.lean`]

Suppose N satisfies the rate bound $c \geq 0$. Then every random word X recoverable for N satisfies $H(X_V) \leq c \sum_v H(X_v)$, and every storage code C for N over a nonempty alphabet α satisfies $\log|C| \leq c\,|V| \log|\alpha|$.

Outline of the formal proof. The first claim applies the rate bound to the polymatroid of Lemma 6.2. For the second, take X uniform on C; then $H(X_V) = \log|C|$, and each $H(X_v) \leq \log|\alpha|$. □

Why it matters. This is the bridge from polymatroids to codes: a rate bound proved for valid polymatroids, by a certificate or by Theorem H, bounds every storage code, linear or not, over every alphabet. For (T), a rate bound c on the side information of the cones gives the crux at rate $1 - c$ (Remark 4.9(b)).

Valid polymatroids inherit the recovery structure as a closure operation, which every certificate below uses.

Lemma 6.4 (Closure)**.** Lean: `IsPolymatroid.eq_of_closure`, `IsPolymatroid.insert_eq_of_subset` [`Polymatroid.lean`]

Let h be a polymatroid valid for N, and let $S \subseteq T$ be such that $N(v) \subseteq S$ for every $v \in T \setminus S$. Then $h(T) = h(S)$.

Outline of the formal proof. Submodularity gives diminishing returns: if adding v to A does not increase h, then adding v to any $B \supseteq A$ does not either. Apply this with $A = N(v)$ and $B \supseteq S$, adding the vertices of $T \setminus S$ one at a time. □

6.2 Theorem H: rate bounds pull back along homomorphisms

Theorem 6.5 (Theorem H)**.** Lean: `pushRank`, `IsPolymatroid.pushRank`, `IsValid.pushRank`, `pushRank_univ`, `sum_pushRank_singleton_le`, `RateBound.of_hom`, `RateBound.of_graphHom`, `RateBound.of_digraphHom` [`Polymatroid.lean`]

Let N_G and N_T be side information on V and W, and let $\varphi\colon V \to W$ map $N_G(u)$ into $N_T(\varphi(u))$ for every u. For a polymatroid h on V put $h'(Z) = h(\varphi^{-1}(Z))$. Then h' is a polymatroid on W, valid for N_T if h is valid for N_G, with $h'(W) = h(V)$ and $\sum_t h'(\{t\}) \le \sum_v h(\{v\})$. Consequently, if N_T satisfies the rate bound $c \ge 0$, so does N_G. In particular this holds for every graph homomorphism $G \to T$ and every digraph homomorphism (arcs to arcs).

Outline of the formal proof. Preimages preserve unions, intersections and inclusions, so h' is a polymatroid. Validity: if u lies in the fibre of t, then $N_G(u) \subseteq \varphi^{-1}(N_T(t))$, so by Lemma 6.4 adding the whole fibre of t to $\varphi^{-1}(N_T(t))$ does not increase h. No looplessness or injectivity is needed. The fibres partition V, and subadditivity bounds h of each fibre by the sum over its vertices. Finally $h(V) = h'(W) \le c\sum_t h'(\{t\}) \le c\sum_v h(\{v\})$. □

The formal theorem is about all polymatroids, so it concerns the Shannon bound. The same argument shows that the push-forward of an entropic polymatroid is entropic: if $h(S) = H(X_S)$, then $h'(Z) = H(Y_Z)$ for the random word $Y_t = X_{\varphi^{-1}(t)}$. We do not need this, because every target below is certified by Shannon inequalities.

Why it matters. Theorem H turns one finite certificate into a statement about infinitely many graphs: a rate bound for a fixed target T holds for every side information that maps to T, whatever its size or degree. This is the mechanism behind every all-codes result of this section about infinitely many graphs, and the reason a target's structure, not the size of the side information, decides goodness. For (T), it underlies the homomorphism route to the crux (Section 6.10). The theorem is folklore-level (the proof is one line); related monotonicity statements for index coding appear in [33], and blow-up lemmas in [4, 22].

6.3 Symmetrization

Theorem 6.6 (Symmetrization)**.** Lean: `IsAut`, `symmetrize`, `IsPolymatroid.symmetrize`, `IsValid.symmetrize`, `symmetrize_invariant`, `symmetrize_univ`, `sum_symmetrize_singleton`, `RateBound.of_invariant` [`Symmetry.lean`]

Let N be side information on V. If every valid polymatroid that is invariant under all automorphisms of N satisfies $h(V) \le c\sum_v h(\{v\})$, then N satisfies the rate bound c.

Outline of the formal proof. Average a valid polymatroid over the finite group of permutations σ of V with $\sigma(N(v)) = N(\sigma(v))$. Each relabelled function $S \mapsto h(\sigma S)$ is a valid polymatroid, so the average is one too. It is invariant, because right multiplication by an automorphism permutes the group, and it has the same value on V and the same sum over singletons as h. □

Why it matters. With symmetrization, a certificate (Section 6.4) needs one elemental inequality per orbit, and each set it mentions can be replaced by an orbit representative through an explicit automorphism. This is what makes the Clebsch certificate below small enough to check.

6.4 Good targets

A certificate for a rate bound c is a nonnegative combination of *elemental inequalities* $h(K \cup \{i\}) + h(K \cup \{j\}) \geq h(K \cup \{i,j\}) + h(K)$ that equals $c\sum_v h(\{v\}) - h(V)$ once every set is replaced by its closure (Lemma 6.4). We found the certificates by linear programming, re-solved them exactly in rational arithmetic on their supports, and checked the Petersen, $K_{14/5}$ and Clebsch certificates in Lean with `linear_combination` and exact coefficients (the five-inequality pentagon certificate is closed by `linarith`); the generator is `tools/certificates.py`. The smallest example shows the mechanism.

Proposition 6.7 (The pentagon is good)**.** Lean: `pentagonNbhd`, `nbhd_cycleGraph_five`, `rateBound_pentagon`, `rateBound_cycleGraph_five` [`Pentagon.lean`]

Every valid polymatroid on the 5*-cycle* 0-1-2-3-4-0 *satisfies* $2h(V) \leq \sum_v h(\{v\})$.

Outline of the formal proof. Writing $I(i;j \mid K)$ for the elemental quantity above, the five non-adjacent inequalities

$$I(0;2) + I(0;3 \mid 1,2) + I(1;3) + I(1;4) + I(2;4 \mid 1) \;\geq\; 0$$

sum to $\sum_v h(\{v\}) - 2h(V)$ after the closures $\{0,1,2\} \to \{0,2\}$, $\{1,2,3\} \to \{1,3\}$, $\{0,1,2,3\} \to V$ and $\{1,2,4\} \to V$. The Lean proof is exactly this computation. □

The *circular clique* $K_{p/q}$ has vertices $\mathbb{Z}/p$, with $i \sim j$ if and only if $q \leq (j-i) \bmod p \leq p-q$.

Proposition 6.8 (Further good targets)**.** Lean: `petersenGraph`, `petersenGraph_adj_iff`, `petersenKneser_card`, `petersenKneser_injective`, `rateBound_petersenGraph` [`Petersen.lean`]

Lean: `circularClique`, `rateBound_circularClique_14_5`, `circularClique_adj_of_lt`, `circularCliqueHom`, `rateBound_circularClique_of_le`, `cycleToCircularClique`, `rateBound_cycleGraph_odd` [`CircularClique.lean`]

The Petersen graph (the Kneser graph $K(5,2)$*) and the circular clique* $K_{14/5}$ *(the Andrásfai graph* And(5)*) are good. If* $p/q \leq p'/q'$ *with* $q' \geq 1$*, the map* $i \mapsto \lfloor p'i/p \rfloor$ *is a homomorphism* $K_{p/q} \to K_{p'/q'}$ *(Bondy and Hell [15]). Hence every circular clique* $K_{p/q}$ *with* $p/q \leq 14/5$ *is good, for example* $K_{8/3}$ *(the Möbius ladder* And(3)*) and* $K_{11/4}$ *(*And(4)*), and so is every odd cycle* C_{2k+1} *with* $k \geq 2$*.*

Outline of the formal proof. The Petersen certificate is a combination of 58 elemental inequalities with weights $\frac{1}{4}$ and $\frac{1}{2}$ and 25 closure equalities (of the 40 that the file proves), and the labelling is identified with $K(5,2)$ by an explicit bijection. For $K_{14/5}$ the polymatroid is first symmetrized over the dihedral group (Theorem 6.6); the certificate then uses 32 orbit rows, 21 explicit automorphisms, 116 invariance equalities and 13 closure equalities. For the homomorphism, if $i < j$ and $q \leq j - i \leq p - q$, then $\lfloor p'i/p \rfloor + q' \leq \lfloor p'j/p \rfloor \leq \lfloor p'i/p \rfloor + p' - q'$ because $pq' \leq p'q$, and adjacency is symmetric. The odd cycle C_{2k+1} maps to $K_{(2k+1)/k}$ by $i \mapsto ki$, which turns the differences ± 1 into $\pm k$, and $(2k+1)/k \leq 14/5$ for $k \geq 2$. Theorem H applies. □

Corollary 6.9 (Corollary H2: rate $\frac{1}{2}$ for every code)**.** Lean: `card_sq_le_of_rateBound_half`, `IsGood.card_sq_le`, `IsGood.codeEntropy_le` [`RateHalf.lean`]

Lean: `IsGood.of_hom` [`Polymatroid.lean`]

Let G *be a good graph, for instance a graph with a homomorphism to* C_5*, to the Petersen graph or to a circular clique* $K_{p/q}$ *with* $p/q \leq 14/5$*. Then every storage code* $C \subseteq \alpha^V$ *on* G *over a nonempty alphabet, linear or not, satisfies* $|C|^2 \leq |\alpha|^{|V|}$*, and every random word* X *recoverable for* G *satisfies* $2H(X_V) \leq \sum_v H(X_v)$.

Outline of the formal proof. Goodness pulls back along homomorphisms (Theorem H); combine with Propositions 6.7 and 6.8 and Theorem 6.3. The integer form takes logarithms. □

The graphs with a homomorphism to some $K_{p/q}$ with $p/q \le 14/5$ are exactly the graphs of circular chromatic number at most $14/5$, since for a finite graph the circular chromatic number is the least p/q such that it maps to $K_{p/q}$ [64, 67].

A graph with a perfect matching carries the pairing code (copy one symbol across each matched edge), of rate exactly $\frac{1}{2}$, so for good graphs with a perfect matching the corollary determines the guessing number: $\mathrm{gn}(G) = |V|/2$. The corollary strictly extends the guessing-number results of Mazumdar, McGregor and Vorotnikova [51, Thms 9–11] (their graphs map to $K_{8/3}$ or to C_5), and contains the odd-cycle theorem of Christofides and Markström [23]. For Petersen it makes rigorous the numerical values of Blasiak, Kleinberg and Lubetzky [13].

Why it matters. Corollary 6.9 is an all-codes answer to the undirected crux, Question 6.1, for every graph that maps to a good target: the graphs need not have bounded degree, and they include graphs of arbitrarily large odd girth, for example long odd cycles. The question is whether high-girth bounded-degree graphs map to good targets. For cubic graphs this is a known open problem:

Corollary 6.10 (Corollary H3: a conditional degree-3 result)**.** Lean: `IsGood.card_sq_le` [`RateHalf.lean`]
Lean: `IsGood.of_hom` [`Polymatroid.lean`]
Lean: `rateBound_cycleGraph_five` [`Pentagon.lean`]
Lean: `rateBound_circularClique_14_5` [`CircularClique.lean`]

If every cubic graph of sufficiently large girth maps to C_5 (Nešetřil's Pentagon Problem [54]), or to $K_{14/5}$, then every storage code on such a graph, nonlinear included, has rate at most $\frac{1}{2}$.

Outline of the formal proof. This is Corollary 6.9 applied under the stated hypothesis. □

Whether high-girth cubic graphs have circular chromatic number below 3 is itself open [27]. Some targets are known to fail: random cubic graphs do not map to C_7 [39], and the Clebsch graph, to which every subcubic graph of girth at least 17 maps by a theorem of DeVos and Šámal [29], is not good (Corollary 6.11). For linear codes, Theorem S below settles degree 3 unconditionally.

6.5 The Clebsch graph: guessing number ten

The Clebsch graph is the folded 5-cube: vertices $0, \dots, 15$, with $i \sim j$ if and only if $i \oplus j$ (bitwise exclusive or) is a power of two or equals 15. It is triangle-free and 5-regular.

Corollary 6.11 (Corollary H4: $\mathrm{gn}(\text{Clebsch}) = 10$)**.** Lean: `rateBound_clebsch`, `rateBound_clebschGraph`, `clebsch_card_le`, `clebsch_guessingNumber`, `clebschCode_eq`, `mem_ker_clebsch_iff`, `finrank_ker_clebsch`, `rank_clebsch` [`ClebschCertificate.lean`]
Lean: `clebschGraph`, `clebschGraph_adj_iff`, `clebschCode`, `clebschCode_card`, `clebschCode_isStorageCode`, `exists_clebsch_storageCode` [`Clebsch.lean`]

Every valid polymatroid on the Clebsch graph satisfies $h(V) \le \frac{5}{8}\sum_v h(\{v\})$. Hence every storage code on the Clebsch graph over an alphabet of size q has at most q^{10} codewords, while the binary linear storage code $\ker(I + A)$ over $\mathbb{F}_2$, where A is the adjacency matrix, has exactly 2^{10}. So $\mathrm{gn}(\text{Clebsch}) = 10$, and $I + A$ has rank 6 over $\mathbb{F}_2$.

Outline of the formal proof. Symmetrize over the automorphism group, of order 1920 (Theorem 6.6). The certificate has 54 orbit rows; unlike the certificates for C_5, the Petersen graph and $K_{14/5}$, it uses elemental inequalities between adjacent vertices as well. It uses 126 explicit automorphisms, 225 invariance equalities and 54 closure equalities, and is checked with exact rational coefficients. For the lower bound, the code is the span of ten explicit vectors, each satisfying $x_v = \sum_{u \sim v} x_u$ over $\mathbb{F}_2$, in systematic form on ten coordinates; injectivity of the encoding gives 2^{10} codewords. Since $\ker(I + A)$ is itself a storage code, the upper bound shows that it has at most 2^{10} words, so it equals the code; its dimension is 10, and the rank of $I + A$ is $16 - 10 = 6$. The identification with the folded 5-cube is checked by decision procedure. □

Proposition 6.12 (The Shannon bound of the Clebsch graph)**.** Lean: `clebsch_rateBound_iff` [`ClebschCertificate.lean`]
Lean: `RateBound.mono` [`Polymatroid.lean`]

The Clebsch graph satisfies the rate bound c if and only if $c \geq \frac{5}{8}$.

Outline of the formal proof. The *if* direction is Corollary 6.11 with monotonicity in c. For the *only if* direction, the entropic polymatroid of the uniform distribution on the binary code has $h(V) = 10 \log 2$ and $h(\{v\}) \leq \log 2$, which violates every rate bound below $\frac{5}{8}$. □

Cameron, Dang and Riis proved $10 \leq \mathrm{gn}(\text{Clebsch}) \leq 11$ [22] and, in the preprint version, asked for the exact guessing numbers of the strongly regular graphs they consider [21, Problem 5.11]; Corollary 6.11 answers this for the Clebsch graph. The binary code $\ker(I + A)$ goes back to Blasiak, Kleinberg and Lubetzky [13].

Why it matters. The Clebsch graph is triangle-free with odd girth 5 and degree 5, and its codes beat the pairing by a constant factor. So bounded degree alone does not force rate $\frac{1}{2}$: any proof of the undirected crux must use local bipartiteness (few short odd cycles), not only sparsity. It also rules out targets: it is not good, although every subcubic graph of girth at least 17 maps to it [29], and by Theorem H no graph that receives it is good, which makes iterated Mycielski graphs bad (Proposition 6.35). On the construction side it shows what a coding gain can look like at small girth: the binary code is the reduction modulo 2 of a rational eigenvalue code with coefficients 1 (Remark 6.23), a mechanism that Theorem A and Corollary 6.22 rule out on locally bipartite families of bounded degree.

6.6 Theorem S: linear codes on subcubic triangle-free graphs

Let F be a field and $M \in F^{V \times V}$ a matrix with unit diagonal. The *recovery set* of v is $S_v = \{u \neq v : M_{vu} \neq 0\}$. The matrix has *no recovery triangle* if $q \notin S_p$ whenever $p, q \in S_r$; this holds whenever the off-diagonal support of M lies in the edges of a triangle-free graph. A vertex satisfies *reciprocity* ($\star\star$) if $|S_v| \leq 2$, or $|S_v| = 3$ and $r \in S_v$ whenever $v \in S_r$.

Theorem 6.13 (Theorem S)**.** Lean: `Star`, `RecTriangleFree`, `ThmSHyp`, `finrank_subCode_le_half`, `thmSHyp_of_cliqueFree`, `star_of_degree_le_three`, `finrank_ker_le_half_of_degree_le_three`, `rank_ge_half_of_degree_le_three` [`Subcubic.lean`]

Let $M \in F^{V \times V}$ have unit diagonal and no recovery triangle. For every vertex set W on which every vertex satisfies ($\star\star$), the kernel of the principal submatrix $M[W, W]$ has dimension at most $\lfloor |W|/2 \rfloor$. In particular, if G is a triangle-free graph of maximum degree at most 3 and the off-diagonal support of M lies in the edges of G, then $\dim \ker M \leq \lfloor |V|/2 \rfloor$ and $\operatorname{rank} M \geq \lceil |V|/2 \rceil$, over every field.

Outline of the formal proof. Codes on vertex sets are subspaces of F^V: the code on W is the set of vectors supported on W that satisfy the rows of W, so deleting vertices is an inclusion of subspaces. The proof is a strong induction over W; suppose W is a minimal counterexample.

1. *Start.* Deleting one vertex costs at most one dimension, so $|W| = 2m + 1$ is odd, the code has dimension exactly $m + 1$, and every singleton deletion is *tight*: it leaves the largest dimension the theorem allows.

2. *Tightness.* On a tight deletion no coordinate vanishes identically, since otherwise a smaller set would be a counterexample. If two coordinates p, q vanish together on a tight code, deleting both is again tight, and the deleted code is the subspace $\{y_p = 0\}$ of the old one.

3. *Maximal chains.* A *good chain* starts at a vertex v_0 with at most two recovery neighbours in W, which exists by a parity argument (otherwise the recovery relation would be a symmetric 3-regular graph on an odd set). It repeatedly adds a pair $\{p, q\}$ that is exactly the part of a chain vertex's recovery set outside the chain. Along a good chain, every deletion is tight and its code lies inside the code of W, so every row of a chain vertex restricted to the rest is a relation on the remaining code. If some chain vertex has exactly one recovery neighbour outside the chain, that coordinate vanishes, contradicting tightness. If one has exactly two, the chain extends. Reciprocity rules out three, because the vertex that added a chain vertex lies in its recovery set.

4. *Final count.* A maximal chain therefore leaves every chain vertex with no recovery neighbours outside it. Call *free* the vertex v_0 and the first vertex of each pair. A codeword that vanishes on the free vertices vanishes on the whole chain, because the row of each pair's parent then forces the second vertex of the pair to vanish. For the last pair $\{p, q\}$ even p is forced: its row involves only earlier chain vertices, since $q \notin S_p$ ($p, q \in S_r$ for the parent r, and M has no recovery triangle). So a codeword is determined, on the chain, by k coordinates for a chain with k pairs, and $m + 1 \le k + \lfloor (|W| - 2k - 1)/2 \rfloor = m$, a contradiction.

□

The proof needs neither Gallai's lemma nor connectivity, and the first part of Theorem 6.13, stated for recovery sets, is slightly stronger than the graph-theoretic statement. The triangle condition is only needed on the recovery sets, and reciprocity only on the vertex set being bounded. Over a field, linear storage codes are exactly the subspaces of the kernels of such matrices:

Lemma 6.14 (Linear storage codes are subspaces of kernels)**.** `Lean`: `IsLinearStorageCode`, `IsLinearStorageCode.exists_coeffs`, `IsLinearStorageCode.exists_pattern`, `isLinearStorageCode_of_le_ker`, `isStorageCode_iff_isLinearStorageCode` [`Subcubic.lean`]

Call a subspace $C \subseteq F^V$ a linear storage code *on G if every $x \in C$ that vanishes on the neighbours of v vanishes at v; over a finite field this is equivalent, by linearity, to the defining property of Definition 2.1. If C is a linear storage code, each coordinate is a fixed linear combination of the neighbouring coordinates on C, so $C \subseteq \ker M$ for a matrix M with unit diagonal and off-diagonal support in the edges of G. Conversely, every subspace of such a kernel is a linear storage code.*

Outline of the formal proof. On C, the evaluation at v vanishes on the common kernel of the evaluations at the neighbours of v, so it lies in their span (Mathlib's `mem_span_of_iInf_ker_le_ker`). Conversely, on $\ker M$ the row of v gives $x_v = -\sum_{u \sim v} M_{vu} x_u$. For the equivalence, $x, y \in C$ agree on a set exactly when $x - y \in C$ vanishes there. □

Corollary 6.15 (Theorem S for linear storage codes)**.** Lean: `IsLinearStorageCode.finrank_le_half`, `IsLinearStorageCode.card_sq_le` [`Subcubic.lean`]

On a triangle-free graph of maximum degree at most 3*, every linear storage code over any field has dimension at most* $\lfloor |V|/2 \rfloor$*; over a finite field* F *it has* $|C|^2 \le |F|^{|V|}$*.*

Outline of the formal proof. By Lemma 6.14, $C \subseteq \ker M$ for a matrix M as in Theorem 6.13; apply Theorem 6.13. Over a finite field, $|C| = |F|^{\dim C}$. □

The exact minrank follows by splitting along separated vertex sets.

Theorem 6.16 (Theorem S, barrier and matching forms)**.** Lean: `finrank_subCode_union_le`, `finrank_subCode_biUnion_le`, `finrank_ker_le_barrier`, `finrank_ker_le_heavy`, `exists_linearStorageCode_of_matching`, `exists_pattern_finrank_ker_ge_of_matching` [`Subcubic.lean`]

Lean: `matchingNumber_eq_min_barrier`, `finrank_ker_le_matchingNumber`, `finrank_ker_le_matchingNumber_of_degree_le_three`, `exists_pattern_matchingNumber_le_finrank_ker`, `minrank_eq_card_sub_matchingNumber` [`MatchingForm.lean`]

Let M *have unit diagonal, no recovery triangle, and off-diagonal support in the edges of a graph* G*. Let* $U \subseteq V$ *be a set outside which every vertex satisfies* $(\star\star)$*, and let the vertices outside* U *be partitioned into blocks with no edges between different blocks (for instance the components of* $G - U$*). Then*

$$\dim \ker M \;\le\; |U| + \sum_{B} \lfloor |B|/2 \rfloor .$$

Taking for U *the vertices violating* $(\star\star)$ *gives* $\dim \ker M \le |U| + \lfloor (|V| - |U|)/2 \rfloor$*. By the Tutte–Berge formula [10], the minimum of the right side over all* U*, with the components of* $G - U$ *as blocks, is the matching number* $\nu(G)$*. Conversely, over every field and for every graph, a matching of* m *edges gives such a matrix with* $\dim \ker M \ge m$*. Hence, for triangle-free* G *of maximum degree at most* 3*,* $\operatorname{minrk}_F(G) = |V| - \nu(G)$ *over every field.*

Outline of the formal proof. Deleting U costs at most $|U|$ dimensions. A code on a union of blocks with no edges between them splits as a sum of the codes on the blocks, because the rows of one block vanish on the others; each block's code obeys Theorem 6.13. The matching code puts one free symbol on each edge of the matching, constant along the edge, and zero elsewhere; it is a linear storage code of dimension m, and Lemma 6.14 turns it into a matrix with $\dim \ker M \ge m$. The Tutte–Berge formula, in the form used here (some U attains $|U| + \sum_B \lfloor |B|/2 \rfloor = \nu(G)$, with the components of $G - U$ as blocks), comes from the repository's shared Lean library `lean-machine`. It ports J. P. Traverso Gianini's derivation of the formula from Mathlib's Tutte theorem [62] and adds the matching number and the blocks. □

The statement appears to be new: we found no statement that the minrank of triangle-free subcubic graphs equals the clique-cover number $|V| - \nu(G)$ over every field, in the minrank, index-coding, storage-code or guessing-game literature. Special cases are known. Tims [61] proves that the minimum rank over every field equals the clique-cover number for all graphs of order at most 10 (by computer), the Petersen graph included, and for the Möbius ladders, cubic triangle-free graphs for which Blasiak, Kleinberg and Lubetzky [13] had shown broadcast rate $|V|/2$ for all codes; Berliner and Langberg [11] prove the same equality over every finite field for outerplanar graphs. None of the hypotheses can be dropped. The Clebsch graph (degree 5, Corollary 6.11) has binary minrank $6 < 8 = |V| - \nu$. The complete graph K_4 shows that triangles must be excluded. On C_5, vector-linear codes with two symbols per vertex reach dimension 5, more than $2\nu(C_5) = 4$, so the theorem is genuinely scalar (although rate $\frac{1}{2}$ still holds for C_5 by Proposition 6.7).

Why it matters. Theorem S settles, for scalar linear codes, the first whole class of bounded degree beyond degree 2, where earlier results covered particular families, and it is exact: in degree 3 no scalar linear code over any field, characteristic 2 included, beats the pairing on any triangle-free graph, locally tree-like or not. With Theorem A it narrows where a coding gain could live: for scalar linear codes, at degree at least 4 over a finite field through p-torsion; for vector-linear codes over finite fields and for nonlinear codes, already at degree 3.

6.7 Bounded complex coefficients: Theorem A

The rate bounds proved so far in this section hold either for all codes on special graphs (Theorem 6.5 and its corollaries) or for all linear codes on triangle-free graphs of maximum degree 3 (Theorem 6.13). The undirected crux (Question 6.1) concerns locally bipartite side information of bounded degree. In this subsection we prove the bound $\frac{1}{2} + o(1)$ on the rate in that setting for one class of codes: linear codes over $\mathbb{R}$ or $\mathbb{C}$ whose recovery coefficients are bounded.

Let G be a finite graph with vertex set V and $N = |V|$. A matrix $M \in F^{V \times V}$ over a field F *fits* G if $M_{vv} = 1$ for every v and $M_{vu} = 0$ whenever $u \neq v$ and $uv \notin E(G)$. By Lemma 6.14, the linear storage codes on G over F are the subspaces of kernels of matrices that fit G; the coefficients R_{vu} of a recovery rule $x_v = \sum_{u \sim v} R_{vu} x_u$ are the negated off-diagonal entries of such a matrix, which we call a *check matrix* of the code. A complex matrix is *B-bounded* if all its entries have modulus at most B; a matrix that fits a graph can be B-bounded only if $B \geq 1$. For $\ell \geq 1$ let $\mathrm{odd}_\ell(G)$ be the set of vertices of G that lie on an odd cycle of length at most ℓ. As in Question 6.1, a sequence of graphs G_n on $N_n \to \infty$ vertices is *locally bipartite* if $|\mathrm{odd}_\ell(G_n)| = o(N_n)$ for every fixed ℓ. Sequences whose girth tends to infinity are locally bipartite, and so are *locally tree-like* sequences, in which, for every fixed ℓ, only $o(N_n)$ vertices lie on cycles of length at most ℓ.

Theorem 6.17 (Theorem A: bounded complex coefficients)**.** Not formalized; a complete proof is given below.

Let $D \geq 1$ and $B \geq 1$.

(i) Let (G_n) be a locally bipartite sequence of graphs of maximum degree at most D on N_n vertices, and let $M_n \in \mathbb{C}^{V(G_n) \times V(G_n)}$ be B-bounded matrices that fit G_n. Then $\liminf_n \mathrm{rank}(M_n)/N_n \geq \frac{1}{2}$.

(ii) (Uniform form.) *For every $\varepsilon > 0$ there are $\delta > 0$ and N_0 such that $\mathrm{rank}\, M \geq (\frac{1}{2} - \varepsilon)N$ for every graph G on $N \geq N_0$ vertices of maximum degree at most D with $|\mathrm{odd}_{\lfloor 1/\delta \rfloor}(G)| \leq \delta N$ and every B-bounded complex matrix M that fits G.*

(iii) (Random regular graphs.) *Let $d \geq 3$ and $\varepsilon > 0$, and let $G_{n,d}$ be a uniformly random d-regular graph on n vertices (nd even). With probability tending to 1 as $n \to \infty$, every B-bounded complex matrix that fits $G_{n,d}$ has rank at least $(\frac{1}{2} - \varepsilon)n$.*

In terms of codes (Corollary 6.21): on a locally bipartite sequence of graphs of bounded degree, a linear storage code over $\mathbb{R}$ or $\mathbb{C}$ whose recovery coefficients have modulus at most B has dimension at most $(\frac{1}{2} + o(1))N_n$.

Why it matters. For the impossibility side, Theorem A proves the analogue of the crux's rate bound (Definition 4.8) for real and complex linear codes with bounded recovery coefficients, scalar or with a fixed number of symbols per vertex (Proposition 6.18), on side information of bounded total degree whose underlying graphs are locally bipartite. The crux itself concerns binary codes; Theorem A bears on them through Corollary 6.22. Only the support of the check matrix enters,

so directed side information is covered through its underlying graph, and locally tree-like side information is covered. For the construction side it is a design rule: on such side information, a code that beats rate $\frac{1}{2}$ by a constant is nonlinear, or vector-linear with a number of symbols per vertex that grows with N, or linear with coefficients that no diagonal rescaling makes bounded (Proposition 6.18(a), for $t = 1$), or linear over a finite field, and then every bounded integer lift of its check matrix has p-torsion of linear rank (Corollary 6.22). Codes over finite alphabets, linear or not, are not covered; they are the open part of the crux.

Local limits. We use the framework of Benjamini and Schramm [9] and of Aldous and Lyons [2]. A *network* is a connected graph of maximum degree at most D with *marks*: a point of a compact metric space Ξ_V at every vertex and a point of a compact metric space Ξ_E at each end of every edge. A *rooted* network has a distinguished vertex o, a *doubly rooted* network an ordered pair of distinguished vertices, and we identify networks that are isomorphic by graph isomorphisms preserving the roots and the marks. Two rooted networks are close if, for a large R, the balls of radius R around their roots are isomorphic by a root-preserving graph isomorphism that changes every mark by less than $1/R$. Since degrees are bounded and the mark spaces are compact, the space of rooted networks is compact (a diagonal argument). A finite marked graph of maximum degree at most D, connected or not, defines a random rooted network: choose o uniformly among its vertices and keep the component of o. A sequence of finite marked graphs *converges locally* to a probability measure μ on rooted networks if these random rooted networks converge to μ in distribution. A probability measure μ on rooted networks is *unimodular* if it satisfies the mass-transport principle

$$\mathbb{E}_\mu \sum_{x \in V(G)} f(G, o, x) \;=\; \mathbb{E}_\mu \sum_{x \in V(G)} f(G, x, o) \tag{4}$$

for every Borel function $f \geq 0$ on doubly rooted networks, where (G, o) denotes a μ-random rooted network. We use four standard facts.

(L1) Every sequence of finite marked graphs as above has a locally convergent subsequence (compactness and Prokhorov's theorem).

(L2) Every local limit of finite marked graphs is unimodular [2, Section 2]. (A finite graph with several components is a convex combination of its components, each rooted uniformly.)

(L3) Let μ be unimodular. A family $T = (T_G)$ of bounded operators T_G on $\ell^2(V(G))$, indexed by networks, is *equivariant* if $\langle T_{G'}\delta_{\varphi x}, \delta_{\varphi y}\rangle = \langle T_G\delta_x, \delta_y\rangle$ for every isomorphism $\varphi\colon G \to G'$ of networks and all vertices x, y, and *measurable* if $(G, x, y) \mapsto \langle T_G\delta_x, \delta_y\rangle$ is Borel on doubly rooted networks. The μ-classes of equivariant, measurable families with $\sup_G \|T_G\| < \infty$ form a von Neumann algebra $\mathrm{Alg}(\mu)$, acting on the direct integral of the spaces $\ell^2(V(G))$, on which $\mathrm{Tr}(T) = \mathbb{E}_\mu\langle T_G\delta_o, \delta_o\rangle$ is a positive linear functional with $\mathrm{Tr}(ST) = \mathrm{Tr}(TS)$ [2, Section 5]. Its elements act fibrewise, and their kernels, ranges and spectral projections are computed fibrewise.

(L4) (Polar decomposition.) For T in a von Neumann algebra $\mathcal{A}$ there is a partial isometry $U \in \mathcal{A}$ such that U^*U is the projection onto $(\ker T)^\perp$ and UU^* is the projection onto the closure of the range of T [42, Theorem 6.1.2 and Proposition 6.1.3].

Proof (not formalized). (i) Let $\nu_n = 1 - \mathrm{rank}(M_n)/N_n$ be the normalized nullity and $\nu = \limsup_n \nu_n$; we show $\nu \leq \frac{1}{2}$. Make G_n a marked graph by putting the mark $M_n[v, u]$, a point of the disc $\{z \in \mathbb{C} : |z| \leq B\}$, at the end v of each edge vu. Pass to a subsequence along which $\nu_n \to \nu$

and, by (L1), the marked graphs converge locally to some μ, which is unimodular by (L2). Write (G, M, o) for a μ-random rooted network. The marks define an operator M_G on $\ell^2(V(G))$ by $(M_G x)_v = x_v + \sum_{u\sim v} M[v,u]\, x_u$. Each row and each column of M_G has at most $D+1$ nonzero entries, of modulus at most B, so $\|M_G\| \le (D+1)B$, and (M_G) is an equivariant measurable family.

Step 1: the limit is bipartite. For $\ell, R \ge 1$ let $A_{\ell,R}$ be the set of rooted networks in which some odd cycle of length at most ℓ meets the ball of radius R around the root. Such a cycle lies in the ball of radius $R+\ell$, so membership in $A_{\ell,R}$ is decided by that ball without its marks, and $A_{\ell,R}$ is open and closed. A uniform root of G_n lies in $A_{\ell,R}$ only if it is within distance R of $\mathrm{odd}_\ell(G_n)$, which has probability at most $(1 + D + \cdots + D^R)|\mathrm{odd}_\ell(G_n)|/N_n \to 0$. Since $A_{\ell,R}$ is open and closed, $\mu(A_{\ell,R})$ is the limit of these probabilities, so $\mu(A_{\ell,R}) = 0$. Every odd cycle of G meets some ball around o, so μ-almost surely G has no odd cycle: G is bipartite.

Step 2: semicontinuity of the nullity. Let $H_n = M_n^* M_n$ and let σ_n be the empirical eigenvalue distribution of H_n (eigenvalues counted with multiplicity, each with mass $1/N_n$); it is supported in $[0, c]$ with $c = ((D+1)B)^2$. For $k \ge 0$, $\int x^k \, d\sigma_n = N_n^{-1} \operatorname{tr} H_n^k$ is the expectation, over a uniform vertex o of G_n, of $\langle H_n^k \delta_o, \delta_o\rangle$. Expanding the product, $\langle (M^*M)^k \delta_o, \delta_o\rangle$ is a sum of at most $(D+1)^{2k}$ products of $2k$ entries of M or their conjugates, taken along walks of length $2k$ from o to o whose steps stay put or move to a neighbour; such walks stay in the ball of radius k. Hence $(G, M, o) \mapsto \langle (M_G^* M_G)^k \delta_o, \delta_o\rangle$ is a bounded continuous function on rooted networks, and $\int x^k \, d\sigma_n \to \mathbb{E}_\mu \langle H^k \delta_o, \delta_o\rangle = \int x^k \, d\sigma$, where $H = M_G^* M_G$ and $\sigma = \mathbb{E}_\mu \sigma_{H,o}$ is the average of the spectral measures $\sigma_{H,o}$ of H at δ_o. All these are probability measures on $[0, c]$, so convergence of moments implies $\sigma_n \to \sigma$ weakly (Weierstrass approximation), and the portmanteau theorem for the closed set $\{0\}$ gives $\limsup_n \sigma_n(\{0\}) \le \sigma(\{0\})$. Since $\ker(M^*M) = \ker M$ for matrices and for bounded operators, $\sigma_n(\{0\}) = \nu_n$ and $\sigma(\{0\}) = \mathbb{E}_\mu \langle P_K \delta_o, \delta_o\rangle$, where P_K is the orthogonal projection onto $K = \ker M_G$. Therefore

$$\nu \;\le\; \mathbb{E}_\mu \langle P_K \delta_o, \delta_o\rangle. \tag{5}$$

This is the easy half of Lück's approximation theorem [49]; for adjacency matrices of random graphs the same argument is used by Bordenave, Lelarge and Salez [16]. The bound B is used here and in (L1).

Step 3: a random colouring. For μ-almost every (G, M, o) the graph G is connected and bipartite, so it has exactly two proper colourings with colours ± 1, each the negative of the other. Let $\hat\mu$ be the law of (G, M, β, o), where (G, M, o) has law μ and β, a vertex mark, is one of the two colourings, chosen by a fair coin independent of everything else. Then $\hat\mu$ is unimodular. Indeed, let $f \ge 0$ be Borel on doubly rooted coloured networks. For a doubly rooted network (G, M, x, y) with G bipartite let $\pi_x(w) = (-1)^{\mathrm{dist}(x,w)}$, and put $\bar f(G, M, x, y) = \frac12 (f(G, M, \pi_x, x, y) + f(G, M, -\pi_x, x, y))$ (and $\bar f = 0$ if G is not bipartite). Since $\{\pi_x, -\pi_x\} = \{\pi_y, -\pi_y\}$ is the set of both colourings, $\bar f$ is the average of f over the two colourings whichever root is used to define it; it is Borel and invariant under isomorphisms. Hence

$$\mathbb{E}_{\hat\mu} \sum_x f(G, M, \beta, o, x) = \mathbb{E}_\mu \sum_x \bar f(G, M, o, x) = \mathbb{E}_\mu \sum_x \bar f(G, M, x, o) = \mathbb{E}_{\hat\mu} \sum_x f(G, M, \beta, x, o),$$

the middle equality being (4) for μ. Moreover $\hat\mu(\beta(o) = -1) = \frac12$. (Without the coin, colouring by the parity of the distance to the root is not unimodular in general: the root would always have colour $+1$ and its neighbours colour -1.)

Step 4: the trace. By (L3) the algebra $\mathrm{Alg}(\hat\mu)$ contains $M = (M_G)$, hence $H = M^*M$ and its spectral projection $P_K = \mathbf{1}_{\{0\}}(H)$, and the projection P_J onto $\ell^2(J)$, where $J = \beta^{-1}(-1)$; let $V_+ = \beta^{-1}(+1)$. Let $T = P_J P_K \in \mathrm{Alg}(\hat\mu)$. If $Tx = 0$, write $x = k + k'$ with $k \in K$ and $k' \in K^\perp$; then $P_J k = Tx = 0$, so k vanishes on J. Every neighbour of a vertex $v \in V_+$ lies in J, so

$0 = (M_G k)_v = k_v + \sum_{u\sim v} M[v,u]\, k_u = k_v$. Hence $k = 0$, and $\ker T = K^\perp$. The range of T lies in $\ell^2(J)$. By (L4) there is a partial isometry $U \in \mathrm{Alg}(\hat\mu)$ with $U^*U = P_K$ and $UU^* \le P_J$, and by (L3)

$$\mathbb{E}_\mu\langle P_K\delta_o, \delta_o\rangle = \mathrm{Tr}(P_K) = \mathrm{Tr}(U^*U) = \mathrm{Tr}(UU^*) \le \mathrm{Tr}(P_J) = \hat\mu(\beta(o) = -1) = \tfrac12.$$

The first equality holds because $\langle P_K\delta_o, \delta_o\rangle$ does not depend on β and the law of (G, M, o) under $\hat\mu$ is μ. With (5), $\nu \le \frac12$.

(ii) If (ii) failed for some ε, then for every m there would be a graph G_m on $N_m \ge m$ vertices of maximum degree at most D with $|\mathrm{odd}_m(G_m)| \le N_m/m$, and a B-bounded matrix M_m fitting G_m with $\operatorname{rank} M_m < (\frac12 - \varepsilon)N_m$. For fixed ℓ and $m \ge \ell$ we have $|\mathrm{odd}_\ell(G_m)| \le |\mathrm{odd}_m(G_m)| \le N_m/m$, so (G_m) is locally bipartite, and (i) gives a contradiction.

(iii) By a theorem of Wormald [66], for each fixed $\ell \ge 3$ the number X_ℓ of cycles of length ℓ in $G_{n,d}$ converges in distribution as $n \to \infty$, to a Poisson law with mean $(d-1)^\ell/(2\ell)$; in particular X_ℓ is bounded in probability. Let δ and N_0 be given by (ii) for $D = d$. Then $|\mathrm{odd}_{\lfloor 1/\delta\rfloor}(G_{n,d})| \le \sum_{\ell\le 1/\delta} \ell X_\ell$ is bounded in probability, so it is at most δn with probability tending to 1, and (ii) applies. □

Proposition 6.18 (Extensions of Theorem A)**.** Not formalized; a complete proof is given below.
Let $D \ge 1$ and $B \ge 1$.

(a) (Tight coefficients and gauge.) *For every $\varepsilon > 0$ there are $\delta' > 0$ and N_0' with the following property. Let G be a graph on $N \ge N_0'$ vertices of maximum degree at most D with $|\mathrm{odd}_{\lfloor 1/\delta'\rfloor}(G)| \le \delta' N$, let $M \in \mathbb{C}^{V\times V}$ fit G, let S be an invertible diagonal matrix, and let X be the set of rows of $S^{-1}MS$ that contain an entry of modulus larger than B. Then $\operatorname{rank} M \ge (\frac12 - \varepsilon)(N - |X|)$.*

(b) (A fixed number of symbols per vertex.) *Let $t \ge 1$. Parts (i) and (ii) of Theorem 6.17 hold for complex matrices M with rows and columns indexed by $V \times [t]$ and entries of modulus at most B whose $t \times t$ blocks satisfy $M_{vv} = I_t$ and $M_{vu} = 0$ whenever $u \ne v$ and $uv \notin E(G)$, with N replaced by tN in the conclusions; the constants δ and N_0 depend on t.*

(c) (Invariant independent sets.) *Let B-bounded matrices M_n fitting graphs G_n of maximum degree at most D, viewed as marked graphs as in the proof of Theorem 6.17, converge locally to μ. Let $\alpha_U(\mu)$ be the supremum of $\hat\mu(\iota(o) = 1)$ over all unimodular laws $\hat\mu$ of networks with an additional vertex mark $\iota \in \{0, 1\}$ such that forgetting ι gives μ and $\iota^{-1}(1)$ is $\hat\mu$-almost surely an independent set. Then $\liminf_n \operatorname{rank}(M_n)/N_n \ge \alpha_U(\mu)$.*

(d) (Sharpness.) *If G has a perfect matching P with adjacency matrix A_P, then $I + A_P$ is 1-bounded, fits G, and has rank $N/2$. In particular the constant $\frac12$ of Theorem 6.17 is attained on every regular bipartite graph of positive degree.*

Proof (not formalized). (a) The matrix $S^{-1}MS$ fits G and has the rank of M, so we may assume $S = I$. Let δ and N_0 be given by Theorem 6.17(ii) for D, B and ε, and put $\delta' = 2\delta/(D+1)$ and $N_0' = \lceil (D+1)N_0/2\rceil$. Let $N' = N - |X|$. If $N' \le 2N/(D+1)$, let Y be an independent set of G with $|Y| \ge N/(D+1)$ (a greedy choice); the principal submatrix $M[Y,Y]$ is the identity, so $\operatorname{rank} M \ge |Y| \ge N'/2$. Otherwise $N' > 2N/(D+1) \ge N_0$, and the principal submatrix $M' = M[V\setminus X, V\setminus X]$ is B-bounded and fits $G' = G[V\setminus X]$. Every odd cycle of G' is an odd cycle of G, and $\lfloor 1/\delta\rfloor \le \lfloor 1/\delta'\rfloor$, so $|\mathrm{odd}_{\lfloor 1/\delta\rfloor}(G')| \le |\mathrm{odd}_{\lfloor 1/\delta'\rfloor}(G)| \le \delta' N < \delta N'$. Theorem 6.17(ii) gives $\operatorname{rank} M \ge \operatorname{rank} M' \ge (\frac12 - \varepsilon)N'$.

(b) Let $G[t]$ be the graph on $V \times [t]$ in which $(v,i) \sim (u,j)$ if and only if $uv \in E(G)$. Such a matrix M fits $G[t]$, whose maximum degree is tD. An odd cycle of length $\ell' \le \ell$ through (v,i)

in $G[t]$ projects to a closed walk of odd length ℓ' through v in G; a closed walk of odd length contains an odd cycle of length at most ℓ' formed by vertices of the walk, all within distance $\ell'/2 \le \ell$ of v. Hence $\mathrm{odd}_\ell(G[t]) \subseteq W \times [t]$, where W is the set of vertices within distance ℓ of $\mathrm{odd}_\ell(G)$, and $|\mathrm{odd}_\ell(G[t])| \le t(1 + D + \cdots + D^\ell)|\mathrm{odd}_\ell(G)|$. So $(G_n[t])$ is locally bipartite when (G_n) is, and Theorem 6.17(i) applies to $G_n[t]$ with D replaced by tD. For (ii), take δ_t and N_0 from Theorem 6.17(ii) for degree tD, put $\ell = \lfloor 1/\delta_t \rfloor$ and $\delta = \min(\delta_t/(1 + D + \cdots + D^\ell), 1/\ell)$. Then $|\mathrm{odd}_{\lfloor 1/\delta \rfloor}(G)| \le \delta N$ gives $|\mathrm{odd}_\ell(G[t])| \le t(1 + D + \cdots + D^\ell)\delta N \le \delta_t\, tN$, and $tN \ge N \ge N_0$.

(c) Steps 2 and 4 of the proof of Theorem 6.17 do not use bipartiteness. Step 2, applied to the whole sequence, gives $\limsup_n(1 - \mathrm{rank}(M_n)/N_n) \le \mathbb{E}_\mu\langle P_K\delta_o, \delta_o\rangle$. Step 4, run in $\mathrm{Alg}(\hat\mu)$ with $\iota^{-1}(1)$ in place of V_+ and $J = \iota^{-1}(0)$ (every neighbour of a vertex of $\iota^{-1}(1)$ lies in J, since $\iota^{-1}(1)$ is independent), gives $\mathbb{E}_\mu\langle P_K\delta_o, \delta_o\rangle \le \hat\mu(\iota(o) = 0) = 1 - \hat\mu(\iota(o) = 1)$. Take the supremum over $\hat\mu$.

(d) Ordering the vertices in matched pairs, $I + A_P$ is block diagonal with one block $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ of rank 1 per edge of P. A regular bipartite graph with at least one edge has a perfect matching, by Hall's theorem. □

Remark 6.19 (On the extensions)**.** Part (a) says that boundedness is needed only up to a diagonal rescaling and outside $o(N)$ rows. A rescaling changes M_{vu} into $s_v^{-1}M_{vu}s_u$, so it preserves the products of entries around closed walks, for instance $M_{vu}M_{uv}$; "bounded in some gauge" is a condition on these products. In part (b) the constants depend on t through the degree tD of $G[t]$, so the argument says nothing about the linear capacity, a supremum over t, or about codes whose number of symbols per vertex grows with N. Part (c) explains why the limit object is needed. A finite independent set Y gives only $\mathrm{rank}\, M \ge |Y|$, since $M[Y, Y]$ is the identity, and graphs of large girth can have small independence ratio: there are cubic graphs of arbitrarily large girth with independence ratio less than $6/13$ [14]. In a limit concentrated on the d-regular tree, by contrast, $\alpha_U = \frac{1}{2}$: the coin-flip colouring of Step 3 gives $\alpha_U \ge \frac{1}{2}$, and the mass transport that sends one unit from each vertex of $\iota^{-1}(1)$ to each of its d neighbours gives $d\,\hat\mu(\iota(o) = 1) \le d\,\hat\mu(\iota(o) = 0)$. By part (d) the bound is attained along locally tree-like sequences as well, for instance on bipartite double covers of d-regular graphs of large girth.

Remark 6.20 (Literature)**.** We have not found Theorem A in the literature; its proof combines the easy half of Lück approximation [49, 16] with a tracial form of the Haemers bound [37], $\mathrm{rank}\, M \ge |Y|$ for an independent set Y, applied to an invariant random independent set of the limit; a tracial version of the Haemers bound for finite graphs was studied by Gao, Gribling and Li [35]. The positive semidefinite analogue is classical and exact: a positive semidefinite matrix with unit diagonal whose off-diagonal support lies in the edges of a triangle-free graph has rank at least $N/2$ (Rosenfeld [59] for real matrices, and Deaett [28]). It needs no degree or coefficient bound, but it does not imply Theorem A, which concerns matrices that need not be symmetric or positive semidefinite. For such matrices triangle-freeness alone does not suffice without a degree bound: Codenotti, Pudlák and Resta [24] construct triangle-free graphs for which $A - 2I$, a matrix that fits the graph after scaling by $-\frac{1}{2}$, has rank $O(N^{3/4})$.

6.8 Codes over fields: torsion and odd characteristic

We record what Theorem A says about codes, and what it implies for codes over finite fields.

Corollary 6.21 (Bounded real and complex linear codes)**.** Not formalized; a complete proof is given below.

Let $F \in \{\mathbb{R}, \mathbb{C}\}$, let $D, t \ge 1$ and $B \ge 1$ be fixed, and let (G_n) be a locally bipartite sequence of graphs of maximum degree at most D on N_n vertices. For $x \in F^{V \times [t]}$ write $x_v \in F^t$ for its block at v.

(i) *Let $C_n \subseteq F^{V(G_n)\times[t]}$ be a linear storage code with t symbols per vertex and bounded coefficients: there are $t \times t$ matrices R_{vu} ($uv \in E(G_n)$) with entries of modulus at most B such that $x_v = \sum_{u\sim v} R_{vu} x_u$ for every $x \in C_n$ and every v. Then $\dim C_n \le (\frac{1}{2} + o(1))\, tN_n$.*

(ii) *Let every vertex v of G_n be a receiver that wants x_v and knows the blocks x_u for u in a set $I_n(v) \subseteq N_{G_n}(v)$ (directed side information). Let a linear map $E\colon F^{V(G_n)\times[t]} \to F^r$ be a linear index code with decoders $x_v = \Lambda_v(Ex) + \sum_{u\in I_n(v)} R_{vu} x_u$, valid for all x, where the maps Λ_v are linear and the matrices R_{vu} have entries of modulus at most B. Then $r \ge (\frac{1}{2} - o(1))\, tN_n$.*

Proof (not formalized). In both cases let M_n be the matrix with blocks $(M_n)_{vv} = I_t$, $(M_n)_{vu} = -R_{vu}$ for the pairs (v,u) that carry a coefficient, and zero blocks elsewhere; it satisfies the hypotheses of Proposition 6.18(b), so $\operatorname{rank} M_n \ge (\frac{1}{2} - o(1))\, tN_n$. (i) $C_n \subseteq \ker M_n$. (ii) For every x, the block of $M_n x$ at v is $\Lambda_v(Ex)$; so $M_n = \Lambda E$, where Λ stacks the maps Λ_v, and $\operatorname{rank} M_n \le r$. □

By Proposition 6.18(a), for $t = 1$ the coefficients need to be bounded only after a diagonal rescaling and outside $o(N_n)$ vertices. The statement does not cover coefficients that no rescaling makes bounded, numbers of symbols per vertex that grow with N_n, codes over finite fields, or nonlinear codes. For finite fields Theorem A still says something: a gain must be a torsion effect.

Corollary 6.22 (Torsion-only principle)**.** Not formalized; a complete proof is given below.

Fix a prime p, integers $D, t \ge 1$, a real $B \ge 1$ and $c > 0$. Let (G_n) be a locally bipartite sequence of graphs of maximum degree at most D on N_n vertices. Let M_n be matrices over $\mathbb{F}_p$ with rows and columns indexed by $V(G_n) \times [t]$ whose $t \times t$ blocks satisfy $(M_n)_{vv} = I_t$ and $(M_n)_{vu} = 0$ whenever $u \ne v$ and $uv \notin E(G_n)$, and suppose that $\dim\ker M_n \ge (\frac{1}{2} + c)\, tN_n$ for all n. Let $\tilde{M}_n$ be integer matrices whose diagonal blocks are I_t, whose off-diagonal blocks vanish at non-edges and whose entries have modulus at most B, that reduce to M_n modulo p (for instance the lifts with these diagonal blocks and entries in $[-p/2, p/2]$, with $B = p/2$). Then p divides at least $(c - o(1))\, tN_n$ of the invariant factors of $\tilde{M}_n$; equivalently, the cokernel $\mathbb{Z}^{tN_n}/\tilde{M}_n\mathbb{Z}^{tN_n}$ contains a subgroup isomorphic to $(\mathbb{Z}/p\mathbb{Z})^{r_n}$ with $r_n \ge (c - o(1))\, tN_n$.

Proof (not formalized). Write $\tilde{M}_n = U\Sigma W$ in Smith normal form, with U, W invertible over $\mathbb{Z}$ and $\Sigma = \operatorname{diag}(s_1, \dots, s_\rho, 0, \dots, 0)$, where $s_i \ge 1$ and $\rho = \operatorname{rank}_{\mathbb{Q}} \tilde{M}_n$. Reducing modulo p, where U and W stay invertible, gives $\operatorname{rank}_{\mathbb{F}_p} M_n = |\{i \le \rho : p \nmid s_i\}|$, so the number of invariant factors divisible by p is $\rho - \operatorname{rank}_{\mathbb{F}_p} M_n$. As a complex matrix, $\tilde{M}_n$ satisfies the hypotheses of Proposition 6.18(b), so $\rho \ge (\frac{1}{2} - o(1))\, tN_n$, while $\operatorname{rank}_{\mathbb{F}_p} M_n = tN_n - \dim\ker M_n \le (\frac{1}{2} - c)\, tN_n$. The cokernel is isomorphic to $\mathbb{Z}^{tN_n - \rho} \oplus \bigoplus_i \mathbb{Z}/s_i\mathbb{Z}$, and its elements of order dividing p form a group $(\mathbb{Z}/p\mathbb{Z})^{r_n}$ with $r_n = |\{i : p \mid s_i\}|$. □

An $\mathbb{F}_{p^e}$-linear storage code becomes, after the choice of a basis of $\mathbb{F}_{p^e}$ over $\mathbb{F}_p$, an $\mathbb{F}_p$-linear code with $t = e$ symbols per vertex and the same rate, because multiplication by an element of $\mathbb{F}_{p^e}$ is $\mathbb{F}_p$-linear; so Corollary 6.22 covers $\mathbb{F}_{p^e}$ for fixed e.

Remark 6.23 (The Clebsch code is inherited from $\mathbb{Q}$)**.** The Clebsch graph is strongly regular with parameters $(16, 5, 0, 2)$, so its adjacency matrix satisfies $A^2 = 5I + 2(J - I - A) = 3I - 2A + 2J$, where J is the all-ones matrix. Hence $(I + A)^2 \equiv 0 \pmod 2$, and over $\mathbb{F}_2$ the image of $I + A$ lies in its kernel. In fact $\operatorname{rank}_{\mathbb{F}_2}(I + A) = 6$, which Corollary 6.11 establishes formally; directly, the Clebsch graph is the Cayley graph of $\mathbb{Z}_2^4$ with connection set $\{e_1, \dots, e_4, e_1 + \cdots + e_4\}$, and in the group algebra $\mathbb{F}_2[\mathbb{Z}_2^4] = \mathbb{F}_2[y_1, \dots, y_4]/(y_1^2, \dots, y_4^2)$, with $e_i \mapsto 1 + y_i$, the matrix $I + A$ acts as multiplication by $a = \sigma_2 + \sigma_3 + \sigma_4$, where σ_k is the k-th elementary symmetric polynomial in the y_i. The product of a with a monomial of degree 2 is $y_1y_2y_3y_4$, and with a monomial of higher degree it is 0, so the

image is spanned by a, $y_1a, \dots, y_4a$ and $y_1y_2y_3y_4$, which are linearly independent. The kernel of $I + A$, of dimension $16 - 6 = 10 > 8$, is the code behind Corollary 6.11, and it is not a torsion effect. Over $\mathbb{Q}$ the eigenvalues of A are 5, 1 and -3, with multiplicities 1, 10 and 5, so $\operatorname{rank}_{\mathbb{Q}}(I - A) = 6$. The integer kernel of $I - A$ is a direct summand of $\mathbb{Z}^{16}$ of rank 10, so its reduction modulo 2 is a 10-dimensional subspace of $\ker_{\mathbb{F}_2}(I - A) = \ker_{\mathbb{F}_2}(I + A)$, hence all of it. The binary code is therefore the reduction of the rational eigenvalue code of the eigenvalue 1 (Proposition 6.30), whose recovery coefficients are 1; Corollary 6.22 rules out exactly this mechanism on locally bipartite families of bounded degree. The other lift $I + A$, with eigenvalues 6, 2 and -2, has full rank and ten invariant factors divisible by 2: the torsion counted in Corollary 6.22 depends on the lift, and its conclusion holds for every bounded lift. A single graph of girth 4 is not a locally bipartite family; the Clebsch graph shows that local bipartiteness cannot be dropped from Theorem A and Corollary 6.22, even for bounded rational codes.

The nilpotent identity $(I + W)^2 = 0$, which the Clebsch check matrix satisfies modulo 2, gives no gain once the girth is at least 5.

Lemma 6.24 (Index-two nilpotence at girth five)**.** Not formalized; a complete proof is given below.
Let G be a graph of girth at least 5, and let $W \in F^{V \times V}$, over any field F, have zero diagonal and $W_{vu} = 0$ whenever $uv \notin E(G)$. If $W^2 = I$, then there is a perfect matching P of G such that $W_{vu} \neq 0$ exactly when $uv \in P$, and $W_{vu}W_{uv} = 1$ for $uv \in P$; in particular $\operatorname{rank}(I+W) = \operatorname{rank}(I-W) = N/2$. Over a field of characteristic 2, $(I + W)^2 = I + W^2$, so every matrix $I + W$ that fits G and satisfies $(I + W)^2 = 0$ has rank exactly $N/2$.

Proof (not formalized). For $v \neq w$, $(W^2)_{vw}$ is a sum over the common neighbours of v and w, of which there is at most one, as G has no cycle of length 3 or 4. Since $(W^2)_{vv} = \sum_{u \sim v} W_{vu}W_{uv} = 1$, some neighbour $m(v)$ of v has $W_{v\,m(v)}W_{m(v)\,v} \neq 0$. If $W_{vu'} \neq 0$ for a neighbour $u' \neq m(v)$, then $m(v)$ and u' are non-adjacent with unique common neighbour v, and $(W^2)_{m(v)\,u'} = W_{m(v)\,v}W_{vu'} \neq 0$, contradicting $W^2 = I$; so row v of W is supported on $\{m(v)\}$. Likewise, if $W_{u'v} \neq 0$ for a neighbour $u' \neq m(v)$, then $(W^2)_{u'\,m(v)} = W_{u'v}W_{v\,m(v)} \neq 0$; so column v is supported on $\{m(v)\}$. As $W_{m(v)\,v} \neq 0$, row $m(v)$ is supported on $\{v\}$, so $m(m(v)) = v$, the pairs $\{v, m(v)\}$ form a perfect matching, and $W_{v\,m(v)}W_{m(v)\,v} = (W^2)_{vv} = 1$. Ordering the vertices in matched pairs, $I \pm W$ is block diagonal with blocks $\left(\begin{smallmatrix} 1 & \pm a \\ \pm a^{-1} & 1 \end{smallmatrix}\right)$ of rank 1. □

In odd characteristic there is a different constraint, which compares a code with its sign twin.

Proposition 6.25 (Sign twins in odd characteristic)**.** Not formalized; a complete proof is given below.
Let F be a field of characteristic different from 2 and let $W \in F^{V \times V}$. Then

$$\operatorname{rank}(I + W) + \operatorname{rank}(I - W) \;=\; \operatorname{rank}\begin{pmatrix} I & W \\ W & I \end{pmatrix} \;\geq\; N.$$

Hence $\dim\ker(I + W) + \dim\ker(I - W) \leq N$, and at most one of $\ker(I + W)$ and $\ker(I - W)$ has dimension larger than $N/2$. If W has zero diagonal and is supported on the edges of a graph G, the block matrix fits the bipartite double cover $G \times K_2$, the graph on $V \times \{0, 1\}$ in which $(v, i) \sim (u, j)$ if and only if $uv \in E(G)$ and $i \neq j$.

Proof (not formalized). Let $Q = \left(\begin{smallmatrix} I & I \\ I & -I \end{smallmatrix}\right)$; then $Q^2 = 2I$, so Q is invertible. A direct computation gives $Q\left(\begin{smallmatrix} I & W \\ W & I \end{smallmatrix}\right)Q^{-1} = \left(\begin{smallmatrix} I+W & 0 \\ 0 & I-W \end{smallmatrix}\right)$, which proves the equality. The block matrix contains I as a principal $N \times N$ submatrix, so its rank is at least N. The last claim holds because the off-diagonal entries of the block matrix are the entries W_{vu} at positions $((v, 0), (u, 1))$ and $((v, 1), (u, 0))$. □

Remark 6.26 (Gauge invariance)**.** Over a field of characteristic different from 2, if G is bipartite with a proper colouring $s\colon V \to \{\pm 1\}$, the diagonal matrix $S = \operatorname{diag}(s)$ satisfies $SWS = -W$ for W supported on the edges, so $\ker(I+W)$ and $\ker(I-W)$ have the same dimension, at most $N/2$ by Proposition 6.25. On a locally bipartite sequence the same conclusion up to $o(N_n)$, for all matrices over a field of characteristic different from 2 and uniformly in the number of symbols per vertex, would follow from approximate gauge invariance: $\dim\ker(I+W_n) - \dim\ker(I-W_n) = o(N_n)$. In the bipartite limit object of the proof of Theorem 6.17 the coin-flip colouring does conjugate $I+W$ into $I-W$, although that proof uses the colouring differently, through the independent set $\beta^{-1}(+1)$. In characteristic 2, $I+W = I-W$ and Proposition 6.25 says nothing; on the Clebsch graph $I+A = I-A$ has a kernel of dimension $10 > 8$ over $\mathbb{F}_2$.

Why it matters. For constructions, Corollary 6.22 is the finite-field part of the design rule stated after Theorem 6.17; in particular, a gain cannot be inherited by reduction modulo p from a code over $\mathbb{Q}$ with bounded coefficients, as the Clebsch code is (Remark 6.23). In characteristic 2, a gain also cannot come from the identity $(I+W)^2 = 0$ once the girth is at least 5 (Lemma 6.24). For the impossibility side, Proposition 6.25 shows that in odd characteristic a gain on one sign twin is paid for on the other. In characteristic 2 only Corollary 6.22 applies.

6.9 Limits of local and spectral arguments

Theorem S settles all scalar linear codes on triangle-free graphs of maximum degree 3, and Theorem A settles bounded complex linear codes on locally bipartite graphs of every bounded degree by a spectral argument. The undirected crux needs finite fields (indeed all codes), at degree at least 4 for scalar linear codes and already at degree 3 for vector-linear and nonlinear ones. This subsection records three limits: inside tree-like regions, the local necessary conditions of a smallest counterexample are compatible with rate above $\frac{1}{2} + o(1)$ at degree at least 4 (Proposition 6.28); real eigenvalue codes, the mechanism behind several known codes that beat $\frac{1}{2}$ at bounded degree, cannot reach rate $\frac{1}{2}$ at girth 7 or more (Proposition 6.30); and girth 5 does not replace low degree (Proposition 6.31).

A class $\mathcal{K}$ of pairs (G, M), where M fits G over some field, is *hereditary* if $(G[W], M[W, W]) \in \mathcal{K}$ whenever $(G, M) \in \mathcal{K}$ and $W \subseteq V(G)$ is nonempty; for example, triangle-free graphs of maximum degree at most 4 with all matrices that fit them, over all fields. For $C = \ker M$ and $U \subseteq V$ let $r(U) = \dim C|_U$ be the dimension of the restriction of C to the coordinates in U (the rank function of the matroid of C), and let $C^\perp = \{y \in F^V : \sum_v y_v x_v = 0 \text{ for all } x \in C\}$, the row space of M. For a hereditary class, the floor bound $\dim\ker M \le \lfloor N/2 \rfloor$ for all members implies $\dim\ker M \le \nu(G)$, the matching number, by the argument of Theorem 6.16: for every vertex set U, $\dim\ker M \le |U| + \sum_H \lfloor |H|/2 \rfloor$ over the components H of $G - U$, and the Tutte–Berge formula turns the minimum over U into $\nu(G)$. The floor bound is therefore the statement to prove.

Lemma 6.27 (Smallest counterexamples)**.** Not formalized; a complete proof is given below.

Let $\mathcal{K}$ be hereditary, and let $(G, M) \in \mathcal{K}$ satisfy $\dim\ker M > \lfloor N/2 \rfloor$ with N as small as possible. Then

(a) N is odd and $\dim\ker M = (N+1)/2$;

(b) $r(U) \ge \lfloor |U|/2 \rfloor + 1$ for every nonempty $U \subseteq V$;

(c) for every nonempty $X \subseteq V$, the vectors of $C^\perp$ supported in X, in particular the rows of M supported in X, span a space of dimension at most $\lceil |X|/2 \rceil - 1$.

Proof (not formalized). A 1×1 matrix that fits a graph is (1), so $N \geq 2$. For $U \subseteq V$, the codewords vanishing on U form a subspace of dimension $\dim C - r(U)$, and their restrictions to $V \setminus U$ lie in $\ker M[V\setminus U, V\setminus U]$ and determine them; so for $U \neq V$ the pair $(G-U, M[V\setminus U, V\setminus U]) \in \mathcal{K}$ has a kernel of dimension at least $\dim C - r(U)$. (a) If N were even, deleting one vertex ($r \leq 1$) would leave a kernel of dimension at least $N/2 > \lfloor (N-1)/2 \rfloor$ on fewer vertices; if $\dim C \geq (N+3)/2$, deleting one vertex would leave a kernel of dimension at least $(N+1)/2 > \lfloor (N-1)/2 \rfloor$. (b) By minimality for $U \neq V$, and trivially for $U = V$, $\dim C - r(U) \leq \lfloor (N-|U|)/2 \rfloor$, so $r(U) \geq (N+1)/2 - \lfloor (N-|U|)/2 \rfloor = \lfloor |U|/2 \rfloor + 1$, as N is odd. (c) The vectors of $C^\perp$ supported in X form the annihilator of the restriction $C|_X$ in F^X, of dimension $|X| - r(X) \leq |X| - \lfloor |X|/2 \rfloor - 1$. □

One way to reach a contradiction in a smallest counterexample is to exhibit a set X with $\lceil |X|/2 \rceil$ independent vectors of $C^\perp$ supported in X, for instance rows of M. The proof of Theorem S does this with a maximal chain grown from a single vertex along recovery neighbourhoods; such a chain closes only through cycles or the boundary of the vertex set, never inside a tree-like region. The next proposition shows that inside a tree-like region of degree $d \geq 4$ no such set exists, however the region is explored, and that the local conditions there are met by a code of rate close to $(d-2)/(d-1)$.

Proposition 6.28 (Proposition L: tree codes are locally consistent)**.** Not formalized; a complete proof is given below.

Let $d \geq 3$ and $r \geq 1$, let B be the ball of radius r around the root of the d-regular tree, $B^\circ \subseteq B$ the set of vertices at distance less than r from the root, and F a field. For each $v \in B^\circ$ let $z_v \in F^B$ have $z_v(v) = 1$, $z_v(u) \neq 0$ for every neighbour u of v, and $z_v(w) = 0$ otherwise (over $\mathbb{F}_2$ these are the rows of $I + A$ indexed by B°). Let $C_B = \{x \in F^B : \sum_w z_v(w) x_w = 0 \text{ for all } v \in B^\circ\}$ and $r(U) = \dim C_B|_U$.

- *(i) $\dim C_B = d(d-1)^{r-1}$, the number of vertices at distance r, and $\dim C_B/|B| \to (d-2)/(d-1)$ as $r \to \infty$: this is $\frac{1}{2}$ for $d = 3$ and $\frac{2}{3}$ for $d = 4$.*
- *(ii) $r(U) \geq |U| - \lfloor |U|/(d-1) \rfloor$ for every $U \subseteq B$.*
- *(iii) If $d \geq 4$, then $r(U) \geq \lfloor |U|/2 \rfloor + 1$ for every nonempty $U \subseteq B$, and for every nonempty $X \subseteq B$ the vectors of $C_B^\perp$ supported in X, in particular the vectors z_v supported in X, span a space of dimension at most $\lceil |X|/2 \rceil - 1$.*

Proof (not formalized). Orient the tree away from the root; for $v \in B^\circ$ let $\mathrm{ch}(v)$ be its set of children, of size d at the root and $d-1$ elsewhere. (i) Order B° by distance from the root. A child of v lies in the support of z_w, for $w \in B^\circ$, only if $w = v$, or w is that child or one of its children, which come after v; so each z_v is nonzero at a coordinate where all earlier vectors vanish, the vectors z_v are linearly independent, and $\dim C_B = |B| - |B^\circ|$ is the number of vertices at distance r. Since $|B| = 1 + d((d-1)^r - 1)/(d-2)$, the ratio tends to $(d-2)/(d-1)$.

(ii) The annihilator of $C_B|_U$ in F^U is $C_B^\perp \cap F^U$, where $C_B^\perp$ is spanned by the vectors z_v, so $r(U) = |U| - \dim(C_B^\perp \cap F^U)$. Let $D_U = \{v \in B^\circ : \mathrm{ch}(v) \subseteq U\}$. If $y = \sum_v \lambda_v z_v$ is supported in U and $\lambda \neq 0$, let v be a vertex of maximal depth with $\lambda_v \neq 0$. For a child c of v, the only vectors z_w with $z_w(c) \neq 0$ are z_v, z_c and the z_w for the children w of c, and $\lambda_c = \lambda_w = 0$ because c and w are deeper than v; so $y_c = \lambda_v z_v(c) \neq 0$, $c \in U$, and $v \in D_U$. Since the vectors z_v are independent by (i), each element of $C_B^\perp \cap F^U$ has a unique coefficient vector λ, and $\lambda \mapsto \lambda|_{D_U}$ is injective on these coefficient vectors; so $\dim(C_B^\perp \cap F^U) \leq |D_U|$. The children sets of distinct vertices are disjoint subsets of size at least $d-1$, so $|D_U| \leq \lfloor |U|/(d-1) \rfloor$.

(iii) For $d \geq 4$, (ii) gives $r(U) \geq |U| - \lfloor |U|/3 \rfloor$, which is at least $\lfloor |U|/2 \rfloor + 1$: check $|U| \leq 5$ directly, and for $|U| \geq 6$ use $|U| - \lfloor |U|/3 \rfloor \geq 2|U|/3 \geq |U|/2 + 1$. The second claim follows as in Lemma 6.27(c). □

Why it matters. In a d-regular graph of girth at least $2r+2$, every ball of radius r is a copy of B. By Proposition 6.28, inside such a ball the constraints that a smallest counterexample satisfies, namely (b) and (c) of Lemma 6.27 for subsets of the ball and the recoverability of every interior vertex from its neighbours, are all met by a code of rate close to $(d-2)/(d-1)$, which exceeds $\frac{1}{2}$ when $d \geq 4$. So an argument that uses only these local necessary conditions inside tree-like regions cannot give even the asymptotic bound $\frac{1}{2} + o(1)$ at degree at least 4. At degree 3 the ball code C_B has rate $\frac{1}{2} + 1/|B|$, so such conditions are compatible with the asymptotic bound there; the exact floor bound needs global information even at degree 3, as the closing of the maximal chain in the proof of Theorem S shows. For such arguments, large girth is the hardest case, not the easiest. And the degree-4 statement over $\mathbb{F}_2$ on 4-regular graphs of large girth is the scalar-linear, characteristic-2 case of the undirected crux at degree 4, for which no characteristic-p analogue of Theorem A is known.

Remark 6.29 (Computational evidence at degree four)**.** Exhaustive exact searches with SAT solvers, not independently verified and not used in any proof, found no counterexample to the floor bound for triangle-free graphs of maximum degree 4: over $\mathbb{F}_2$ none of the 193,900 connected 4-regular triangle-free graphs on 17 vertices has a linear storage code of dimension 9; over $\mathbb{F}_3$ none of the connected 4-regular triangle-free graphs on 13 and 15 vertices, and none of the non-bipartite connected triangle-free graphs with degrees between 2 and 4 on 9 vertices, or on 11 vertices with a vertex of degree 4, has a code of dimension larger than its matching number.

Several known codes that beat $\frac{1}{2}$ on graphs of bounded degree, for instance on the Clebsch graph over $\mathbb{Q}$ and on the Hoffman–Singleton graph, come from eigenvalues. For a graph G with adjacency matrix A and a real number $\theta \neq 0$, the matrix $I - A/\theta$ fits G, has coefficients of modulus $1/|\theta|$, and its kernel, the *eigenvalue code* of θ, is the θ-eigenspace of A, of dimension $\mathrm{mult}(\theta)$ (zero if θ is not an eigenvalue). The next proposition, a linear-programming bound in the spirit of Nozaki's bounds for regular graphs [55], shows where this mechanism stops. Let

$$p_0 = 1, \quad p_1(x) = \frac{x}{\sqrt{d}}, \quad p_2(x) = \frac{x^2 - d}{\sqrt{d(d-1)}}, \quad p_3(x) = \frac{x(x^2 - 2d + 1)}{\sqrt{d}\,(d-1)}, \qquad K_n(\theta) = \sum_{k=0}^{n} p_k(\theta)^2;$$

these are the first orthonormal polynomials of the Kesten–McKay law, the spectral measure of the d-regular tree at a vertex [45, 52].

Proposition 6.30 (Eigenvalue codes and girth)**.** Not formalized; a complete proof is given below.
Let $d \geq 3$ and let G be a d-regular graph on N vertices with girth at least g.

(i) For $n \in \{1, 2, 3\}$ with $2n + 1 \leq g$ and every real θ, $\mathrm{mult}(\theta) \leq N/K_n(\theta)$.

(ii) $K_1(\theta) < 2$ if and only if $\theta^2 < d$; and $K_2(\theta) < 2$ if and only if $1 < \theta^2 < d$.

(iii) $K_3(\theta) > 2$ for every real θ. Hence, if $g \geq 7$, every eigenvalue has multiplicity less than $N/2$.

(iv) If $g \geq 7$, no eigenvalue code of G has dimension $N/2$ or more. If $g \geq 5$ and an eigenvalue code has dimension larger than $N/2$, then θ is an integer with $2 \leq |\theta| < \sqrt{d}$, so $d \geq 5$.

Equality holds in (i) for the Petersen graph ($d = 3$, $g = 5$, $n = 2$, $\theta = 1$, multiplicity 5) and for the Hoffman–Singleton graph ($d = 7$, $g = 5$, $n = 2$, $\theta = 2$, multiplicity 28).

Proof (not formalized). Orthonormality. Let A_T be the adjacency operator of the d-regular tree, o its root, S_k the sphere of radius k around o, and $e_k = |S_k|^{-1/2}\mathbf{1}_{S_k}$. Counting neighbours in

adjacent spheres gives $A_T e_0 = \sqrt{d}\, e_1$, $A_T e_1 = \sqrt{d}\, e_0 + \sqrt{d-1}\, e_2$ and $A_T e_k = \sqrt{d-1}\,(e_{k-1} + e_{k+1})$ for $k \geq 2$. A direct computation from these relations gives $p_k(A_T)e_0 = e_k$ for $k \leq 3$; for instance $(A_T^2 - d)e_0 = \sqrt{d(d-1)}\, e_2$. Hence $\langle p_j(A_T)e_0, p_k(A_T)e_0\rangle = \delta_{jk}$ for $j, k \leq 3$.

Moments. For $k < g$ and every vertex v of G, $(A^k)_{vv} = \langle A_T^k e_0, e_0\rangle$. Indeed, let $\pi\colon T \to G$ be a covering of the component of v by the d-regular tree with $\pi(o) = v$. Walks from v lift uniquely to walks from o of the same length, and a lift of a closed walk of length $k < g$ is closed: otherwise the path in T between its distinct endpoints would project to a closed non-backtracking walk of positive length at most k, which contains a cycle of length at most $k < g$. Conversely, closed walks from o project to closed walks from v.

(i) Let $q = \sum_{k \leq n} c_k p_k$ with $c_k = p_k(\theta)/K_n(\theta)$, so that $q(\theta) = 1$. The symmetric matrix $q(A)^2$ has eigenvalue 1 on the θ-eigenspace and nonnegative eigenvalues elsewhere, so $\mathrm{mult}(\theta) \leq \operatorname{tr} q(A)^2$. Since q^2 has degree $2n < g$, the moments give $\operatorname{tr} q(A)^2 = N\|q(A_T)e_0\|^2 = N\sum_k c_k^2 = N/K_n(\theta)$.

(ii) $K_1(\theta) = 1 + \theta^2/d$. With $s = \theta^2$, $d(d-1)(K_2(\theta) - 2) = (d-1)s + (s-d)^2 - d(d-1) = (s-1)(s-d)$.

(iii) With $s = \theta^2$, $d(d-1)^2(K_3(\theta) - 2) = f(s) = s^3 - (3d-1)s^2 + (3d^2 - 4d + 2)s + d(d-1)$, and $f(d+y) = y^3 + y^2 - 2(d-1)y + d(d-1)^2$. The derivative $3y^2 + 2y - 2(d-1)$ has roots $y_\pm = (-1 \pm \sqrt{6d-5})/3$, a local maximum at $y_- < 0$ and a local minimum at $y_+ > 0$. On $[-d, 0]$ the minimum of $f(d+y)$ is therefore attained at an endpoint, where the values are $f(0) = d(d-1) > 0$ and $f(d) = d(d-1)^2 > 0$. On $[0, \infty)$ the minimum is $f(d + y_+) = d(d-1)^2 - y_+^2(2y_+ + 1)$, using $3y_+^2 + 2y_+ = 2(d-1)$. From $y_+ < \sqrt{6d}/3$ we get $y_+^2(2y_+ + 1) < \frac{2d}{3}\left(\frac{2\sqrt{6d}}{3} + 1\right)$, which is less than $d(d-1)^2$ because $(d-1)^2 > \frac{4}{9}\sqrt{6d} + \frac{2}{3}$ for $d \geq 3$ (at $d = 3$, $4 > 2.56$, and the left side grows faster). So $f(s) > 0$ for all $s \geq 0$. The second claim follows from (i) with $n = 3$.

(iv) The first claim is (iii). For the second: A has integer entries, so the roots of the minimal polynomial of θ over $\mathbb{Q}$ are eigenvalues of the same multiplicity; if this multiplicity exceeds $N/2$, the minimal polynomial has degree 1, and θ, an algebraic integer, is an integer. By (i) with $n = 2$ and (ii), $1 < \theta^2 < d$, so $|\theta| \geq 2$ and $d > 4$.

Equality cases. $K_2(1) = 1 + \frac{1}{3} + \frac{4}{6} = 2$ for $d = 3$, and $K_2(2) = 1 + \frac{4}{7} + \frac{9}{42} = \frac{25}{14}$ for $d = 7$, with $50 \cdot \frac{14}{25} = 28$; the multiplicities are those of the Petersen graph (spectrum 3, 1^5, $(-2)^4$) and of the Hoffman–Singleton graph (computed in the proof of Proposition 6.31). □

Why it matters. Proposition 6.30 bounds the girth up to which the spectral mechanism can work: real eigenvalue codes, which give the Clebsch graph a code of dimension 10 over $\mathbb{Q}$ (eigenvalue 1) and the Hoffman–Singleton graph one of dimension 28 (Proposition 6.31), never reach rate $\frac{1}{2}$ at girth 7 or more, at any degree, and beat it at girth 5 or 6 only when $d \geq 5$; whether they beat it at girth 6 at all is open. It is a finite-girth counterpart of Theorem A for eigenvalue codes, with explicit thresholds. It does not see p-torsion: over $\mathbb{F}_p$ the rank of $A - \theta I$ can be smaller than over $\mathbb{Q}$, and over $\mathbb{F}_2$ a code need not come from an eigenvalue at all, and by Corollary 6.22 the finite-field gains that matter here are torsion effects.

Proposition 6.31 (Proposition HS: Hoffman–Singleton beats one half outside characteristic two)**.** Not formalized; a complete proof is given below.

Let A be the adjacency matrix of the Hoffman–Singleton graph (50 vertices, 7-regular, girth 5). Over every field F of characteristic different from 2, the matrix $I - A/2$ fits the graph and has rank at most 22, with equality if the characteristic is 0. So $\ker(I - A/2)$ is a linear storage code over F of dimension at least $28 > 25$, with recovery coefficients $\frac{1}{2}$, and disjoint unions of copies of the graph form a 7-regular family of girth 5 that carries such codes of rate at least 0.56.

Proof (not formalized). By [40] the Hoffman–Singleton graph is strongly regular with parameters $(50, 7, 0, 1)$: adjacent vertices have no common neighbour and non-adjacent vertices exactly one.

Hence $A^2 = 7I + (J - I - A)$, where J is the all-ones matrix, that is, $(A - 2I)(A + 3I) = J$. The all-ones vector is an eigenvector with eigenvalue 7, and J vanishes on its orthogonal complement, which is invariant under A; so the other eigenvalues are 2 and -3, with multiplicities $f + g = 49$ and $7 + 2f - 3g = \operatorname{tr} A = 0$, and $f = 28$. Thus $\operatorname{rank}_{\mathbb{Q}}(2I - A) = 22$. Minors of an integer matrix that vanish over $\mathbb{Z}$ vanish modulo p, so $\operatorname{rank}_{\mathbb{F}_p}(2I - A) \le 22$, and $I - A/2 = (2I - A)/2$ has the same rank whenever 2 is invertible. The rank over a field depends only on its characteristic. □

Remark 6.32 (Computations on the Hoffman–Singleton graph)**.** By exact computation (not used in any proof), the rank of $2I - A$ is 22 over $\mathbb{F}_3$ and $\mathbb{F}_7$ but 21 over $\mathbb{F}_5$, an example of the p-torsion that Proposition 6.30 does not see. Over $\mathbb{F}_2$ the matrix $I + A$ has rank 29, giving only dimension 21, and we do not know whether the Hoffman–Singleton graph carries an $\mathbb{F}_2$-linear storage code of dimension larger than 25.

Why it matters. Theorem S needs maximum degree 3 (and no triangles), and Theorem A needs local bipartiteness. One might hope that large girth alone gives the bound $\frac{1}{2}$ at every bounded degree, over every field. Proposition 6.31 shows that girth 5 does not suffice: at degree 7, over $\mathbb{Q}$ and over every field of odd characteristic, there are linear codes of rate 0.56 with bounded coefficients. By Proposition 6.30, eigenvalue codes need girth at most 6 to beat $\frac{1}{2}$; whether, for each d, some girth forces rate at most $\frac{1}{2} + o(1)$ for all linear codes over a fixed finite field is open, and in characteristic 2 even the Hoffman–Singleton graph is undecided.

Taken together, on side information of bounded degree $d \ge 4$ that is locally bipartite, an $\mathbb{F}_p$-linear code with a fixed number of symbols per vertex that beats $\frac{1}{2}$ by a constant must have p-torsion of linear rank in every bounded integer lift of its check matrix (Corollary 6.22); in characteristic 2 it gains nothing from the identity $(I + W)^2 = 0$ once the girth is at least 5 (Lemma 6.24); it does not come from a real eigenvalue code once the girth is at least 7 (Proposition 6.30); and the local necessary conditions of a smallest counterexample cannot refute its existence inside tree-like regions (Proposition 6.28). These are design constraints for constructions, and they say where a characteristic-p proof of the crux would have to work.

6.10 Targets, Mycielskians and Question G

By Theorem H (Theorem 6.5), a homomorphism from side information to a good graph T certifies the rate bound $\frac{1}{2}$ for every code on the side information, linear or not. If the homomorphism is defined only after the deletion of a vertex set U, conditioning on the symbols at U weakens the bound by at most $|U|/2$ symbols: for a valid polymatroid h, the function $S \mapsto h(S \cup U) - h(U)$ on subsets of $V \setminus U$ is a valid polymatroid for $G - U$, so $h(V) \le h(U) + \frac{1}{2}\sum_{v \notin U} h(\{v\}) \le \frac{1}{2}\sum_v h(\{v\}) + \frac{1}{2}\sum_{u \in U} h(\{u\})$. A single good target that receives the side information of the crux after the deletion of $o(N)$ vertices would therefore prove the crux for all codes. Two facts constrain the targets. Good graphs are triangle-free: the triangle carries the code $\{(a, b, a + b)\}$ of rate $\frac{2}{3}$, and a graph that contains a triangle receives it (Theorem 6.5). And homomorphisms do not decrease the fractional chromatic number χ_f, the least total weight of a family of weighted independent sets that covers every vertex with weight at least 1, because preimages of independent sets are independent. So side information H with a deleted set U needs a target with $\chi_f(T) \ge \chi_f(H - U) \ge (|V(H)| - |U|)/\alpha(H)$, where α is the independence number.

The classical way to raise χ_f while keeping a graph triangle-free is Mycielski's construction. For a graph G on V, the *Mycielskian* $\mathrm{M}(G)$ has vertex set $V \cup V' \cup \{z\}$, where $V' = \{v' : v \in V\}$, and edges uv, uv' and $u'v$ for $uv \in E(G)$, and $v'z$ for $v \in V$. Thus $\mathrm{M}(K_2)$ is the 5-cycle and

$\mathrm{M}^2(K_2)$ is the Grötzsch graph. The Mycielskian of a triangle-free graph is triangle-free [53], and $\chi_f(\mathrm{M}(G)) = \chi_f(G) + 1/\chi_f(G)$ [46], so $\chi_f(\mathrm{M}^k(K_2)) \to \infty$.

Lemma 6.33 (Homomorphisms into a Mycielskian)**.** Not formalized; a complete proof is given below.

For graphs H and G, H admits a homomorphism to $\mathrm{M}(G)$ if and only if H has an independent set Z such that the set $N_H(Z)$ of vertices outside Z with a neighbour in Z is independent and $H - Z$ admits a homomorphism to G.

The 'if' direction is the case $t = 1$ of [36, Lemma 2.3]; we include the short proof of both directions.

Proof (not formalized). Let $\varphi\colon H \to \mathrm{M}(G)$, and put $Z = \varphi^{-1}(z)$ and $Y = \varphi^{-1}(V')$. Both are independent, because z has no loop and V' is independent in $\mathrm{M}(G)$, and $N_H(Z) \subseteq Y$ because z is adjacent only to V'. Define ψ on $H - Z$ by $\psi(w) = \varphi(w)$ if $\varphi(w) \in V$ and $\psi(w) = v$ if $\varphi(w) = v'$. An edge of $H - Z$ has at most one end in Y; if it has none, φ maps it to an edge of G; if it has one, φ maps it to an edge uv' with $uv \in E(G)$, and ψ maps it to uv. So ψ is a homomorphism to G. Conversely, given Z and $\psi\colon H - Z \to G$, map Z to z, each $w \in N_H(Z)$ to $\psi(w)'$, and every other vertex w to $\psi(w)$. There are no edges inside Z or inside $N_H(Z)$, and none between Z and the vertices outside $Z \cup N_H(Z)$; edges between Z and $N_H(Z)$ go to edges zv'; and every other edge ww'' goes to $\psi(w)\psi(w'')$, $\psi(w)'\psi(w'')$ or $\psi(w)\psi(w'')'$, which are edges of $\mathrm{M}(G)$ because $\psi(w)\psi(w'') \in E(G)$. □

Corollary 6.34 (Odd cycle transversals)**.** Not formalized; a complete proof is given below.

If H is triangle-free and $H - X$ is bipartite for a set X of k vertices, then H admits a homomorphism to $\mathrm{M}^k(K_2)$.

Proof (not formalized). Induction on k; a bipartite graph maps to K_2. For $k \geq 1$ pick $x \in X$. The set $Z = \{x\}$ is independent, $N_H(x)$ is independent because H is triangle-free, and $H - x$ is triangle-free and becomes bipartite after the deletion of $X \setminus \{x\}$, so $H - x \to \mathrm{M}^{k-1}(K_2)$ by induction. Lemma 6.33 gives $H \to \mathrm{M}^k(K_2)$. (A similar iteration is used in [36].) □

Proposition 6.35 (Mycielskians leave the good graphs)**.** Not formalized; a complete proof is given below.

(i) The Clebsch graph admits a homomorphism to $\mathrm{M}^6(K_2)$. Hence $\mathrm{M}^6(K_2)$ is not good.

(ii) K_2 and $\mathrm{M}(K_2)$, the 5-cycle, are good. Hence there is $j \in \{1, \dots, 5\}$ such that $\mathrm{M}^j(K_2)$ is good and $\mathrm{M}^{j+1}(K_2)$ is not: the Mycielski construction does not preserve goodness.

(iii) No good graph receives every triangle-free graph of maximum degree 5. More generally, if $T_1, T_2, \dots$ are graphs and every finite triangle-free graph admits a homomorphism to some T_k, then some T_k is not good; by Corollary 6.34 this applies to the graphs $\mathrm{M}^k(K_2)$, which are universal for triangle-free graphs by a theorem of Cropper, Gyárfás and Lehel [26].

Proof (not formalized). (i) The Clebsch graph Γ is triangle-free and 5-regular on 16 vertices. For an edge uv, the neighbourhoods $N(u)$ and $N(v)$ are disjoint independent sets (as Γ is triangle-free), so $\Gamma[N(u) \cup N(v)]$ is bipartite on 10 vertices and the remaining set X of 6 vertices has $\Gamma - X$ bipartite. By Corollary 6.34, $\Gamma \to \mathrm{M}^6(K_2)$. The Clebsch graph is not good: by Corollary 6.11 it has guessing number $10 > 8$, while a good graph on 16 vertices has guessing number at most 8 by Theorem 6.3. By Theorem 6.5, a graph that receives a graph that is not good is not good. (ii) The 5-cycle is good (Proposition 6.7), and K_2 maps to it, so K_2 is good by Theorem 6.5. With (i), the least k for which

$\mathrm{M}^k(K_2)$ is not good satisfies $2 \le k \le 6$; take $j = k-1$. (iii) The Clebsch graph is triangle-free of maximum degree 5, and a graph that receives it is not good. □

Remark 6.36 (Which iterate fails; computational)**.** The Grötzsch graph $\mathrm{M}^2(K_2)$ is good by an exact certificate, a nonnegative rational combination of elemental Shannon inequalities found by computer and checked in exact arithmetic; the certificate is not published here, not formalized and not used in any proof. With it, $j \in \{2,3,4,5\}$ in Proposition 6.35(ii). A SAT computation (not used in any proof) indicates that the Clebsch graph does not map to $\mathrm{M}^5(K_2)$, so the argument of (i) cannot locate the first iterate that is not good; which iterate it is remains open. At the level of Shannon inequalities the construction may fail at the first step: numerically, by a floating-point linear program without an exact certificate, the Mycielskian of the circular clique $K_{8/3}$ (which is good by Proposition 6.8) has Shannon bound about $7/13 > \frac{1}{2}$.

Why it matters. Mycielski's construction is the standard way to build triangle-free graphs of large fractional chromatic number, and its iterates are universal for triangle-free graphs [26]; Corollary 6.34 refines this, bounding the number of iterations by the size of an odd cycle transversal. Proposition 6.35 shows that this universality forces them out of the good graphs. Targets of growing χ_f for the homomorphism route to the crux must therefore be built differently, and they must avoid receiving the Clebsch graph and the other graphs known not to be good.

Remark 6.37 (Good graphs with $\chi_f > 3$; computational)**.** Write $C_n(S)$ for the Cayley graph of $\mathbb{Z}_n$ with connection set $\pm S$; for vertex-transitive graphs $\chi_f = N/\alpha$. Exact rational certificates, nonnegative combinations of elemental Shannon inequalities found by computer and checked in exact arithmetic, show that the triangle-free circulants $C_{13}(1,5)$ (4-regular, $\chi_f = 13/4$), $C_{16}(1,3,8)$ (5-regular, $\chi_f = 16/5$) and $C_{20}(1,6,10)$ (5-regular, $\alpha = 6$, $\chi_f = 10/3$) are good. The certificate for $C_{20}(1,6,10)$ consists of 777 inequalities; $10/3$ is the largest value of χ_f among good graphs known to us. The certificates are not published here, and none of this is formalized or used in any proof. These examples show that $\chi_f \le 3$ is no barrier, and that χ_f does not determine goodness: $C_{16}(1,3,8)$ has the same number of vertices, degree, independence number and fractional chromatic number as the Clebsch graph, which is not good. Nearby circulants are not good: for instance $C_{20}(2,5,6)$, with $\chi_f = 10/3$, carries the $\mathbb{F}_2$-linear code $\ker(I+A)$ of dimension $12 > 10$.

Remark 6.38 (Cayley graphs of elementary abelian 2-groups; partly computational)**.** Proof sketch only: every non-bipartite Cayley graph $\Gamma = \mathrm{Cay}(\mathbb{Z}_2^k, S)$ is not good. The graph Γ is bipartite if and only if some linear functional equals 1 on S, and by linear duality over $\mathbb{F}_2$ this fails exactly when some subset of S with an odd number of elements sums to 0. A minimal such subset S' consists of $m-1$ linearly independent elements and their sum, with m odd (and $m \ge 5$ if Γ is triangle-free), so $\mathrm{Cay}(\langle S' \rangle, S')$ is the folded m-cube, a subgraph of Γ. In the group algebra the matrix $I+A$ of a folded cube satisfies $(I+A)^2 = 0$ over $\mathbb{F}_2$, so its kernel has dimension at least half; exact computation shows that the dimension is $(2^{m-1} + 2^{(m-1)/2})/2$ for $m \le 11$, and we have not proved this formula in general. Where it holds, the folded m-cube is not good, and neither is Γ, by Theorem 6.5. The argument is complete for $k \le 10$, where $m \le 11$; it would exclude, as targets, all non-bipartite triangle-free Cayley graphs of elementary abelian 2-groups, a family that contains the Clebsch graph and the folded cubes.

These examples lead to the following question.

Question 6.39 (Question G: the fractional chromatic number of good graphs)**.** *Is χ_f unbounded on good graphs? That is, is $\sup\{\chi_f(T) : T \text{ a good graph}\}$ infinite?*

Good graphs are triangle-free, and triangle-free graphs have unbounded χ_f; the question is whether goodness bounds it. The analogous question for digraphs, with the fractional dichromatic number (fractional covers by vertex sets inducing acyclic subgraphs), is also open.

Why it matters. A negative answer, $\chi_f(T) \le C$ for every good graph T, would rule out the homomorphism route for every family of side information whose underlying graphs keep fractional chromatic number above C after the deletion of $o(N)$ vertices, for instance families whose independence ratio stays below $1/C$. A positive answer is necessary for the route but not sufficient, since the targets must also receive the side information. With the methods available to us, certifying a single target by the Shannon linear program is limited to about 20 vertices, while a triangle-free graph with large χ_f must be large, so Question G needs a structural argument in either direction.

7 What a linear-time transposer would need

Our aim is to decide (T), and we would prefer a construction. This section collects what the results above require of any linear-time transposer, and of any oblivious one in particular. It contains no new theorems: every claim is one of the results above or the first two papers, a citation, or explicitly labelled as a heuristic.

7.1 Necessary conditions

A machine witnessing (T) must have all of the following properties, in addition to those of the first two papers [18, 20] ($\Omega(\log d)$ passes, folding macroscopic motion, and a coding gain of order $\log d$ over routing on the block network of a slightly modified machine; within the abstract local-coding models of the second paper, a fixed fractional saving is also excluded when the interaction graph has treewidth $o(d)$).

(C1) *Curved motion.* In every macroscopic limit, the work heads travel a total distance of at least half a block unit, a linear number of cells, along motion that is not countably-piecewise affine (Theorem Cq, Theorem 3.14), and a common change of clock does not remove it (Corollary 3.21). The curved motion must be head-dependent, and at least two work-head limits must be non-monotone and not all equal (Corollary 3.24). Curved monotone expanders of the Bourgain–Yehudayoff type [17] remain possible.

(C2) *Coding on bounded-depth side information, or depth-robust dependencies.* For an oblivious machine, Theorem D (Theorem 4.10) leaves two ways out. Either some genie of εL bits cuts the time–space dependency DAG to bounded depth, and then the side information of the cones admits index codes of broadcast length εL, defeating the crux; or no such genie exists, and the walks of the work heads are depth-robust. Home-based heads always admit such a genie (Theorem B, Theorem 4.20), so for them only the first way is open (Corollary B, Corollary 4.23), and by the trade-off (Proposition 4.13) the depth is at least about β/ε.

(C3) *Codes that beat the pairing, of a particular kind.* If, as in the hardest instances we know, the relevant side information is sparse, symmetric and locally tree-like, a construction needs storage codes of rate at least $\frac{1}{2}+\delta$ on it, for a fixed $\delta > 0$ (a heuristic reading of Question 6.1). The classes of codes that our results exclude are listed in Section 7.3. Over a finite field, with a fixed number of symbols per vertex, a gain must come from p-torsion (Corollary 6.22). The binary code of rate $5/8$ on the Clebsch graph (Corollary 6.11) is not a torsion effect: it is the reduction of a rational eigenvalue code with coefficients 1 (Remark 6.23), a mechanism that

Corollary 6.22 rules out on locally bipartite families of bounded degree. In odd characteristic at most one of the two signed versions $I \pm W$ of a code can gain (Proposition 6.25).

(C4) *If the dependencies are depth-robust: the deep-nesting template.* The only candidate we know for depth-robust walk pairs is the deep-nesting pair (Definition 5.9). Proof sketches indicate that per-head band rules fail already on one deeply nested walk (Remarks 5.6 and 5.7), although one such walk alone is easy to cut (Lemma 5.8); and with one level of nesting, labels exact on one head pay a price per hop on the cover independent of the depth (Theorem E_1, Theorem 5.22). Whether the pairs are depth-robust as m grows is open (Conjecture 5.25).

7.2 An architecture compatible with these conditions

We describe, as a design and not as a result, how the first paper's geometric recursion [18, §13] would have to look in the light of (C1)–(C4). The recursion runs $\Theta(\log d)$ levels of geometrically decreasing volume, each with $O(1)$ passes of an encoder and a decoder. Condition (C1) forces the heads of each level to follow curved rulers, for example near-identity Möbius maps realizing a monotone expander, and the second paper excludes the affine variants [20]. The encoder of each level must then save a constant fraction of the data by coding across the expander's relations. By (C3), the code it uses must be nonlinear, or linear over a finite field, with a p-torsion gain if it has a fixed number of symbols per vertex, and a scalar linear code needs degree at least 4; its side-information graph must not map to any good target. Codes that beat $\frac{1}{2}$ exist on single graphs with short odd cycles, such as the Clebsch graph (girth 4) and, outside characteristic 2, the Hoffman–Singleton graph (girth 5, Proposition 6.31). We do not know any family of bounded degree and growing odd girth that carries one; such a family would answer Question 6.1 in the negative. For an oblivious design, Corollary B applies if its work heads are all home-based, and the design then stands or falls with the crux; otherwise, if the crux holds, its dependency structure must be depth-robust, and the deep-nesting pair is the template.

7.3 Mechanisms that fail

The following classes of constructions fail for reasons proved here, in addition to those listed in the first two papers.

- *Home-based motion of all work heads with legs of boundedly many constant-rate runs*, obliviously or adaptively, with any number of legs at any scales (Theorem C).
- *One expanding tape, or a common change of clock*: no single work tape can carry the expansion, and a common reparametrization of time adds nothing (Corollaries 3.20 and 3.21).
- *Stroke trees for all work heads*, the recursive schedules of Definition 3.22: their limits are affine wherever the heads move (Proposition 3.23).
- *Linear codes with bounded real or complex coefficients*, scalar or with a bounded number of symbols per vertex, on locally bipartite side information of bounded degree (Theorem A and Proposition 6.18), *scalar linear codes of degree* 3 on triangle-free side information (Theorem S), and *all codes* on side information that is C_5-, $K_{14/5}$- or Petersen-colourable (Corollary 6.9).

8 Roadmap and open problems

8.1 Where the decision of (T) stands

For oblivious machines, Theorem D reduces ¬(T) to two statements, and this paper narrows both; the last row of the table records the unconditional results on motion.

	proved here	**open**
Depth reduction (TCP)	one walk (by planarity) and all couplings of home-based walks (Theorem B); packing certificates cannot show depth-robustness (Proposition 5.14); on the deep-nesting pair with one level of nesting, labels exact on one head pay a fixed price (Theorems E, E_1)	TCP(2) at fixed ε; its sharpest instance is the deep-nesting pair (Conjecture 5.25), where per-head band rules appear to fail (sketch only, Remarks 5.6 and 5.7)
The crux (index coding on bounded-depth side information)	undirected form, rate $\frac{1}{2}$: bounded complex linear codes on locally bipartite graphs (Theorem A); scalar linear codes over every field at degree 3 (Theorem S); all codes on C_5-, $K_{14/5}$- and Petersen-colourable graphs (Corollary 6.9)	on locally bipartite graphs, scalar linear codes over finite fields at degree at least 4 (vector-linear ones already at degree 3) and nonlinear codes at degree at least 3; the directed, depth-bounded form
Motion	machines whose work heads are all home-based with legs of boundedly many constant-rate runs fail unconditionally (Theorem C); half a block unit of travel must be curved (Theorem Cq)	whether curved motion of the Bourgain–Yehudayoff type can be coupled to a coding scheme

For home-based oblivious machines only the crux remains (Corollary B), and only in a weak form: a rate $\rho(D)$ with $\rho(D)\log D \to \infty$ suffices. Theorem C also covers adaptive machines; for them the decomposition needs further hypotheses, which we do not treat here.

8.2 Open problems

1. *Decide (T).* Construct a machine transposing $d \times d$ matrices with $\lceil \log_2 d \rceil$-bit entries in time $O(d^2 \log d)$, or prove that none exists. The first paper's frozen checker accepts either answer [18].

2. *Depth reduction for interacting walks* (Problem 5.1). Does every coupling of k walks admit an εT-edge cut that leaves depth $D(k, \varepsilon)$? A positive answer for every k, with the crux, refutes oblivious (T) (Theorem D). The first open case is $k = 2$, and its sharpest instance is the deep-nesting pair. There, the missing step is a robust bound, logarithmic in the depth, on how many levels a label of small cost can serve on both heads at once (Conjecture 5.25, stated for labels on a cyclic cover and for a suitable odd shift); the same statement for labels on the walk DAG would make these pairs depth-robust as $m \to \infty$, give TCP(2) a negative answer, and point constructions toward deep nesting.

3. *The undirected crux* (Question 6.1). Do storage codes on locally bipartite graphs of bounded degree have rate at most $\frac{1}{2} + o(1)$? Open for scalar linear codes over finite fields at degree at least 4 (and for vector-linear ones already at degree 3), where a gain would need p-torsion (Corollary 6.22), and for nonlinear codes at every degree at least 3. Arguments that use only the local necessary conditions of a smallest counterexample inside tree-like regions cannot give rate $\frac{1}{2} + o(1)$ at degree at least 4 (Proposition 6.28), and eigenvalue codes cannot reach rate $\frac{1}{2}$ at girth at least 7 (Proposition 6.30).

4. *Targets* (Question 6.39). Is the fractional chromatic number unbounded on good graphs? Theorem H proves rate $\frac{1}{2}$ on every graph that maps to a good target, so a family of good targets with growing fractional chromatic number would extend Corollary 6.9 to side information that needs more colours; a negative answer, $\chi_f \le C$ on all good graphs, would close this route for side information whose fractional chromatic number stays above C after the deletion of $o(N)$ vertices. Mycielski's construction does not preserve goodness (Proposition 6.35).

5. *Cubic graphs of large girth.* Do all storage codes on cubic graphs of large girth have rate at most $\frac{1}{2}$? By Corollary 6.10 this would follow from a positive answer to Nešetřil's Pentagon Problem [54], or from a proof that such graphs have circular chromatic number at most 14/5, which is open. For scalar linear codes, Theorem S answers it.

6. *The crux in full.* Prove an index-coding lower bound of any rate $\rho(D) \gg \beta/D$ for the side-information systems of bounded-depth cones of oblivious schedules (Definition 4.8), linear or nonlinear; by Proposition 4.13, no depth reduction can use a smaller rate. For home-based heads, where depth reduction is proved, the rate $\rho(D) \gg 1/\log D$ suffices (Corollary B).

7. *Formalization.* Formalize the results of Sections 3 to 5 and 6.7 to 6.10 that are not formalized.

A Index of Lean declarations

Every declaration of this paper's library cited in the text, grouped by the Lean module that contains it, relative to the namespace `MotionDepthCodes`. All of them depend only on the axioms `propext`, `Classical.choice` and `Quot.sound` (Appendix B). The declarations of the first paper's library cited in Section 3 are verified there [18].

`MotionDepthCodes.CircularClique` `circularClique`, `circularCliqueHom`, `circularClique_adj_of_lt`, `cycleToCircularClique`, `rateBound_circularClique_14_5`, `rateBound_circularClique_of_le`, `rateBound_cycleGraph_odd`.

`MotionDepthCodes.Clebsch` `clebschCode`, `clebschCode_card`, `clebschCode_isStorageCode`, `clebschGraph`, `clebschGraph_adj_iff`, `exists_clebsch_storageCode`.

`MotionDepthCodes.ClebschCertificate` `clebschCode_eq`, `clebsch_card_le`, `clebsch_guessingNumber`, `clebsch_rateBound_iff`, `finrank_ker_clebsch`, `mem_ker_clebsch_iff`, `rank_clebsch`, `rateBound_clebsch`, `rateBound_clebschGraph`.

`MotionDepthCodes.DepthLabels` `card_descents_eq`, `card_violated_eq`, `congr_of_exact`, `cost_ge_of_negative_cycles`, `cycleDefect`, `cycle_errors_congr`, `le_of_modEq_neg`, `lemma_Zq`.

MotionDepthCodes.MatchingForm exists_pattern_matchingNumber_le_finrank_ker, finrank_ker_le_matchingNumber, finrank_ker_le_matchingNumber_of_degree_le_three, matchingNumber_eq_min_barrier, minrank_eq_card_sub_matchingNumber.

MotionDepthCodes.Pentagon nbhd_cycleGraph_five, pentagonNbhd, rateBound_cycleGraph_five, rateBound_pentagon.

MotionDepthCodes.Petersen petersenGraph, petersenGraph_adj_iff, petersenKneser_card, petersenKneser_injective, rateBound_petersenGraph.

MotionDepthCodes.Polymatroid IsGood, IsGood.of_hom, IsPolymatroid, IsPolymatroid.eq_of_closure, IsPolymatroid.insert_eq_of_subset, IsPolymatroid.pushRank, IsValid, IsValid.pushRank, RateBound, RateBound.mono, RateBound.of_digraphHom, RateBound.of_graphHom, RateBound.of_hom, pushRank, pushRank_univ, sum_pushRank_singleton_le.

MotionDepthCodes.RateHalf IsGood.card_sq_le, IsGood.codeEntropy_le, card_sq_le_of_rateBound_half.

MotionDepthCodes.StorageCode IsRecoverable, IsStorageCode, codeEntropy, codeEntropy_isPolymatroid, codeEntropy_isValid, codeEntropy_univ_le_of_rateBound, codeEntropy_univ_uniform, log_card_le_of_rateBound.

MotionDepthCodes.Subcubic IsLinearStorageCode, IsLinearStorageCode.card_sq_le, IsLinearStorageCode.exists_coeffs, IsLinearStorageCode.exists_pattern, IsLinearStorageCode.finrank_le_half, RecTriangleFree, Star, ThmSHyp, exists_linearStorageCode_of_matching, exists_pattern_finrank_ker_ge_of_matching, finrank_ker_le_barrier, finrank_ker_le_half_of_degree_le_three, finrank_ker_le_heavy, finrank_subCode_biUnion_le, finrank_subCode_le_half, finrank_subCode_union_le, isLinearStorageCode_of_le_ker, isStorageCode_iff_isLinearStorageCode, rank_ge_half_of_degree_le_three, star_of_degree_le_three, thmSHyp_of_cliqueFree.

MotionDepthCodes.Symmetry IsAut, IsPolymatroid.symmetrize, IsValid.symmetrize, RateBound.of_invariant, sum_symmetrize_singleton, symmetrize, symmetrize_invariant, symmetrize_univ.

Also named in the text emod_of_bounds, nbhd, outNbhd.

B Building and verifying the formalization

The Lean project is the directory `lean/` next to this paper. It uses Lean and Mathlib, both at version v4.34.1 (`lean-toolchain`, `lakefile.toml`, `lake-manifest.json`), and imports the library of the second paper [20], which imports that of the first [18], and the repository's shared library `lean-machine`, through relative paths. Keep the repository's directory layout, so that these sibling directories stay in place. `lean-machine` contains a port of J. P. Traverso Gianini's formalization of the Tutte–Berge formula [62], used for the matching form of Theorem S; its README records the source, the commit and the changes.

Build. With `elan` installed, run in `lean/`:

```
lake exe cache get && lake build
```

The first command downloads the compiled Mathlib; the second compiles the library and the parts of the earlier libraries that it uses. Before the first build, fetch the second paper's pinned external dependency as its README describes (`python3 verification/fetch_openai_metric.py`, run in that paper's directory).

Axioms. The file `PaperAxioms.lean`, generated from the LaTeX sources of this paper by `tools/lean\_index.py`, runs `#print axioms` on every declaration of this library named here (Appendix A):

```
lake env lean PaperAxioms.lean
```

Every declaration depends at most on `propext`, `Classical.choice` and `Quot.sound`. No Lean file of this library uses `sorry`, `admit` or `native_decide`, or declares an `axiom`. The heartbeat limit is raised once, for the Clebsch certificate, which is a long but routine computation.

Certificates. The certificates of Sections 6.4 and 6.5 are Lean proofs generated by `tools/certificates.py` (which needs `numpy` and `scipy`). For example,

```
python3 tools/certificates.py clebsch 5 8 clebsch clebschNbhd -all-pairs
```

prints the automorphisms and the proof used in `ClebschCertificate.lean`. Different solver versions may produce different certificates; any certificate printed is checked by Lean, and the files in the library are the record.

Where things are. `Polymatroid.lean` (polymatroids, closure, Theorem H), `StorageCode.lean` (entropic polymatroids of codes), `Symmetry.lean` (symmetrization), `Pentagon.lean`, `Petersen.lean` and `CircularClique.lean` (good targets; the Bondy–Hell homomorphisms and odd cycles), `RateHalf.lean` (Corollary H2), `Clebsch.lean` and `ClebschCertificate.lean` (Corollary H4), `Subcubic.lean` (Theorem S, linear storage codes and the matching code), `MatchingForm.lean` (the matching form of Theorem S), `DepthLabels.lean` (Lemma Zq, the exact cases of the chain and transfer identities, and Theorem E), and `FinSums.lean` (sums over small index sets used by the certificates).

References

[1] Micah Adler, Nicholas J. A. Harvey, Kamal Jain, Robert D. Kleinberg, and April Rasala Lehman. On the capacity of information networks. In *Proceedings of the Seventeenth Annual ACM-SIAM Symposium on Discrete Algorithms (SODA 2006)*, pages 241–250, 2006.

[2] David Aldous and Russell Lyons. Processes on unimodular random networks. *Electronic Journal of Probability*, 12:1454–1508, 2007. Paper no. 54; arXiv:math/0603062.

[3] Fatemeh Arbabjolfaei and Young-Han Kim. Three stories on a two-sided coin: Index coding, locally recoverable distributed storage, and guessing games on graphs. In *53rd Annual Allerton Conference on Communication, Control, and Computing*, 2015. arXiv:1511.01050.

[4] Rahil Baber, Demetres Christofides, Anh N. Dang, Emil R. Vaughan, and Søren Riis. Graph guessing games and non-Shannon information inequalities. *IEEE Transactions on Information Theory*, 63(7):4257–4267, 2017. arXiv:1410.8349, doi:10.1109/TIT.2016.2628819.

[5] Stefan Banach. Sur les lignes rectifiables et les surfaces dont l'aire est finie. *Fundamenta Mathematicae*, 7:225–236, 1925.

[6] Ziv Bar-Yossef, Yitzhak Birk, T. S. Jayram, and Tomer Kol. Index coding with side information. *IEEE Transactions on Information Theory*, 57(3):1479–1494, 2011.

[7] Alexander Barg and Gilles Zémor. High-rate storage codes on triangle-free graphs. *IEEE Transactions on Information Theory*, 68(12):7787–7797, 2022. doi:10.1109/TIT.2022.3191309.

[8] Bachir Bekka, Pierre de la Harpe, and Alain Valette. *Kazhdan's Property (T)*, volume 11 of *New Mathematical Monographs*. Cambridge University Press, 2008. doi:10.1017/CBO9780511542749.

[9] Itai Benjamini and Oded Schramm. Recurrence of distributional limits of finite planar graphs. *Electronic Journal of Probability*, 6:1–13, 2001. Paper no. 23. doi:10.1214/EJP.v6-96.

[10] Claude Berge. Sur le couplage maximum d'un graphe. *Comptes Rendus de l'Académie des Sciences, Paris*, 247:258–259, 1958.

[11] Y. Berliner and Michael Langberg. Index coding with outerplanar side information. In *Proceedings of the 2011 IEEE International Symposium on Information Theory (ISIT)*, 2011. doi:10.1109/ISIT.2011.6034247.

[12] Yitzhak Birk and Tomer Kol. Coding on demand by an informed source (ISCOD) for efficient broadcast of different supplemental data to caching clients. *IEEE Transactions on Information Theory*, 52(6):2825–2830, 2006.

[13] Anna Blasiak, Robert Kleinberg, and Eyal Lubetzky. Broadcasting with side information: Bounding and approximating the broadcast rate. *IEEE Transactions on Information Theory*, 59(9):5811–5823, 2013. Preprint: Index coding via linear programming, arXiv:1004.1379.

[14] Béla Bollobás. The independence ratio of regular graphs. *Proceedings of the American Mathematical Society*, 83(2):433–436, 1981. doi:10.1090/S0002-9939-1981-0624948-6.

[15] J. A. Bondy and Pavol Hell. A note on the star chromatic number. *Journal of Graph Theory*, 14(4):479–482, 1990. doi:10.1002/jgt.3190140412.

[16] Charles Bordenave, Marc Lelarge, and Justin Salez. The rank of diluted random graphs. *Annals of Probability*, 39(3):1097–1121, 2011. doi:10.1214/10-AOP567.

[17] Jean Bourgain and Amir Yehudayoff. Expansion in $\mathrm{SL}_2(\mathbb{R})$ and monotone expanders. *Geometric and Functional Analysis*, 23(1):1–41, 2013. doi:10.1007/s00039-012-0200-9.

[18] Hans Brende and Claude Opus 5.5 (Anthropic). Matrix transposition and the tape coding conjecture. Manuscript and Lean formalization, repository HansBrende/papers, October 2026. Pinned revision 09759470fade9736c13a7f0047409362fe303d93. URL: https://github.com/HansBrende/papers/tree/09759470fade9736c13a7f0047409362fe303d93/Matrix-Transposition-and-the-Tape-Coding-Conjecture.

[19] Hans Brende and GPT-6 Astra (OpenAI). Bit and word transposition on tapes. Manuscript and Lean formalization, repository HansBrende/papers, October 2026. Pinned revision 09759470fade9736c13a7f0047409362fe303d93. URL: https://github.com/HansBrende/papers/tree/09759470fade9736c13a7f0047409362fe303d93/Bit-and-Word-Transposition-on-Tapes.

[20] Hans Brende and GPT-6 Astra (OpenAI). Local coding barriers and the limits of tape geometry for matrix transposition. Manuscript and Lean formalization, repository HansBrende/papers, October 2026. Pinned revision 09759470fade9736c13a7f0047409362fe303d93. URL: https://github.com/HansBrende/papers/tree/09759470fade9736c13a7f0047409362fe303d93/Local-Coding-Barriers-and-the-Limits-of-Tape-Geometry-for-Matrix-Transposition.

[21] Peter J. Cameron, Anh N. Dang, and Søren Riis. Guessing games on triangle-free graphs, 2014. Preprint, version 2 (2015); published without Problem 5.11 as [22]. arXiv:1410.2405.

[22] Peter J. Cameron, Anh N. Dang, and Søren Riis. Guessing games on triangle-free graphs. *Electronic Journal of Combinatorics*, 23(1):Paper 1.48, 2016. doi:10.37236/4731.

[23] Demetres Christofides and Klas Markström. The guessing number of undirected graphs. *Electronic Journal of Combinatorics*, 18(1):Paper 192, 2011.

[24] Bruno Codenotti, Pavel Pudlák, and Giovanni Resta. Some structural properties of low-rank matrices related to computational complexity. *Theoretical Computer Science*, 235(1):89–107, 2000.

[25] Alain Connes, Jacob Feldman, and Benjamin Weiss. An amenable equivalence relation is generated by a single transformation. *Ergodic Theory and Dynamical Systems*, 1(4):431–450, 1981.

[26] M. Cropper, András Gyárfás, and Jenő Lehel. Hall ratio of the Mycielski graphs. *Discrete Mathematics*, 306(16):1988–1990, 2006. doi:10.1016/j.disc.2005.09.020.

[27] Amir Daneshgar and Meysam Madani. On the odd girth and the circular chromatic number of generalized Petersen graphs. *Journal of Combinatorial Optimization*, 33(3):897–923, 2017. Preprint arXiv:1501.06551, Problem 2. doi:10.1007/s10878-016-0013-0.

[28] Louis Deaett. The minimum semidefinite rank of a triangle-free graph. *Linear Algebra and its Applications*, 434(8):1945–1955, 2011.

[29] Matt DeVos and Robert Šámal. High-girth cubic graphs are homomorphic to the Clebsch graph. *Journal of Graph Theory*, 66(3):241–259, 2011. doi:10.1002/jgt.20580.

[30] Martin Dietzfelbinger and Wolfgang Maass. The complexity of matrix transposition on one-tape off-line Turing machines with output tape. *Theoretical Computer Science*, 108(2):271–290, 1993.

[31] Martin Dietzfelbinger, Wolfgang Maass, and Georg Schnitger. The complexity of matrix transposition on one-tape off-line Turing machines. *Theoretical Computer Science*, 82(1):113–129, 1991.

[32] R. Dougherty, S. Jackson, and A. S. Kechris. The structure of hyperfinite Borel equivalence relations. *Transactions of the American Mathematical Society*, 341(1):193–225, 1994. doi:10.1090/S0002-9947-1994-1149121-0.

[33] Javad B. Ebrahimi and Mahdi Jafari Siavoshani. On index coding and graph homomorphism. In *IEEE Information Theory Workshop (ITW)*, 2014. arXiv:1408.7019.

[34] Herbert Federer. *Geometric Measure Theory*, volume 153 of *Die Grundlehren der mathematischen Wissenschaften*. Springer-Verlag, New York, 1969.

[35] Li Gao, Sander Gribling, and Yinan Li. On a tracial version of Haemers bound. *IEEE Transactions on Information Theory*, 68(10):6585–6604, 2022. doi:10.1109/TIT.2022.3176935.

[36] Lior Gishboliner, Eoin Hurley, and Yuval Wigderson. Asymmetric results about graph homomorphisms, 2025. arXiv:2502.20278.

[37] Willem Haemers. On some problems of Lovász concerning the Shannon capacity of a graph. *IEEE Transactions on Information Theory*, 25(2):231–232, 1979.

[38] David Harvey and Joris van der Hoeven. Integer multiplication is at least as hard as matrix transposition. arXiv:2503.22848; also in Proc. FOCS 2025, pp. 1366–1381 (numbering as in arXiv), 2025. URL: https://arxiv.org/abs/2503.22848.

[39] Hamed Hatami. Random cubic graphs are not homomorphic to the cycle of size 7. *Journal of Combinatorial Theory, Series B*, 93(2):319–325, 2005.

[40] Alan J. Hoffman and Robert R. Singleton. On Moore graphs with diameters 2 and 3. *IBM Journal of Research and Development*, 4(5):497–504, 1960. doi:10.1147/rd.45.0497.

[41] Stasys Jukna and Georg Schnitger. Min-rank conjecture for log-depth circuits. *Journal of Computer and System Sciences*, 77(6):1023–1038, 2011.

[42] Richard V. Kadison and John R. Ringrose. *Fundamentals of the Theory of Operator Algebras, Volume II: Advanced Theory*, volume 16 of *Graduate Studies in Mathematics*. American Mathematical Society, Providence, RI, 1997.

[43] Alexander S. Kechris. Amenable versus hyperfinite Borel equivalence relations. *Journal of Symbolic Logic*, 58(3):894–907, 1993. doi:10.2307/2275102.

[44] Alexander S. Kechris. *Classical Descriptive Set Theory*, volume 156 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1995. doi:10.1007/978-1-4612-4190-4.

[45] Harry Kesten. Symmetric random walks on groups. *Transactions of the American Mathematical Society*, 92:336–354, 1959. doi:10.1090/S0002-9947-1959-0109367-6.

[46] Michael Larsen, James Propp, and Daniel Ullman. The fractional chromatic number of Mycielski's graphs. *Journal of Graph Theory*, 19(3):411–416, 1995. doi:10.1002/jgt.3190190313.

[47] Richard J. Lipton and Robert Endre Tarjan. A separator theorem for planar graphs. *SIAM Journal on Applied Mathematics*, 36(2):177–189, 1979.

[48] Richard J. Lipton and Robert Endre Tarjan. Applications of a planar separator theorem. *SIAM Journal on Computing*, 9:615–627, 1980.

[49] Wolfgang Lück. Approximating L^2-invariants by their finite-dimensional analogues. *Geometric and Functional Analysis*, 4(4):455–481, 1994.

[50] Arya Mazumdar. On a duality between recoverable distributed storage and index coding. In *Proceedings of the IEEE International Symposium on Information Theory (ISIT)*, 2014. arXiv:1401.2672.

[51] Arya Mazumdar, Andrew McGregor, and Sofya Vorotnikova. Storage capacity as an information-theoretic vertex cover and the index coding rate. *IEEE Transactions on Information Theory*, 65(9):5580–5591, 2019.

[52] Brendan D. McKay. The expected eigenvalue distribution of a large regular graph. *Linear Algebra and its Applications*, 40:203–216, 1981. doi:10.1016/0024-3795(81)90150-6.

[53] Jan Mycielski. Sur le coloriage des graphs. *Colloquium Mathematicum*, 3:161–162, 1955.

[54] Jaroslav Nešetřil. Aspects of structural combinatorics (graph homomorphisms and their use). *Taiwanese Journal of Mathematics*, 3(4):381–423, 1999. doi:10.11650/twjm/1500407157.

[55] Hiroshi Nozaki. Linear programming bounds for regular graphs. *Graphs and Combinatorics*, 31(6):1973–1984, 2015. doi:10.1007/s00373-015-1613-7.

[56] Donald S. Ornstein and Benjamin Weiss. Ergodic theory of amenable group actions. I. The Rohlin lemma. *Bulletin of the American Mathematical Society (N.S.)*, 2(1):161–164, 1980. doi:10.1090/S0273-0979-1980-14702-3.

[57] Wolfgang J. Paul, Nicholas Pippenger, Endre Szemerédi, and William T. Trotter. On determinism versus non-determinism and related problems. In *24th Annual Symposium on Foundations of Computer Science (FOCS 1983)*, pages 429–438, 1983.

[58] Søren Riis. Information flows, graphs and their guessing numbers. *Electronic Journal of Combinatorics*, 14:Research Paper R44, 2007.

[59] Moshe Rosenfeld. Almost orthogonal lines in E^d. In Peter Gritzmann and Bernd Sturmfels, editors, *Applied Geometry and Discrete Mathematics: The Victor Klee Festschrift*, volume 4 of *DIMACS Series in Discrete Mathematics and Theoretical Computer Science*, pages 489–492. American Mathematical Society, 1991.

[60] Rahul Santhanam. On separators, segregators and time versus space. ECCC Report TR01-022, Electronic Colloquium on Computational Complexity, 2001. Also in Proc. 16th IEEE Conference on Computational Complexity, 2001, pp. 286–294.

[61] Geoff Tims. *Haemers' Minimum Rank*. PhD thesis, Iowa State University, 2013. doi:10.31274/etd-180810-3393.

[62] Juan Pablo Traverso Gianini. Berge's theorem, the Tutte–Berge formula and the Gallai–Edmonds theorem in Lean 4. Lean formalization, repository jtraverso/berge-matching, 2026. Commit f01d042c071c778fb3f432312356136a92369076. URL: https://github.com/jtraverso/berge-matching.

[63] Leslie G. Valiant. Graph-theoretic arguments in low-level complexity. In *Mathematical Foundations of Computer Science (MFCS)*, volume 53 of *Lecture Notes in Computer Science*, pages 162–176. Springer, 1977.

[64] Andrew Vince. Star chromatic number. *Journal of Graph Theory*, 12(4):551–559, 1988.

[65] John von Neumann. Zur allgemeinen Theorie des Masses. *Fundamenta Mathematicae*, 13(1):73–116, 1929. doi:10.4064/fm-13-1-73-116.

[66] Nicholas C. Wormald. The asymptotic distribution of short cycles in random regular graphs. *Journal of Combinatorial Theory, Series B*, 31:168–182, 1981. doi:10.1016/S0095-8956(81)80022-6.

[67] Xuding Zhu. Circular chromatic number: A survey. *Discrete Mathematics*, 229(1–3):371–410, 2001. doi:10.1016/S0012-365X(00)00217-X.

[68] Robert J. Zimmer. Hyperfinite factors and amenable ergodic actions. *Inventiones Mathematicae*, 41(1):23–31, 1977. doi:10.1007/BF01390162.